

Health Education

& *Sterilization and Disinfection*

UNIT I Health Education

Principles · Ethics · Attributes of the health educator · Essential steps · Introduction to the main methods · History, development and growth of health education in India · Various methods of health education

UNIT II Sterilization & Disinfection

Physical, chemical and mechanical methods · Disposal of contaminated media · Sterilization of syringes, glasswares and apparatus · Biomedical waste management

Complete theory notes — no question bank, pure study material

18 chapters · 140+ tables and summary boxes · A4

Compiled from standard references: Park's Textbook of Preventive and Social Medicine; Ananthanarayan & Paniker's Textbook of Microbiology;

Bio-Medical Waste Management Rules, 2016; WHO and CDC guidelines on disinfection and sterilization.

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How to Use These Notes

The notes are organised as a two-part book that follows the syllabus wording exactly. Everything is explained in running prose first and then compressed into a table, so that the first reading builds understanding and every later reading is a rapid revision from the tables.

Visual cue	What it means
Blue chapter banner	Start of a new chapter. A page break occurs only at the end of a chapter, so each chapter reads continuously.
Teal underlined heading	A major section of the chapter.
Green heading with a triangle	A sub-section.
Blue-shaded box marked DEFINITION	A definition or principle to be reproduced verbatim.
Yellow-shaded box marked MNEMONIC	A memory aid or a string of numbers worth memorising.
Pink-shaded box marked HIGH-YIELD FOR EXAM	Facts that are asked most often. If time is short, read only these.
Violet-shaded box marked REMEMBER	A clarification, a common confusion, or a practical tip.
Purple bold text	A key term or the name of a concept.
Dark red bold text	A number, date, temperature or value that must be recalled exactly.
Bold text	Emphasis on the examinable part of a sentence.
<i>Italic text</i>	Scientific names, foreign words and book titles.

REMEMBER

- The two parts are independent — they may be read in either order.
- **Chapter 18 is a pure revision chapter**; read it last, and read it again on the day before the examination.
- No multiple-choice questions are included: this is a study text, by design.

HEALTH EDUCATION

Principles · Ethics · The Health Educator · Essential Steps · Methods · Growth in India

CHAPTER 1 HEALTH, DISEASE AND THE FOUNDATIONS OF HEALTH EDUCATION

Health education is an applied science. Before its principles, ethics and methods can make sense, the vocabulary it works with — health, disease, prevention, health promotion — must be precise. This chapter builds that base, and almost every definition in it is independently examinable.

1.1 The Concept of Health

DEFINITIONS OF HEALTH

- **WHO definition (1948, from the Preamble to the WHO Constitution):** Health is a state of **complete physical, mental and social well-being** and not merely the absence of disease or infirmity.
- **1978 (Alma-Ata) addition:** ... and the ability to lead a *socially and economically productive life*.
- **Operational definition (WHO):** health is a condition in which there is (a) no obvious evidence of disease and the person is functioning normally, and (b) no minimal or hidden impairment of the several organs.

- The WHO definition has been criticised as an **idealistic goal rather than a realistic proposition** — the word “complete” makes it unattainable and unmeasurable; hence the operational definition was framed for practical use.
- Health is **relative, multidimensional, a positive concept** (not merely absence of disease), a **fundamental human right**, and an **essential component of development**.
- WHO's slogan “**Health for All by the Year 2000 AD**” was adopted at the **30th World Health Assembly, 1977** and the means to achieve it — **Primary Health Care** — was declared at **Alma-Ata (Almaty, USSR), 6–12 September 1978**.

► 1.1.1 Dimensions of Health

TABLE 1.1 DIMENSIONS OF HEALTH

Dimension	What it covers	Key markers / exam points
Physical	Perfect functioning of the body; all organs in harmony; the “biological” dimension.	Good complexion, clean skin, bright eyes, firm flesh, sweet breath, good appetite, sound sleep, regular bowel/bladder, coordinated movements, normal pulse, BP and weight for height.
Mental	Ability to respond to the many varied experiences of life with flexibility and a cheerful temper; absence of mental illness.	Free from internal conflict, self-respecting, knows self, adjusts to environment, accepts criticism, controls anger/fear. Mental health is not merely absence of mental illness.
Social	Quantity and quality of interpersonal ties; harmony and integration within and between the individual and the community.	Social skills, social functioning, ability to see oneself as a member of a society. Measured through social adjustment and social support.
Spiritual	That part of the individual which reaches out for meaning and purpose in life; integrity, principles and ethics.	Included as a dimension by WHO; covers purpose in life, transcendent values, faith, commitment.
Emotional	Recognising and appropriately expressing feelings; earlier merged with mental health, now treated separately.	Mental = “cognition / knowing”; Emotional = “affective / feeling”.
Vocational	Work, occupation and economic self-sufficiency as a source of self-esteem and health.	Becomes especially significant when a person suddenly loses a job or retires.
Others	Philosophical, cultural, socio-economic, environmental, educational, nutritional, curative and preventive dimensions.	Often listed together as “other dimensions” in textbooks.

MNEMONIC

The four **core WHO dimensions** = P M S S — **Physical, Mental, Social, Spiritual**. Add E V (**Emotional, Vocational**) for the six

commonly asked dimensions.

► 1.1.2 Concepts of Health (Evolution of Thought)

TABLE 1.2 CHANGING CONCEPTS OF HEALTH

Concept	Core idea	Limitation
Biomedical (germ theory)	Health = absence of disease; the body is a machine and disease is its breakdown; doctor's task is repair.	Ignored environmental, social, psychological and cultural determinants.
Ecological	Health is a dynamic equilibrium between man and his environment ; disease is maladjustment. Introduced the ideas of ecology and human ecology.	Focuses on adaptation, less on social organisation.
Psychosocial	Health is determined not only by biological factors but also by social, psychological, cultural, economic and political factors.	Needed integration with biological understanding.
Holistic	A synthesis of all the above — health is a unified, multidimensional process involving the well-being of the whole person in the context of the environment. Recognises the influence of agriculture, education, housing, industry and public health.	The currently accepted concept; emphasises health promotion and intersectoral coordination .

► 1.1.3 Spectrum and Determinants of Health

Spectrum of health: health and disease lie along a continuum (“health–disease spectrum”) — positive health → better health → freedom from sickness → unrecognised sickness → mild sickness → severe sickness → death. There is no single cut-off point; the lowest point is death and the highest point is optimum or positive health.

TABLE 1.3 DETERMINANTS OF HEALTH

Group of determinants	Examples
Biological / genetic	Genetic constitution, sex, age, race; inborn errors of metabolism, haemoglobinopathies, chromosomal anomalies.
Behavioural & socio-cultural	Lifestyle: diet, smoking, alcohol, physical activity, sexual behaviour, customs, beliefs, religion, cultural practices.
Environment	“Internal” (tissues, organs) and “external” — physical, biological and psychosocial; water, air, housing, waste, noise, radiation.
Socio-economic	Economic status, education (especially female literacy), occupation, political system. Education is regarded as a key determinant — the infant mortality rate is closely linked to female literacy.
Health services	Availability, accessibility, affordability, acceptability and quality of care; immunisation, antenatal care, safe delivery.
Ageing of population, gender, other factors	Growing elderly population, gender discrimination, information technology, globalisation and trade.

1.2 Disease, Illness and Sickness

Disease — A physiological or psychological dysfunction — the **objective, pathological** state, as diagnosed by the physician. Literally *dis-ease*, the opposite of ease.

Illness — The **subjective** state of the person who feels aware of not being well.

Sickness — A **state of social dysfunction** — the role the individual assumes when ill (“sickness role”).

► 1.2.1 Epidemiological Triad and the Iceberg of Disease

- **Epidemiological triad** = **Agent + Host + Environment**. Disease results from an interaction of the three; this is the basis of the multifactorial causation concept.
- **Iceberg phenomenon of disease:** what is visible above the waterline is clinical disease; hidden below are pre-symptomatic, sub-clinical, latent, carrier and undiagnosed states. The submerged portion constitutes the **hidden mass of disease** and is of great epidemiological importance — it is the target of screening and of health education.

► 1.2.2 Natural History of Disease and Levels of Prevention

TABLE 1.4 LEVELS OF PREVENTION AND MODES OF INTERVENTION

Level of prevention	Phase of disease	Mode of intervention	Examples
Primordial	Before risk factors appear (in childhood)	Discouraging the emergence of risk-factor lifestyles	Health education to prevent children from starting smoking; promoting healthy eating habits in schools; national policy on tobacco and salt.
Primary	Pre-pathogenesis (susceptibility)	Health promotion and specific protection	Health education, nutrition, safe water, housing; immunisation, chemoprophylaxis, iodisation of salt, use of helmets, fluoridation.
Secondary	Early pathogenesis / pre-symptomatic	Early diagnosis and prompt treatment	Screening (cancer cervix, diabetes), case finding, DOTS for tuberculosis, treating hypertension early.
Tertiary	Late pathogenesis / disability	Disability limitation and rehabilitation	Preventing complications of diabetes; physiotherapy, prosthesis, reconstructive surgery, vocational and social rehabilitation.

HIGH-YIELD FOR EXAM

- **Health education is used at EVERY level of prevention**, but it is the dominant tool of **primordial** and **primary** prevention (health promotion).
- The **five modes of intervention** (Leavell & Clark) are: health promotion, specific protection, early diagnosis & treatment, disability limitation, rehabilitation.
- Rehabilitation has **four components**: **medical, vocational, social and psychological** rehabilitation.

1.3 Health Promotion and the Ottawa Charter

Health promotion (WHO, Ottawa 1986) — The **process of enabling people to increase control over, and to improve, their health**. It is not confined to health care; health is a resource for everyday life, not the objective of living.

Ottawa Charter, 1986	Content
Five action areas	1. Build healthy public policy 2. Create supportive environments 3. Strengthen community action 4. Develop personal skills (this is where health education sits) 5. Reorient health services towards prevention and promotion.
Three basic strategies	Advocate (for health), Enable (empower people, reduce inequity), Mediate (between differing interests in society).
Prerequisites for health	Peace, shelter, education, food, income, a stable ecosystem, sustainable resources, social justice and equity.

- The **First International Conference on Health Promotion** was held at **Ottawa, Canada in 1986** and produced the **Ottawa Charter**.
- The **Bangkok Charter for Health Promotion in a Globalized World** was adopted in **2005** (6th Global Conference); it added advocacy, investment, capacity building, regulation/legislation and partnership as key actions.
- **Health promotion approaches** (Park): (1) medical/preventive, (2) behaviour change, (3) educational, (4) empowerment, (5) social change.

1.4 Health Education and its Cousins — Distinguishing the Terms

TABLE 1.5 HEALTH EDUCATION VERSUS RELATED CONCEPTS

Term	Meaning	Relationship
Health education	Planned learning experiences that help people voluntarily change behaviour to improve health.	The educational core of health promotion.
Health promotion	The broad process of enabling control over health; includes policy, environment and services besides education.	Wider than health education . Health education ⊂ health promotion.

Term	Meaning	Relationship
IEC — Information, Education & Communication	A strategy combining information giving, education and communication to increase awareness and promote behaviour change; the term used in almost all Indian national health programmes.	The operational form of health education in Indian public health.
BCC — Behaviour Change Communication	An interactive, evidence-based, audience-segmented process aimed specifically at changing and sustaining behaviour.	The newer, more focused successor to IEC ; emphasises two-way dialogue.
Health propaganda / publicity	One-way dissemination of slogans and messages to persuade, without concern for understanding or voluntary choice.	Not health education — education respects free, informed choice.
Social marketing	Applying commercial marketing techniques (product, price, place, promotion) to promote socially beneficial behaviour — e.g. subsidised condoms, ORS.	A tool used within health promotion.
Health literacy	The cognitive and social skills that determine the motivation and ability of individuals to gain access to, understand and use information to promote health.	The outcome of effective health education.

EXAM POINT

- Health education relies on **voluntary, informed decision-making**; propaganda relies on persuasion and repetition. This distinction is a classic MCQ.
- Education informs and enables; propaganda persuades; indoctrination compels.**

1.5 Health Education in Primary Health Care

The Alma-Ata Declaration (1978) listed **eight essential components of primary health care**. The very first is education:

- Education concerning prevailing health problems and the methods of preventing and controlling them** — i.e. health education.
- Promotion of food supply and proper nutrition.
- An adequate supply of safe water and basic sanitation.
- Maternal and child health care, including family planning.
- Immunisation against the major infectious diseases.
- Prevention and control of locally endemic diseases.
- Appropriate treatment of common diseases and injuries.
- Provision of essential drugs.

HIGH-YIELD FOR EXAM

Health education is the FIRST of the eight elements of primary health care and is described as the **most cost-effective single intervention in public health**. The four principles (pillars) of PHC are **equitable distribution, community participation, intersectoral coordination and appropriate technology**.



2.1 Definitions of Health Education

Several definitions are quoted in examinations. Learn the author with the definition.

TABLE 2.1 STANDARD DEFINITIONS

Source	Definition
WHO Expert Committee, 1969	Health education, like general education, is concerned with change in knowledge, feelings and behaviour of people . In its most usual form it concentrates on developing such health practices as are believed to bring about the best possible state of well-being.
Alma-Ata / WHO (1978)	A process aimed at encouraging people to want to be healthy, to know how to stay healthy, to do what they can individually and collectively to maintain health, and to seek help when needed.
Grout	Health education is the translation of what is known about health into desirable individual and community behaviour patterns by means of the educational process.
Dorothy Nyswander	Health education is the process of providing learning experiences which favourably influence understanding, attitudes and conduct in regard to individual and community health.
Lawrence W. Green (1980)	Any combination of learning experiences designed to facilitate voluntary actions conducive to health.
Joint Committee on Health Education Terminology (2001)	Any combination of planned learning experiences based on sound theories that provide individuals, groups and communities the opportunity to acquire information and the skills needed to make quality health decisions.
National Conference on Preventive Medicine (USA)	A process that informs, motivates and helps people to adopt and maintain healthful practices and lifestyles, advocates environmental changes as needed and conducts professional training and research to the same end.

HIGH-YIELD FOR EXAM

- Two words recur in almost every definition and are the key to MCQs: **voluntary** and **behaviour change**.
- Health education aims at change in the **three domains of learning** — Knowledge (cognitive), Attitude/feelings (affective) and Practice/behaviour (psychomotor): the basis of the famous **KAP** studies (Knowledge–Attitude–Practice).

2.2 Aim and Objectives of Health Education

► 2.2.1 Aim

The ultimate aim of health education is **not merely to give information but to bring about a change in behaviour**, so that people attain and maintain positive health through their own effort and action. It seeks to make health an asset valued by the community.

► 2.2.2 The Three Classical Objectives (WHO)

Objective	What it involves	Level of learning
1. Informing people	Disseminating scientific knowledge about health, diseases and their prevention; the dissemination of information is the first and most basic objective.	Cognitive — knowledge
2. Motivating people	Persuading people to change and adopt healthy practices; motivation involves arousing interest, creating a felt need and sustaining desire.	Affective — attitude
3. Guiding into action	Helping people to take action, to use health services wisely and to sustain the new behaviour — by working with them, not for them.	Psychomotor — practice

► 2.2.3 Specific Objectives

- To make people **aware** of their own health problems and the resources available.
- To help people **understand** that health is a valuable community asset.
- To promote the **correct use of health services** — to seek care early and avoid quackery and self-medication.

- To develop a **sense of responsibility** for one's own health and that of others.
- To bring about a **change in attitudes and values**, not just knowledge.
- To develop and maintain **healthful behaviour and skills** (hand-washing, oral rehydration, breast-feeding, exercise).
- To encourage **community participation** and self-reliance in solving local health problems.
- To secure the **adoption and maintenance of healthy lifestyles**.
- To create demand for and support healthy **public policy** and environmental change.

REMEMBER

Objectives should be written in **SMART** form — **S**pecific, **M**easurable, **A**chievable/Attainable, **R**elevant/Realistic, **T**ime-bound.

2.3 Scope and Content of Health Education

The scope of health education is as wide as health itself. The WHO Expert Committee grouped the content under the following heads:

TABLE 2.2 CONTENT OF HEALTH EDUCATION

Area	Content taught
Human biology	Structure and functioning of the body; how to keep it fit; personal responsibility for one's body.
Nutrition	Balanced diet, locally available low-cost foods, nutritional deficiency disorders, infant and young child feeding, breast-feeding, supplementary feeding, food hygiene and adulteration.
Hygiene	Personal hygiene — bathing, care of skin, hair, teeth, eyes, ears, nose, hands and feet, clothing, rest, sleep, exercise, posture. Environmental hygiene — safe water, excreta and refuse disposal, housing, ventilation, lighting, vector control.
Family health care	Antenatal, natal and postnatal care, family planning, spacing of children, care of the newborn, growth monitoring, immunisation, adolescent health, care of the elderly.
Control of communicable diseases	Modes of spread and prevention of tuberculosis, malaria, dengue, diarrhoeal disease, hepatitis, HIV/AIDS, COVID-19, vaccine-preventable diseases; notification and isolation.
Control of non-communicable diseases	Hypertension, diabetes, cancer, cardiovascular disease, obesity, COPD; risk-factor reduction — tobacco, alcohol, salt, sugar, fat, physical inactivity, stress.
Mental health	Coping with stress, sleep hygiene, substance abuse prevention, suicide prevention, reduction of stigma.
Prevention of accidents	Road, home, occupational, agricultural and industrial accidents; first aid; poisoning; drowning; burns.
Use of health services	What services exist, where, when and how to use them; referral; health insurance; entitlements; rights of the patient.

2.4 Approaches and Models in Health Education

Approach (Park)	Description
Regulatory / managed prevention approach	Legislation and enforcement — compulsory immunisation, seat-belt and helmet laws, food standards, ban on smoking in public places. Effective but may be resented as compulsion.
Service approach	Providing health services at the doorstep on the assumption that people will use them. Often fails because it is based on the planners' assumptions and not on the felt needs of the people.
Health education approach	Providing information and education so that people act on their own; slow but produces permanent, self-sustaining change. The approach of choice.
Primary health care approach	Community participation and involvement of people in planning and implementation, with health education as its backbone.

► 2.4.1 Models of Health Education

Model	Assumption	Comment
Medical model	Ill-health is due to disease; information from an expert will make people comply.	Expert-centred, paternalistic, one-way; limited success.
Motivation model	Behaviour changes when the individual is motivated and internalises the value of health.	Change is gradual but lasting; needs interpersonal work.
Social intervention / social change model	Behaviour is determined by the social environment; change society and behaviour follows.	Requires policy, legislation and environmental modification.

2.5 Health Education and the Pharmacist

For a **Junior Pharmacist** health education is a daily duty, not a theoretical subject. The pharmacist is frequently the most accessible health professional a patient meets, and the last one before the medicine is used.

- **Patient counselling** on the dose, frequency, duration, route, timing in relation to food, expected effects and common adverse effects of each medicine.
- Ensuring **adherence/compliance** — explaining why a full course of antibiotics or anti-tubercular therapy must be completed; demonstrating inhaler, insulin pen, ORS and eye-drop technique.
- Promoting **rational drug use** and discouraging self-medication, irrational antibiotic use and **antimicrobial resistance**.
- Advising on **storage** of medicines (cool, dry, dark, out of reach of children; cold chain for vaccines and insulin) and safe **disposal of expired medicines**.
- Explaining **look-alike / sound-alike** hazards, expiry dates, generic equivalence and cost-effective generic substitution (Jan Aushadhi).
- Counselling on **lifestyle** — tobacco cessation, diet, exercise, salt restriction — and on **ORS, immunisation, family planning and iron-folic acid**.
- Reporting and educating about **adverse drug reactions** (pharmacovigilance, PvPI) and drug–food/drug–drug interactions.
- Health education in the pharmacy through **posters, leaflets, display boards** and participation in health days and camps.

2.6 Benefits and Limitations

Benefits	Limitations / difficulties
Cheapest and most cost-effective public-health intervention; needs no costly technology.	Results are slow and not immediately visible or measurable.
Produces permanent, self-sustaining change because it is voluntary.	Behaviour is deeply rooted in culture, custom and religion — difficult to change.
Empowers people; builds self-reliance and community participation.	Illiteracy, poverty, superstition and lack of felt need obstruct it.
Improves utilisation of existing health services and reduces the load of preventable disease.	Knowledge alone does not guarantee practice — the KAP gap .
Applicable at all levels of prevention and in all settings.	Shortage of trained health educators and of evaluation of programmes.



CHAPTER 3 PRINCIPLES OF HEALTH EDUCATION

These principles are the single most frequently examined part of the health education syllabus. Learn the **name** of each principle together with one concrete example.

TABLE 3.1 PRINCIPLES OF HEALTH EDUCATION

Principle	Meaning	Illustration
1. Credibility	The message must be scientifically accurate , consistent with facts and compatible with the beliefs of the community. It must come from a trusted, respected source . Once credibility is lost it is almost impossible to regain.	A pharmacist who gives correct, unbiased drug information is believed; exaggerated claims destroy trust.
2. Interest	People will listen only to what interests them. Health teaching must be related to the felt needs of the people — a need actually perceived by them, not merely assumed by the educator.	Talking about malaria during an outbreak; teaching a diabetic about foot care after an ulcer has frightened him.
3. Participation	Active involvement of the learner produces far better results than passive listening. The highest degree of participation is the hallmark of good health education — it is based on the psychological principle of active learning.	Group discussion, panel discussion, workshops, demonstrations by the learner, involving village leaders in planning.
4. Motivation	Every person has a fundamental desire to learn; awakening this desire is motivation. Primary motives are inborn (sex, hunger, survival); secondary motives are created by outside forces (praise, rivalry, rewards, punishment, recognition).	Incentives and rewards in family-planning and immunisation programmes; appreciation of a mother who completes her child's immunisation.
5. Comprehension	Teach at the level of understanding of the audience. Never use words or concepts the audience cannot follow; assess the audience's literacy and existing knowledge first.	Saying "raised blood sugar" instead of "hyperglycaemia"; using the local dialect.
6. Reinforcement	Few people learn everything in a single exposure. Repetition at intervals , through different channels, fixes the message.	Repeating the pulse-polio message by radio, poster, school children and house visits.
7. Learning by doing	Learning is far more effective when the learner performs the act. "If I do it, I know it."	Making the mother actually prepare and taste the ORS solution rather than only hearing about it.
8. Known to unknown	Proceed from what the people already know to what they do not — from the simple to the complex, and from the concrete to the abstract. Use the existing knowledge as a peg on which to hang new knowledge.	Explaining germs by starting from the familiar idea of "dirt causes disease".
9. Setting an example	The educator must practise what he preaches . Example is more powerful than precept.	A health worker who smokes cannot run a tobacco-cessation clinic credibly.
10. Good human relations	Sharing information, warmth, sympathy, kindness and a non-judgemental manner build the rapport on which all education rests. Approachability of the health worker is vital.	Listening patiently to an anxious mother before advising her.
11. Feedback	The educator must obtain the reaction of the audience and modify the message, channel and method accordingly. Feedback converts one-way communication into two-way communication .	Asking the patient to repeat back the dosage instructions — the "teach-back" method.
12. Leaders / community leaders	Work through the local leaders — opinion leaders, teachers, panchayat members, religious heads, ASHA — because people imitate those they respect. Leaders are the channel to the community.	Training village opinion leaders as change agents for sanitation.
13. Soil, seed and sower	The soil = the community/learner, the seed = the message/facts, and the sower = the health educator. Good seed must be sown by a good sower in well-prepared soil.	Community must be prepared (rapport, felt need) before the message is delivered.
14. Total personality / whole person	Health education must be directed to the whole person and to all dimensions of health, not to one disease or one organ in isolation.	Counselling a hypertensive about diet, exercise, stress and tobacco, not only about tablets.
15. Use of multiple channels / media	No single method is adequate; combining individual, group and mass approaches reinforces the message through several senses.	A camp using a film show, a demonstration and personal counselling together.

MNEMONIC

- A convenient string for the core principles: C I P M C R L K S G F L – **C**redibility, **I**nterest, **P**articipation, **M**otivation, **C**omprehension, **R**einforcement, **L**earning by doing, **K**nown to unknown, **S**etting an example, **G**ood human relations, **F**eedback, **L**eaders.
- Remember the trio **soil – seed – sower** = community – message – educator.

3.1 Motivation in Detail

Type	Definition	Examples
Primary (biological) motives	Inborn, unlearned drives common to all human beings.	Hunger, thirst, sex, sleep, self-preservation, maternal drive.
Secondary (social) motives	Acquired through learning and social interaction; created by external forces.	Praise, recognition, rivalry, competition, love of children, fear of disease, desire for prestige, money, punishment.
Incentives	Objects or conditions that arouse or sustain motivated behaviour; may be positive (reward) or negative (punishment).	Cash incentive under Janani Suraksha Yojana; free supply of spectacles; certificates to “open-defecation-free” villages.

► 3.1.1 Maslow's Hierarchy of Needs (Basis of Motivation)

A need lower in the hierarchy must be reasonably satisfied before a higher need motivates behaviour — which is why health advice fails in a starving family.

Level	Need	Health-education implication
5 (highest)	Self-actualisation — realising one's potential	Appeals to personal excellence, fitness goals, being a role model.
4	Esteem — self-respect, recognition, status	Awards to model villages/mothers; praise for compliance.
3	Love and belonging — affection, acceptance by the group	Group activities, peer educators, mothers' meetings, self-help groups.
2	Safety and security — protection, freedom from fear	Immunisation, accident prevention, insurance, safe workplaces.
1 (lowest)	Physiological — air, water, food, shelter, sleep, sex	Nutrition and safe water must come first; teaching hygiene to a hungry community will not work.

3.2 Principles of Learning Applied to Health Education

- **Learning is an active process** — the learner must participate.
- **Learning proceeds from the known to the unknown**, simple to complex, concrete to abstract, whole to part.
- **Readiness** — physical, mental and emotional readiness of the learner (maturation) is essential.
- **Reinforcement and practice** strengthen learning; **recency** and **primacy** effects make the first and the last points best remembered.
- **Satisfaction and success** promote learning; repeated failure discourages it.
- **Multi-sensory learning** is superior — the more senses involved, the better the retention.
- **Transfer of learning** — what is learnt should be applicable to real life.
- **Individual differences** — people learn at different rates; the learning curve is not uniform (plateaus occur).

HIGH-YIELD FOR EXAM

- Approximate contribution of the senses to learning: **sight ≈ 83%**, hearing ≈ 11%, smell ≈ 3.5%, touch ≈ 1.5%, taste ≈ 1%. Hence **visual aids are the most effective single class of aid**.
- Approximate retention: we remember about **10% of what we read, 20% of what we hear, 30% of what we see, 50% of what we see and hear, 70% of what we say, and 90% of what we say and do** — the rationale for “learning by doing” (Edgar Dale's Cone of Experience).

3.3 Barriers to Effective Health Education

Barrier	Examples
Physiological	Deafness, poor vision, defective speech, fatigue, illness, pain, hunger.

Barrier	Examples
Psychological	Low intelligence, poor memory, emotional disturbance, anxiety, fear, prejudice, neurosis, hostility to the educator.
Environmental	Noise, poor lighting, overcrowding, invisibility of the speaker, uncomfortable seating, distance, bad weather.
Cultural	Illiteracy, language and dialect differences, customs, taboos, religious beliefs, food fads, superstition, blind faith, caste and gender norms.
Social	Poverty, social class distance between educator and learner, unemployment, migration, lack of leadership.
Programme-related	Poorly trained educators, unsuitable methods and media, inadequate funds, no follow-up, no evaluation, message not in local language.



CHAPTER 4 ETHICS IN HEALTH EDUCATION

Health education deals with people's beliefs, bodies and private lives. Ethics defines the limits within which the educator may work. Ethical practice is what separates **health education** from **manipulation**.

4.1 The Core Ethical Principles

TABLE 4.1 CORE ETHICAL PRINCIPLES

Principle	Meaning	Application in health education
Autonomy (respect for persons)	The right of a competent individual to make his or her own decisions, free from coercion.	Present balanced facts and let the person decide; never coerce a woman into sterilisation or a patient into a trial. Voluntariness is the essence of health education.
Beneficence	The duty to do good and to act in the best interest of the individual and community.	Choose messages that genuinely benefit the audience; promote proven interventions.
Non-maleficence (primum non nocere)	The duty to do no harm, including psychological and social harm.	Avoid messages that frighten unnecessarily, stigmatise, or create guilt; avoid exaggerated fear appeals.
Justice / equity	Fair distribution of benefits and burdens; the most needy must not be the least served.	Reach the illiterate, the poor, the disabled, tribal and remote populations; materials in local languages and accessible formats.
Veracity (truthfulness)	Obligation to tell the truth and not to deceive.	Never overstate benefits or conceal risks of a drug, vaccine or procedure; correct one's own errors.
Fidelity	Faithfulness to promises and to the trust reposed in the professional.	Keep appointments and commitments made to the community.
Confidentiality / privacy	Information obtained in a professional relationship must not be disclosed without consent.	Never name an HIV-positive or TB patient in a community meeting; secure records and digital data.

MNEMONIC

The **four classical principles of bioethics** (Beauchamp & Childress) = **A B N J** → Autonomy, Beneficence, Non-maleficence, Justice.

4.2 Informed Consent in Health Education and Research

- Consent must be **informed** (full disclosure in understandable language), **voluntary** (free of coercion or undue inducement) and given by a person **competent** to consent.
- The participant must be told the purpose, procedure, risks, benefits, alternatives, confidentiality safeguards and the **right to withdraw at any time without penalty**.
- Consent for photography, filming or the use of a patient's story or image in teaching material must be **separate and explicit**.
- For minors and persons unable to consent, **assent** of the individual plus consent of the parent/legal guardian is required.
- Research involving human participants requires clearance by an **Institutional Ethics Committee**; in India the governing document is the **ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants (2017)**.

4.3 Code of Ethics for the Health Education Profession

The **Coalition of National Health Education Organizations (CNHEO)** code is the internationally quoted standard. It is arranged in six articles, each describing a set of responsibilities.

TABLE 4.2 CNHEO CODE OF ETHICS FOR THE HEALTH EDUCATION PROFESSION

Article	Responsibility to ...	Key obligations
I	The public	Support the right of individuals to make informed decisions; respect diversity, values and dignity; act on issues that can adversely affect health.
II	The profession	Maintain and improve competence; promote the profession's integrity; give credit to others' work; avoid conflicts of interest.
III	Employers	Accept only responsibilities for which one is qualified; be accountable and honest about capabilities and outcomes; use resources properly.

Article	Responsibility to ...	Key obligations
IV	The delivery of health education	Use appropriate, evidence-based strategies; be truthful about the effectiveness of programmes; respect the rights of participants.
V	Research and evaluation	Conduct research ethically and honestly; protect privacy and dignity; report results accurately, including negative findings; avoid plagiarism and fabrication.
VI	Professional preparation	Select students fairly; provide quality training and supervision; act as a role model for trainees.

4.4 Common Ethical Dilemmas and Pitfalls

Issue	The ethical problem	Correct practice
Victim blaming	Blaming the individual for a disease caused largely by poverty, environment or genetics — e.g. blaming a malnourished mother for her child's stunting.	Address structural determinants too; use supportive, non-accusatory language.
Fear appeals / scare tactics	Graphic images and threats may cause anxiety, denial or avoidance rather than behaviour change, especially if no realistic solution is offered.	Use moderate fear only when an effective, feasible action is simultaneously offered (efficacy message).
Stigmatisation and labelling	Messages about HIV, tuberculosis, leprosy, mental illness or obesity may reinforce discrimination.	Use person-first language ("a person with leprosy", not "a leper"); involve affected people in designing messages.
Paternalism	Deciding for people "for their own good", withholding information.	Inform fully and respect the decision even if the educator disagrees.
Conflict of interest / commercial sponsorship	Accepting support from tobacco, alcohol, infant-formula or pharmaceutical companies may bias messages.	Disclose all funding; refuse sponsorship that compromises the message. Note the IMS Act, 1992 (Infant Milk Substitutes Act) which restricts promotion of infant formula.
Cultural insensitivity	Imposing outside norms; ignoring religion, caste, gender or local custom.	Adapt messages to local culture; use local idiom, folk media and local leaders.
Data privacy in digital health education	Mobile apps, teleconsultation and social media generate identifiable health data.	Obtain consent for data use; anonymise; store securely; comply with applicable data-protection law.
Misinformation and unverified claims	Forwarding unverified remedies, especially on social media, can cause direct harm.	Verify from authoritative sources before communicating; correct misinformation promptly and respectfully.

4.5 Ethics for the Pharmacist as Health Educator

In India the conduct of a registered pharmacist is governed by the **Pharmacy Act, 1948** and the **Pharmacy Practice Regulations, 2015** framed by the Pharmacy Council of India, besides the Drugs and Cosmetics Act, 1940.

- Dispense only against a **valid prescription** for prescription-only (Schedule H, H1, X) medicines; never encourage self-medication with antibiotics.
- Maintain **patient confidentiality**; prescription records must not be disclosed except as required by law.
- Give **unbiased drug information**; do not let trade incentives influence advice or substitution.
- Do not **claim to diagnose or to be a physician**; refer when the condition is beyond the pharmacist's scope.
- Keep **professional competence** current through continuing education.
- Display the **registration certificate**, observe conditions of the drug licence, and maintain prescribed registers (including Schedule H1 register).
- Never advertise medicines in a manner prohibited by the **Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954**.

HIGH-YIELD FOR EXAM

- The single most quoted ethical requirement of health education is that behaviour change must be **voluntary and informed** — any element of force converts education into coercion.
- Confidentiality may be **legally overridden** in narrow situations such as notification of a notifiable disease or a court order — recognised as "privileged communication".



CHAPTER 5 THE HEALTH EDUCATOR — ATTRIBUTES, ROLES AND RESPONSIBILITIES

Health educator — A person who, by training and function, helps individuals, families and communities to acquire the knowledge, attitudes and skills needed to protect and promote their health. **Every health worker is a health educator**, but some — health education officers, extension educators, counsellors — are specialists in it.

5.1 Attributes / Qualities of a Good Health Educator

Attributes are conventionally grouped as **personal, professional and social**. This grouping makes the long list easy to reproduce in an examination.

► 5.1.1 Personal Attributes

Attribute	Why it matters
Good personality and pleasant appearance	Creates the first favourable impression; neat, clean, appropriately dressed.
Sound physical and mental health	The educator is a living example of what he teaches; personal hygiene and healthy habits carry more weight than words.
Sincerity, honesty and integrity	The basis of credibility; never misleads, never exaggerates.
Patience, tolerance and perseverance	Behaviour change is slow; resistance, ridicule and failure must be absorbed without irritation.
Empathy and sympathy	The ability to feel with the people and to see the problem from their point of view; empathy, not pity .
Emotional stability, self-control and cheerfulness	Anger, sarcasm or discouragement destroys rapport.
Humility and a non-judgemental attitude	Never talks down to people, never ridicules beliefs however irrational.
Enthusiasm, initiative and resourcefulness	Improvises aids from locally available material; works with limited resources.
Sense of humour	Holds attention, relieves tension and makes the message memorable.
Punctuality, reliability and self-discipline	Keeping appointments with the community builds trust.

► 5.1.2 Professional Attributes

Attribute	Content
Sound and up-to-date knowledge	Of health sciences, the local disease pattern, national health programmes and the services available; must keep learning continuously.
Mastery of the subject to be taught	Able to answer questions accurately and to admit frankly when he does not know.
Skill in communication	Clear speech, correct use of the local language and idiom , good listening, effective use of non-verbal cues.
Skill in teaching methods and media	Able to select and use individual, group and mass methods and to prepare and handle audio-visual aids.
Ability to plan, organise and evaluate	Needs assessment, objective setting, budgeting, record-keeping, monitoring and evaluation.
Leadership and organising ability	Can mobilise the community, form committees, coordinate with other sectors.
Research and analytical ability	Can carry out a KAP survey, interpret data and use evidence.
Professional ethics	Confidentiality, respect for autonomy, truthfulness (Chapter 4).
Willingness to work as a team member	Health education is a team activity involving doctors, nurses, pharmacists, ASHA, teachers and voluntary agencies.

► 5.1.3 Social Attributes

- **Knowledge of the community** — its culture, customs, beliefs, taboos, social structure, caste and religious composition, festivals and economy.
- **Acceptance by the community** — preferably drawn from or resident in the community; ability to identify and work through **local leaders**.
- **Good human relations and social skills** — friendliness, approachability, the ability to mix freely with all sections without discrimination.

- **Respect for local culture** — modifies the message to fit the culture rather than attacking the culture.
- **Democratic outlook** — encourages participation and shared decision-making rather than issuing orders.

MNEMONIC

Qualities of a health educator in one line — “**K-A-S**”: **K**nowledge (sound and current), **A**ttitude (empathy, patience, honesty, non-judgemental), **S**kills (communication, teaching, planning, evaluation, leadership).

5.2 Areas of Responsibility of a Health Educator

The **National Commission for Health Education Credentialing (NCHEC)** framework is the standard statement of what a health educator is responsible for. The classical form lists **seven areas**; the current (2020) model lists **eight**.

TABLE 5.1 NCHEC AREAS OF RESPONSIBILITY

Area	Responsibility	Typical tasks
I	Assessment of needs and capacity	Collect and analyse data on health status, behaviours, determinants, existing resources; identify priority needs (KAP surveys, community diagnosis).
II	Planning	Involve stakeholders, set SMART objectives, select theory-based strategies, prepare a work plan, budget and timeline.
III	Implementation	Deliver the programme, train staff and volunteers, monitor progress, adapt to field realities.
IV	Evaluation and research	Design evaluation, collect and interpret data, measure process, impact and outcome; disseminate findings.
V	Advocacy	Influence policy, legislation and resource allocation for health; build coalitions (this is a separate area in the 2020 model).
VI	Communication	Determine the message and audience, select channels, develop and pre-test materials, deliver and evaluate the communication.
VII	Leadership and management	Administer programmes, manage people, budgets, materials and partnerships; ensure accountability.
VIII	Ethics and professionalism	Practise within the code of ethics, maintain competence, serve as a resource person and mentor, contribute to the profession.

HIGH-YIELD FOR EXAM

The classical **seven areas of responsibility** are often asked as: assess needs; plan; implement; conduct evaluation/research; **administer and manage**; **serve as a health education resource person**; **communicate and advocate**. The credential awarded in the USA is **CHES** (Certified Health Education Specialist), instituted in **1988** by NCHEC.

5.3 Functions of the Health Educator in the Field

- Carrying out **community diagnosis** and identifying felt needs.
- Planning, conducting and evaluating **health education programmes and campaigns**.
- Preparing, pre-testing and distributing **IEC material** in the local language.
- **Training** health workers, ASHA, anganwadi workers, teachers, dais and volunteers in health education skills.
- Acting as a **resource person and consultant** to other departments and voluntary agencies.
- Organising **school health education**, health exhibitions, health melas, camps and observance of health days.
- **Intersectoral coordination** with education, agriculture, rural development, water supply, ICDS and panchayat raj institutions.
- **Advocacy** with administrators and elected representatives for health-supportive policies.
- **Research and record-keeping**: KAP studies, operational research, reports and documentation.
- Providing **feedback** to programme managers about community response and misconceptions.

5.4 The Health Education Workforce in India

TABLE 5.2 HEALTH EDUCATION FUNCTIONARIES IN INDIA

Functionary	Level	Health education role
ASHA (Accredited Social Health Activist)	Village — one per 1000 population (relaxed in tribal/hilly areas)	The first port of call and the principal change agent for health education at village level; created under NRHM in 2005 ; an honorary volunteer paid by performance-based incentives; holds the drug kit and promotes institutional delivery, immunisation, ORS and family planning.
ANM / MPW (Female) — Auxiliary Nurse Midwife	Sub-centre — one per 5000 (3000 in hilly/tribal)	Antenatal, natal and postnatal education, immunisation, family planning counselling, nutrition and hygiene education.
MPW (Male) / MHW	Sub-centre	Education on communicable disease control, environmental sanitation, vector control, safe water.
Anganwadi Worker (AWW)	Anganwadi — one per 400–800 population (ICDS, launched 1975)	Nutrition and health education (NHED) to women 15–45 years, supplementary nutrition, pre-school education, growth monitoring.
Health Worker / Health Assistant (LHV)	Sub-centre / PHC	Supervision and on-the-job training of field workers in health education.
Block Extension Educator (BEE)	Block / PHC	The designated health education specialist at block level ; plans and conducts IEC activities, trains workers, maintains aids and equipment.
District Extension and Media Officer (DEMO) / District Health Education Officer	District	Coordinates all district IEC activity, media liaison, exhibitions, supply of material, supervision of BEEs.
State Health Education Bureau / State IEC Bureau	State	Plans state-level IEC strategy, produces material, trains district staff.
Central Health Education Bureau (CHEB)	National	Apex national body for health education (see Chapter 9).
Pharmacist	PHC / CHC / hospital / community pharmacy	Patient counselling, drug adherence, rational drug use, ADR reporting, storage and disposal advice, display of IEC material.

5.5 Difficulties Faced by the Health Educator

- **Illiteracy, poverty and superstition** in the community; blind faith in quacks and magico-religious remedies.
- **Absence of a felt need** — people do not perceive the problem as important.
- **Cultural and language barriers**; resistance to change deep-rooted customs.
- **Too few trained health educators**; health education added as an extra duty to already overburdened staff.
- **Inadequate funds, transport and audio-visual equipment**; material not adapted to the local language or culture.
- **Lack of coordination** between health and other departments.
- **Absence of evaluation** — programmes continue without knowing whether they work.
- **Misinformation on social media** competing with scientific messages.



CHAPTER 6 COMMUNICATION — THE VEHICLE OF HEALTH EDUCATION

Health education is delivered through communication; therefore the theory of communication is an examinable part of this unit.

Communication — The **two-way process of exchanging or sharing of ideas, facts, feelings and impressions** between a source and a receiver in such a way that there is common understanding. Derived from the Latin *communis* = “common”.

6.1 Aims and Functions of Communication

- To **inform** — transfer of knowledge and skills.
- To **educate** — produce understanding and change in attitude.
- To **motivate and persuade** — create the desire to act.
- To **change behaviour** and to sustain it.
- To **change social norms** and to advocate policy.
- To provide **entertainment with education** (“edutainment”, folk media).

6.2 Elements of the Communication Process

TABLE 6.1 ELEMENTS OF COMMUNICATION

Element	Description	Requirement for effectiveness
Sender / Source / Communicator	The originator of the message — the health educator.	Credibility, knowledge of the subject and of the audience, communication skill, positive attitude.
Message	The information, idea or attitude to be transmitted; its content, treatment and code.	Must be meaningful, clear, simple, specific, timely, accurate, culturally acceptable and interesting ; fitted to the felt need.
Channel / Medium	The physical route by which the message travels.	Interpersonal (face to face), mass media (radio, TV, press), traditional/folk media, digital media. Use more than one channel .
Receiver / Audience	The individual, group or community for whom the message is meant.	Message must be matched to age, education, culture, felt needs and readiness of the audience (audience segmentation).
Feedback	The flow of information from the receiver back to the sender.	Converts one-way into two-way communication ; permits correction of the message. Its absence is the biggest weakness of mass media.
Noise	Any distortion or interference that reduces fidelity.	Physical (traffic sound), physiological (deafness), psychological (prejudice), semantic (wrong words). Must be minimised.
Effect / Outcome	The observable change produced in the receiver.	Change in knowledge, attitude or practice; the real measure of success.

MNEMONIC

- **SMCRF** → **S**ender → **M**essage → **C**hannel → **R**eceiver → **F**eedback (plus **noise** acting throughout).
- **Berlo's model (1960)** is the classic S-M-C-R model. **Lasswell's formula (1948)** asks: **Who says What in Which channel to Whom with What effect?** **Shannon-Weaver (1949)** added the concept of **noise**.

6.3 Types of Communication

Basis	Types	Notes
Direction	One-way (didactic) vs two-way (Socratic)	One-way = lecture, radio, poster: speedy, reaches many, but no feedback, little learning, listener passive. Two-way = discussion, counselling: slower but produces real learning. Two-way is superior for health education.
Mode	Verbal vs Non-verbal	Non-verbal (gestures, posture, facial expression, eye contact, touch, silence, dress, distance) may carry more weight than words and can contradict them.
Number of persons	Intrapersonal, interpersonal, group, mass	Interpersonal is the most effective for behaviour change; mass is the most efficient for awareness.

Basis	Types	Notes
Formality	Formal (through official channels) vs informal (grapevine)	Informal channels spread fast and are often more believed — useful but also the route of rumours.
Sensory channel	Visual, auditory, audio-visual, tactile, olfactory	Audio-visual is most effective because it engages two senses.
Purpose	Didactic, Socratic, therapeutic, mass, group	Therapeutic communication is used in counselling and patient care.

6.4 Barriers to Communication

TABLE 6.2 BARRIERS TO COMMUNICATION AND THEIR REMEDIES

Barrier	Examples	Remedy
Physiological	Deafness, poor vision, defective speech, fatigue, pain, hunger, difficulty in understanding due to low intelligence.	Speak clearly and loudly, use visual aids, keep sessions short, treat the underlying problem.
Psychological	Emotional disturbance, anxiety, fear, prejudice, hostility, low motivation, neurosis, poor memory, levels of perception .	Build rapport first, reduce anxiety, be non-judgemental, repeat and reinforce.
Environmental	Noise, poor lighting, overcrowding, distance, invisibility of the speaker, congested room, bad weather.	Choose a quiet, well-lit, adequately sized venue; use a microphone; seat the audience so all can see.
Cultural	Illiteracy, language and dialect, customs, taboos, religious beliefs, food fads, blind faith, caste and gender norms, economic and social class differences.	Use local language and idiom, folk media and local leaders; adapt the message to culture.
Semantic / linguistic	Technical jargon, medical terminology, abbreviations, words with different local meanings.	Use simple everyday words; test the material on a sample audience (pre-testing).
Social	Poverty, unemployment, social distance between educator and community, lack of leadership.	Work through community structures; address underlying social needs.

6.5 The Seven C's of Effective Communication

C	Requirement
Clear	The message must be unambiguous and easily understood.
Concise	Short, to the point, free of unnecessary detail.
Concrete	Specific and vivid, with facts and figures rather than vague generalities.
Correct	Scientifically accurate, free of error in fact and language.
Coherent	Logically organised and consistent; one idea leads to the next.
Complete	Contains everything the receiver needs in order to act.
Courteous	Respectful, considerate of the receiver's feelings and culture.

6.6 Health Communication and BCC

- **Health communication** is the study and use of communication strategies to inform and influence individual and community decisions that enhance health.
- **BCC (Behaviour Change Communication)** is an interactive, research-based process that develops tailored messages through a mix of channels to encourage and sustain positive behaviour. Its steps: situation analysis → audience segmentation → message design → channel selection → pre-testing → implementation → monitoring → evaluation.
- **Counselling** is a two-way, non-directive, confidential process that helps a person to understand a problem and arrive at his own decision. Widely used in HIV (pre-test and post-test counselling), family planning, tobacco cessation and medication adherence.
- **Pre-testing** of every message and material on a sample of the target audience is mandatory good practice — it checks comprehension, attraction, acceptability, involvement and persuasion.

REMEMBER

In counselling the pharmacist should use the **teach-back (show-back) method**: after explaining, ask the patient to repeat the instructions or demonstrate the technique. This is the practical application of the principle of **feedback**.



CHAPTER 7 ESSENTIAL STEPS IN HEALTH EDUCATION

The syllabus phrase “essential steps” is used in two senses, and both are examined. **(A)** the psychological steps through which an individual passes on the way from ignorance to sustained action, and **(B)** the managerial steps in planning and running a health education programme. Learn both.

7.1 Part A — The Psychological / Behavioural Steps

TABLE 7.1 ESSENTIAL STEPS IN HEALTH EDUCATION (BEHAVIOURAL SEQUENCE)

Step	What happens	What the educator does
1. Gaining entry and building rapport	The educator becomes known and accepted; confidence is established.	Meet community leaders, be introduced by a trusted person, be courteous, listen first.
2. Capturing attention / creating awareness	The individual becomes aware that the problem and a solution exist. Awareness is created mainly through mass media .	Use posters, radio, TV, exhibitions, folk media, health days.
3. Arousing interest and creating a felt need	The person begins to feel that the problem concerns him personally and seeks more information.	Relate the message to his own life and family; supply details; group discussion.
4. Motivation and desire (evaluation stage)	The person weighs the advantages and disadvantages and develops the desire to act. This is the crucial and most difficult step.	Interpersonal counselling, testimonials of adopters, demonstration of benefit, incentives.
5. Decision-making	The individual makes up his mind to try the new practice.	Remove doubts, answer objections, make the action easy and available.
6. Action / trial and adoption	He actually performs the behaviour — first on trial, then permanently.	Provide the means (ORS packets, latrine, contraceptive, vaccine), teach the skill by learning by doing .
7. Reinforcement, follow-up and maintenance	The new behaviour is repeated until it becomes a habit; relapse is prevented.	Repeat visits, praise, support groups, continuing reminders, family support.
8. Evaluation	The educator measures whether knowledge, attitude and practice actually changed.	KAP survey, records, coverage data, feedback for redesign.

MNEMONIC

Core sequence to recite: **Awareness – Interest – Motivation/Desire – Decision – Action – Maintenance**. Several textbooks compress it into **four essential steps: awareness, motivation, decision-making and action**.

► 7.1.1 Rogers' Stages of Adoption of an Innovation

Stage	Description
Awareness	The person learns for the first time that the new idea or practice exists.
Interest	He wants to know more and actively seeks information.
Evaluation	He mentally weighs the pros and cons and applies it to his own situation.
Trial	He tries the practice on a small scale.
Adoption	Satisfied with the trial, he accepts it permanently and becomes an advocate for it.

TABLE 7.2 ROGERS' ADOPTER CATEGORIES

Adopter category	Share of population	Character
Innovators	≈ 2.5%	Venturesome, cosmopolite, first to try; often not respected locally.
Early adopters	≈ 13.5%	Local opinion leaders — the most important group to target because others imitate them.
Early majority	≈ 34%	Deliberate, adopt just before the average person.
Late majority	≈ 34%	Sceptical, adopt under peer pressure or economic necessity.
Laggards	≈ 16%	Traditional, suspicious of change, last to adopt.

7.2 Part B — Steps in Planning a Health Education Programme

- Community diagnosis / needs assessment.** Collect data on demography, disease pattern, existing KAP, felt needs, resources, culture and leadership. Methods: survey, records, interviews, focus-group discussion, observation.
- Setting priorities.** Rank problems by magnitude, severity, feasibility of control, community concern and cost. One cannot do everything at once.
- Formulating objectives.** Write **SMART** objectives in the three domains (knowledge, attitude, practice), stating who will do what, how much and by when.
- Identifying and segmenting the target audience.** Primary audience (those whose behaviour must change), secondary audience (those who influence them — husbands, mothers-in-law, teachers) and tertiary audience (policy makers).
- Selecting the content.** Accurate, relevant, need-based, culturally acceptable, limited to a few key messages.
- Selecting methods and media.** Match the method to the objective, audience size, literacy, resources and time (Chapter 8). Combine individual, group and mass approaches.
- Preparing and pre-testing the material.** Develop aids, then pre-test on a sample of the audience for comprehension, attractiveness, acceptability and persuasiveness; revise accordingly.
- Planning the resources.** Manpower, money, material, transport, time; prepare a budget and a time schedule (Gantt chart).
- Training the personnel.** Orient and train the workers, volunteers and leaders who will deliver the programme.
- Implementation.** Carry out the plan, supervise, maintain records, and co-ordinate with other sectors.
- Monitoring.** Continuous checking during implementation that activities are going as planned; make mid-course corrections.
- Evaluation.** Measure achievement against the objectives — see §7.4.
- Feedback and replanning.** Use the evaluation findings to modify the programme; the cycle then repeats.

HIGH-YIELD FOR EXAM

The **first and most essential step** in planning any health education programme is **assessment of the needs of the community (community diagnosis)**, and the needs must be the **felt needs** of the people. The **last step** is **evaluation with feedback**, which makes the process a continuous cycle.

7.3 Theories and Models Used to Plan Health Education

TABLE 7.3 BEHAVIOUR CHANGE MODELS

Model / theory	Key constructs	Use
Health Belief Model (Hochbaum, Rosenstock, Kegels — 1950s)	Perceived susceptibility , perceived severity , perceived benefits , perceived barriers , cues to action , self-efficacy .	Explains why people do or do not take a preventive action such as screening or immunisation. The oldest and most widely used model in health education.
PRECEDE-PROCEED model (Lawrence Green)	PRECEDE = P redisposing, R einforcing and E nabling Constructs in Educational D iagnosis and E valuation; PROCEED = P olicy, R egulatory and O rganizational Constructs in Educational and E nvironmental D evelopment.	The standard framework for planning and evaluating a health education programme; works backwards from the desired outcome.
Transtheoretical / Stages of Change model (Prochaska & DiClemente)	Pre-contemplation — Contemplation — Preparation — Action — Maintenance (± relapse and termination).	Matching the intervention to the person's stage of readiness; widely used in tobacco, alcohol and obesity work.
Theory of Reasoned Action / Planned Behaviour (Ajzen & Fishbein)	Attitude towards the behaviour + subjective norm (+ perceived behavioural control) → intention → behaviour.	Shows that intention is the immediate determinant of behaviour and that social norms matter.
Social Cognitive / Social Learning Theory (Bandura)	Observational learning (modelling), self-efficacy , reciprocal determinism, reinforcement, outcome expectations.	Basis of peer education, demonstrations and role models; the source of the important concept of self-efficacy .
Diffusion of Innovations (Everett Rogers)	Innovation, communication channels, time, social system; adopter categories.	Explains how new practices spread through a community and identifies opinion leaders.

Model / theory	Key constructs	Use
Social ecological model	Individual → interpersonal → organisational → community → public policy levels.	Reminds the educator that behaviour is nested in an environment; intervene at several levels.

7.4 Evaluation of Health Education

Type of evaluation	Question answered	Indicators
Formative	Is the material suitable <i>before</i> full-scale launch?	Pre-testing results, comprehension and acceptability of messages.
Process (monitoring)	Was the programme delivered as planned?	Number of sessions held, people reached, materials distributed, quality of sessions, cost incurred, coverage.
Impact	Did knowledge, attitude and behaviour change?	KAP scores before and after, proportion practising the behaviour, service utilisation rates.
Outcome	Did health status improve?	Incidence and prevalence of disease, morbidity, mortality, nutritional status, immunisation coverage.
Summative	Was the whole programme worth it?	Overall achievement of objectives, cost-effectiveness, sustainability.

- Evaluation must be **built into the programme from the planning stage**, not added at the end.
- **Baseline data** collected before the intervention are essential for measuring change.
- Common designs: before–after (pre-test/post-test) in one group; comparison with a control group; and time-trend analysis.
- Evaluation should be **participatory** — involving the community itself.



CHAPTER 8 METHODS OF HEALTH EDUCATION

This is the largest and most heavily examined chapter of the unit. The methods of health education are classified by the **size of the audience reached** into three main approaches.

HIGH-YIELD FOR EXAM

- The **three main methods (approaches) of health education** are: (1) **Individual approach**, (2) **Group approach**, (3) **Mass approach**.
- **Individual** methods are the **most effective for behaviour change** but reach the fewest people. **Mass** methods reach the largest numbers but are least effective in changing behaviour. **Group** methods lie in between and are the most widely used in practice.

TABLE 8.1 THE THREE APPROACHES COMPARED

Approach	Methods included	Reach	Effectiveness for behaviour change
Individual	Personal contact, health counselling, patient interview, home visit, personal letter	Very few	Highest — permits full two-way communication and tailoring
Group	Lecture, demonstration, group discussion, panel discussion, symposium, workshop, seminar, conference, role play, buzz session, brainstorming, case study, field trip	Moderate	High — peer interaction reinforces learning
Mass	Television, radio, newspapers, printed matter, films, posters, exhibitions, health museums, folk media, internet and social media, mHealth	Very large	Lowest — good for awareness, poor for behaviour change; no feedback

8.1 Individual Approach

► 8.1.1 Personal Contact / Individual Health Teaching

- Takes place whenever a health worker meets a person — in the clinic, the pharmacy, the home or the field. **Every contact is an opportunity for health education.**
- **Advantages:** full two-way communication; the message can be tailored to the individual's problem, language and level; doubts can be cleared on the spot; confidence and rapport develop; questions can be asked freely; privacy is possible; immediate feedback.
- **Limitations:** reaches only one person at a time; time-consuming and expensive per head; depends entirely on the skill and attitude of the worker; the number of workers available is limited.

► 8.1.2 Health Counselling

- A **face-to-face, two-way, confidential and non-directive** process in which the counsellor helps the client to understand his problem, examine the alternatives and **arrive at his own decision**.
- Essentials: privacy, adequate time, active listening, empathy, an accepting and non-judgemental attitude, confidentiality, and **no imposition of the counsellor's own values**.
- Steps: establishing rapport → identifying the problem → exploring feelings and alternatives → helping the client decide → supporting action → follow-up.
- **Uses:** genetic counselling, marital and pre-marital counselling, family planning, HIV pre-test and post-test counselling, tobacco and alcohol cessation, nutrition, adolescent health, and **medication counselling by the pharmacist**.
- **GATHER** approach in counselling: **G**reet, **A**sk, **T**ell, **H**elp, **E**xplain, **R**eturn/refer.

► 8.1.3 Home Visit

- Education in the family's own surroundings, where the real problems (water storage, kitchen, latrine, ventilation, vector breeding) can be seen and discussed.
- **Advantages:** builds excellent rapport; allows the whole family to be involved; practical problems become visible; ideal for the housebound, the elderly and postnatal mothers.
- **Limitations:** very time-consuming, costly, requires transport; reaches few families; may be resented as intrusion if not properly announced.

8.2 Group Approach

A group is an audience assembled for a common purpose. Group methods use the **dynamics of the group** — peer influence, discussion and shared decision-making — to reinforce learning. The composition of the group (mothers, school children, industrial workers, patients) decides the method and the message.

► 8.2.1 Lecture

- A talk given by one person to an audience; the oldest and most widely misused method.
- The audience should **not exceed about 30 people**, and the lecture should not exceed **15–20 minutes** of effective attention span.
- **Essentially one-way communication** — hence its greatest defect: the audience is passive and there is little feedback and little learning.
- **Advantages:** economical of time; covers much information; suitable for presenting new knowledge to a literate audience; good for introducing a topic.
- **Limitations:** no participation; retention poor; unsuitable for illiterate or uninterested audiences; attention falls rapidly.
- **Improved by:** using audio-visual aids, examples and humour, asking questions, allowing discussion at the end, and good voice modulation and eye contact.

► 8.2.2 Demonstration

- A **carefully prepared presentation showing how to perform a skill or procedure**, with an explanation of the underlying principle — “seeing is believing”.
- It should be followed by a **return demonstration (re-demonstration) by the learner** — this converts it into learning by doing and is the single most effective group method for teaching skills.
- **Examples:** preparation of ORS, hand-washing steps, use of a metered-dose inhaler or insulin pen, breast-feeding position, preparation of weaning food, cooking demonstration, use of a condom, bandaging and first aid.
- **Requirements:** all must be able to see; keep the group small; use real articles; perform slowly, step by step; explain while doing; repeat; check understanding.
- **Advantages:** holds attention, uses several senses, teaches skills, highly convincing. **Limitations:** needs preparation, equipment and a small group; time-consuming.

► 8.2.3 Discussion Methods

TABLE 8.2 GROUP METHODS OF HEALTH EDUCATION

Method	Group size / structure	How it is conducted	Value and limits
Group discussion	6–12 participants (ideal); seated in a circle or semicircle so all can see one another; has a group leader (chairman) and a recorder or secretary.	The leader introduces the topic, keeps the discussion on track, encourages the silent and restrains the talkative, and sums up. The recorder notes the points and prepares the report.	Considered one of the most effective methods — maximum participation, free exchange of ideas, group decision. Fails if the group is too large, if a few dominate, or if the leader is autocratic.
Panel discussion	4–8 panellists (experts) with a chairman/moderator , before an audience.	The chairman introduces the subject and the panellists, who discuss the topic among themselves for about 30–45 minutes; the audience then puts questions. There are no set speeches .	Presents several viewpoints on a controversial topic; stimulating. Depends heavily on the chairman and on informed panellists; the audience is largely passive.
Symposium	A series of 2–5 speakers , each an expert, with a chairman.	Each speaker delivers a prepared speech on one aspect of the subject; there is no discussion among the speakers . The chairman sums up at the end and the audience may ask questions.	Thorough, authoritative coverage of a subject. More formal and less interactive than a panel discussion.
Workshop	A large group divided into small working groups of 4–10, each with a leader and recorder; has a chairman and resource persons.	Consists of individual and group work : the participants learn by doing, produce material, solve problems, then report back to the plenary session. Lasts from a day to several weeks.	Excellent for developing skills and for producing a concrete output (e.g. a training module). Needs careful organisation and resources.
Seminar	A group of students or workers with an expert leader.	One member presents a paper on a topic which is then discussed by all.	Good for in-depth academic study; needs a literate, prepared group.
Conference	A large formal assembly of professionals.	Papers are read, sessions and workshops are held, resolutions are passed.	Good for exchange of knowledge and for advocacy; expensive; reaches professionals rather than the community.

Method	Group size / structure	How it is conducted	Value and limits
Role play (socio-drama)	Audience not more than about 25 ; a few volunteers act.	Participants act out a real-life situation (e.g. a mother refusing immunisation, a quack, a pharmacist counselling) without a script; the audience then discusses what happened and why.	Creates deep insight into attitudes and feelings, exposes prejudices, highly interesting; especially useful for illiterate groups and school children. Needs a skilled facilitator; may embarrass shy participants.
Buzz session / buzz group	A large group split into small buzz groups of 4–6 for a few minutes.	Each small group discusses the question simultaneously ('buzzing'), then a spokesman reports its conclusion to the whole gathering.	Converts a passive audience into an active one in a few minutes; useful when time is short and the group is large.
Brainstorming	Usually 6–12 members , for 10–15 minutes.	Members generate as many ideas as possible without any criticism or evaluation ; quantity is encouraged; only afterwards are the ideas judged.	Excellent for generating fresh solutions and for planning; must strictly postpone criticism.
Case study / problem-solving	Small group.	A real or simulated case is presented and analysed by the group.	Develops analytical and decision-making ability.
Field trip / study visit / exhibit tour	A small organised group.	The group visits a site — a water works, a model village, a hospital, an ICDS centre — with a clear objective, preparation and follow-up discussion.	Very convincing — 'seeing for oneself'; costly and needs organisation.

MNEMONIC

- Numbers worth memorising: **Lecture ≤ 30** listeners · **Group discussion 6–12** · **Panel 4–8** panellists · **Symposium 2–5** speakers · **Role play audience ≤ 25** · **Buzz group 4–6** · **Brainstorming 6–12**.
- Difference to remember: in a **panel discussion** the experts **talk to one another** with no prepared speeches; in a **symposium** each expert delivers a **prepared speech** and they do **not** discuss among themselves.

8.3 Mass Approach

Mass media are used when the object is to reach very large numbers quickly, chiefly to **create awareness** and to support interpersonal work. They are **one-way** and provide no feedback, so they seldom change behaviour by themselves.

TABLE 8.3 MASS MEDIA IN HEALTH EDUCATION

Medium	Features	Advantages	Limitations
Television	Audio-visual; the most powerful mass medium; reaches the illiterate.	Sight plus sound; high credibility and impact; wide reach; can demonstrate.	Expensive; one-way; viewing is often for entertainment; message quickly forgotten; electricity and set required.
Radio	Auditory; very wide reach including remote and illiterate populations; local-language broadcasts and community radio.	Cheap, portable, no literacy needed, works without electricity (battery), large coverage.	Only one sense; no visual demonstration; no feedback; listener may not be attentive.
Newspapers, magazines, press	Printed word; reach the literate, urban, opinion-forming public.	Can be re-read and preserved; carry detail; cheap per copy; good for advocacy.	Useless for illiterates; low readership in rural areas; news value may distort the health message.
Printed material — posters, leaflets, pamphlets, folders, booklets, handbills, flip books, banners, hoardings	Cheap, portable, can be taken home and re-read; can be produced locally in the local language.	Inexpensive, widely distributable, reinforces spoken messages, good take-home reminder.	Needs literacy (except pictorial matter); discarded easily; no feedback.
Films, documentaries, video and mobile film vans	Audio-visual, dramatic, emotionally powerful; useful in villages without electricity using mobile vans.	Holds attention, teaches skills, reaches illiterates, entertaining.	Costly to make; needs equipment, power and a darkened space; passive audience.

Medium	Features	Advantages	Limitations
Health museum and health exhibition	Static and moving displays, models, specimens, charts, working models; health melas and fairs.	Several senses involved, self-paced, attractive, suitable for all ages and literacy levels; can be combined with services and check-ups.	Costly; static in one place; requires space, staff and maintenance.
Folk media / traditional media	Puppet shows, street plays (nukkad natak), folk songs, ballads, dances, <i>kathputli, tamasha, jatra, bhavai</i> , drum-beating.	Culturally rooted, entertaining and highly acceptable ; excellent for illiterate rural audiences; cheap; permits some interaction.	Message may be diluted by entertainment; needs local artists; limited audience per show.
Internet, social media and mHealth	Websites, WhatsApp, YouTube, mobile SMS/IVR services, apps, teleconsultation, e-learning.	Very rapid, cheap, interactive, personalised, scalable; permits two-way communication and reminders; useful for adherence reminders.	Digital divide (age, income, language); serious risk of misinformation ; privacy concerns; needs literacy and a device.
Health days and campaigns	Observance of days such as World Health Day (7 April), World TB Day (24 March), World No-Tobacco Day (31 May), World AIDS Day (1 December), National Immunisation Day (Pulse Polio).	Focuses attention intensely for a short period; mobilises many agencies and the press.	Short-lived effect unless followed by sustained work — the classic weakness of the campaign approach .

8.4 Audio-Visual (AV) Aids

Audio-visual aids — Instructional devices that are **used to supplement and reinforce the spoken or written word** by appealing to the senses, thereby making learning clearer, more interesting and better retained. They are **aids to teaching, never a substitute for the teacher**.

► 8.4.1 Classification of AV Aids

TABLE 8.4 CLASSIFICATION OF AUDIO-VISUAL AIDS

Basis of classification	Categories	Examples
By sense organ used	Auditory (hearing) · Visual (sight) · Combined audio-visual	Auditory: radio, tape recorder, microphone, public address system, megaphone, telephone. Visual: posters, charts, graphs, maps, models, specimens, flannelgraph, flash cards, bulletin board, blackboard, slides, OHP transparencies, photographs, printed matter. Combined: television, cinema/film, video, sound filmstrip, drama, puppet show, computer presentation, LCD projection.
By complexity	Simple · Sophisticated	Simple: chalkboard, chart, poster, flannelgraph, flash card, model, specimen — cheap, locally made, need no power. Sophisticated: television, cinema, LCD projector, computer — costly, need power and maintenance.
By projection	Projected · Non-projected	Projected: slides, filmstrips, films, overhead projector, LCD/DLP projector, epidiascope. Non-projected: blackboard, charts, posters, models, flannelgraph, flip chart, bulletin board.
By dimension	Two-dimensional · Three-dimensional	2-D: poster, chart, graph, map, photograph, cartoon. 3-D: models, specimens, objects, dioramas, puppets.

► 8.4.2 The Common Aids in Detail

Aid	Description and correct use
Chalkboard / blackboard / whiteboard	The most common, cheapest and most flexible aid. Write large and legibly; use few words; stand aside while writing; erase before starting a new idea; use coloured chalk for emphasis.
Chart	A sheet presenting an idea in a summarised, logical, visual form (flow chart, tree chart, tabular chart, pull chart, flip chart). One chart = one idea . Letters must be readable from the back of the room.

Aid	Description and correct use
Poster	A pictorial display with a brief, arresting message ; usually about 50 × 75 cm (20 × 30 inches) . Requirements: one idea only , few words (ideally under about seven), bold lettering readable from 3–4 metres, bright contrasting colours, an eye-catching illustration, and a slogan. Must be displayed at eye level where people wait, and changed frequently .
Leaflet, pamphlet, folder, booklet, handbill	Printed take-home matter. A leaflet/handbill is a single sheet with one message; a folder is folded, giving a few panels; a pamphlet/booklet has several pages and more detail. Use simple language, pictures and plenty of white space.
Flannelgraph (khadigraph / flannel board)	A board covered with flannel, khadi or velvet on which cut-outs backed with sandpaper or rough cloth are made to stick. The story is built up step by step as the talk proceeds, which holds attention. Cheap, reusable, needs no electricity.
Flash cards	A set of about 10–12 stiff cards (roughly 25 × 30 cm) carrying a picture on the front and the text of the message on the back for the speaker. They are 'flashed' one by one in sequence at about 1 metre from a small audience (up to 25–30). Excellent for illiterate groups.
Bulletin board / notice board	A board on which notices, cuttings, photographs and posters are pinned. Must be placed where people gather and kept up to date , neat and uncluttered.
Models	Three-dimensional representations: scale models (life-size, enlarged or reduced), cut-away (sectional) models , working models . Useful when the real object cannot be shown — e.g. a sanitary latrine, the human heart, a smokeless chulha, a model of the eye.
Specimens and objects	Real materials — foodstuffs, drug samples, an adulterated sample, a mosquito larva in a bottle, an intra-uterine device, a diseased organ. The most concrete and convincing of all visual aids.
Graphs and diagrams	Line, bar, pie and pictogram forms used to present numerical data. Keep to one comparison per graph; label clearly; use pictograms for illiterates.
Maps	Show the geographical distribution of disease, services and resources; spot maps of cases are used in outbreak investigation.
Puppets	Glove, string (marionette), rod and shadow puppets present a story humorously; very popular with children and rural audiences; can tackle sensitive subjects indirectly.
Overhead projector (OHP) / slides / LCD projector	For larger audiences. Rules: not more than 6–7 lines per slide and a few words per line; large font; high contrast; do not read out the slide; allow enough time for the audience to read.
Cinema, video, television	Best for showing motion, process and emotion; should be preceded by an introduction and followed by a discussion .

► 8.4.3 Criteria of a Good AV Aid and Rules for Use

- **Simple** — conveys one idea at a time, free from unnecessary detail.
- **Accurate** and scientifically correct; **up to date**.
- **Appropriate** to the age, education, culture and language of the audience.
- **Visible / audible** to the whole audience; adequate size and contrast.
- **Interesting and attractive**, arousing curiosity.
- **Cheap, locally available and easily transportable**; preferably made from local material.
- **Durable and easy to maintain**, and safe to use.
- Rules of use: be **familiar with the aid and test the equipment beforehand**; keep the aid **covered until needed and remove it afterwards**; talk to the audience, not to the aid; use **several aids** in one session; never let the aid replace the educator.

HIGH-YIELD FOR EXAM

- An AV aid is only an **aid**; the **health educator remains the key factor**.
- **Edgar Dale's Cone of Experience** arranges learning experiences from the most concrete at the base (**direct purposeful experience**) to the most abstract at the apex (**verbal symbols**). The nearer the base, the more effective the learning.

- The most effective single visual aid for an **illiterate** rural audience is a **demonstration with real objects / flash cards / flannelgraph**; the cheapest and most versatile aid in a classroom is the **chalkboard**.

8.5 Choosing the Right Method

TABLE 8.5 MATCHING THE METHOD TO THE OBJECTIVE

Situation / objective	Method of choice
Creating awareness among a very large population quickly	Mass media — radio, television, press, hoardings, social media.
Changing a deep-rooted personal behaviour	Individual counselling with follow-up.
Teaching a skill (ORS, inhaler, hand-washing)	Demonstration followed by return demonstration by the learner.
Reaching an illiterate rural audience	Folk media, flash cards, flannelgraph, puppets, film, demonstration.
Getting a community to take a collective decision	Group discussion with local leaders.
Exploring a controversial subject with several viewpoints	Panel discussion.
Giving authoritative, comprehensive information on a subject	Symposium or lecture.
Developing practical skills in workers	Workshop with small working groups.
Activating a large, passive audience in a few minutes	Buzz session.
Generating new solutions to a problem	Brainstorming.
Exposing attitudes and prejudices, e.g. about immunisation refusal	Role play / socio-drama.
Educating a patient at the point of dispensing	Personal contact plus a printed leaflet (medication counselling).

8.6 Health Education in Specific Settings

Setting	Key points
School health education	The school is the most cost-effective and important setting , because habits are formed in childhood and children carry messages home ('each child a health messenger'). Content: personal hygiene, nutrition, oral health, exercise, first aid, prevention of tobacco/alcohol/substance use, adolescent and reproductive health, road safety, mental health. Methods: integration with the curriculum, health clubs, school health committees, eco-clubs, rallies, essay/painting competitions, midday meal as a teaching opportunity. Programmes: School Health Programme , now under Ayushman Bharat — School Health and Wellness Ambassadors (two teachers per school).
Community / village	Work through the Gram Panchayat, Village Health Sanitation and Nutrition Committee (VHSNC), Mahila Mandal, self-help groups and ASHA ; use Village Health and Nutrition Days (VHND); folk media; Swachh Bharat and Poshan Abhiyaan platforms.
Hospital and clinic	Waiting-area posters, display boards, television, group talks for patients and attendants, bedside teaching, discharge counselling and pre-operative explanation. Pharmacy counselling counter is a major opportunity.
Workplace / industry	Occupational hazards, use of personal protective equipment, safety drills, first aid, ergonomics, tobacco and alcohol, periodic medical examination, employees' health committees.
Antenatal and postnatal clinic	Diet in pregnancy, iron-folic acid and calcium, immunisation (Td), danger signs, institutional delivery, exclusive breast-feeding, newborn care, contraception, care of the umbilical cord.



9.1 The Ancient and Medieval Foundations

- **Indus Valley Civilisation (c. 3000 BC)** — Mohenjo-daro and Harappa had planned drainage, covered sewers, public baths and wells: evidence of an organised, community-wide idea of hygiene.
- **Vedic period** — the Vedas, especially the **Atharva Veda**, prescribed rules of personal cleanliness, diet, daily routine (*dinacharya*), seasonal regimen (*ritucharya*) and conduct. Health teaching was embedded in **religion and custom**, which made it powerfully effective.
- **Ayurveda — Charaka Samhita and Sushruta Samhita** — laid down *swasthavritta* (the code of healthy living): the concept that **prevention precedes cure**. Sushruta is called the 'Father of Surgery'; Charaka emphasised diet, conduct and mental hygiene.
- **Emperor Ashoka (3rd century BC)** — rock edicts publicised health rules, hospitals (for people and animals), wells, rest-houses and the planting of medicinal herbs along roads: possibly the world's earliest **state-sponsored health communication**.
- **Medieval period** — the Unani system flourished; epidemics of plague, cholera and smallpox were common; organised public health declined and health practices were transmitted through religion, folklore, proverbs and folk songs.

9.2 The Colonial Period — The Beginnings of Organised Health Education

TABLE 9.1 PRE-INDEPENDENCE LANDMARKS

Year	Landmark	Significance
1859–63	Royal Commission on the health of the Army in India (appointed after the 1857 revolt).	Its report led to the creation of public health machinery in India.
1864	Commissions of Public Health / Sanitary Commissioners appointed in Bombay, Madras and Bengal.	First official sanitary organisation; sanitary reports and public health propaganda began.
1873	Births, Deaths and Marriages Registration Act.	Beginning of vital statistics, the factual basis of health education.
1880	Vaccination Act.	Made smallpox vaccination compulsory in certain areas; required public persuasion and education.
1897	Epidemic Diseases Act (after the Bombay plague of 1896).	Provided for notification, inspection and segregation; heavy use of publicity.
1904	Plague Commission report.	Recommended public education about rats, fleas and personal hygiene.
1919	Montague–Chelmsford reforms (Government of India Act, 1919).	Public health, sanitation and vital statistics were transferred to the provinces — health education became a provincial responsibility.
1920	Indian Red Cross Society established.	Became a major agency of voluntary health education, first aid and maternal-child welfare.
1923	The Rockefeller Foundation assisted health and hygiene work; rural health units set up.	Introduced modern public health education methods to India.
1932	All India Institute of Hygiene and Public Health (AIIHPH) , Calcutta established.	The first institution in India to give formal training in public health including health education; created a Department of Health Education.
1935	Government of India Act, 1935.	Further provincial autonomy in health matters.
1939	Health Publicity Bureau (also called the Central Health Publicity Bureau) organised at the centre under the Public Health Commissioner.	The direct forerunner of the Central Health Education Bureau ; produced posters, pamphlets and films.
1939	First Maternity and Child Welfare Bureau / Lady Reading Health School, Delhi.	Trained health visitors who carried out home-based health education.
1943–46	Bhore Committee (Health Survey and Development Committee), chaired by Sir Joseph Bhore; report submitted in 1946 in four volumes.	The most important landmark . It recognised health education as an essential function of the health services, recommended a Central Health Education Bureau, state bureaux and the appointment of trained health educators , the three-month and long-term plans, the 'social physician' concept, and integration of preventive with curative services.

9.3 Post-Independence Growth

TABLE 9.2 POST-INDEPENDENCE GROWTH OF HEALTH EDUCATION IN INDIA

Year	Development	Significance for health education
1951	First Five Year Plan began.	Health education included in planned development; the community development programme used the extension-education approach.
1952	National Family Planning Programme launched — India was the first country in the world to have a state family planning programme.	Created an enormous and continuing demand for mass health education, motivation and IEC.
1956	CENTRAL HEALTH EDUCATION BUREAU (CHEB) established at New Delhi under the Directorate General of Health Services, Ministry of Health (on the recommendation of the Bhole Committee), by reorganising the Health Publicity Bureau.	The single most important landmark in Indian health education — the apex national body. This year is very frequently asked.
1956	Indian Public Health Association (IPHA) and other professional bodies took shape; publication of professional journals.	Professionalisation of public health and health education.
1959–61	Mudaliar Committee (Health Survey and Planning Committee), report 1961.	Found health education weak; recommended strengthening of health education bureaux and training.
1961	School Health Committee (chaired by Renuka Ray) reported.	Recommended a systematic school health service and school health education .
1963	Chadach Committee .	Monthly home visits by basic health workers for malaria surveillance — combined with health education.
1966	Mukherjee Committee .	Separated malaria work from family planning so that each could be pursued properly.
1967	Jungalwalla Committee ('Integration of Health Services').	Recommended an integrated health service and uniform referral system; state health education bureaux strengthened in most states around this period.
1973	Kartar Singh Committee ('Multipurpose Workers').	Created the Multipurpose Health Worker (MPW) scheme — one male and one female worker per 5000 population — who carry health education to every household.
1975	Srivastava Committee (Group on Medical Education and Support Manpower).	Recommended community health volunteers/workers and the ROME (Reorientation of Medical Education) scheme — health education by community-based volunteers.
1975	Integrated Child Development Services (ICDS) launched (2 October 1975).	Nutrition and Health Education (NHED) became one of its six services, delivered by the anganwadi worker to women aged 15–45.
1977	Rural Health Scheme and the Community Health Worker / Volunteer scheme; NIHFW established (9 March 1977) by merging NIHAE (1964) and NIFP (1970).	NIHFW became the apex institute for training, research and evaluation in health and family welfare, including health education and IEC.
1978	Alma-Ata Declaration — 'Health for All by 2000 AD' through Primary Health Care ; India was a signatory.	Health education declared the first of the eight essential elements of PHC — the greatest single boost that health education has received.
1983	First National Health Policy of India.	Gave a central place to health education, community participation and the training of health volunteers.
1986	Ottawa Charter (international) and the Bajaj Committee (health manpower planning) in India; National Policy on Education, 1986 .	Shift from health education to the wider concept of health promotion ; health education strengthened in schools.
1992	Child Survival and Safe Motherhood (CSSM) programme.	Large-scale IEC for immunisation, ORS and safe delivery.
1997	Reproductive and Child Health (RCH) Programme , Phase I (RCH-II from 2005).	Adopted the target-free, client-centred approach; IEC replaced by BCC with emphasis on counselling and informed choice.
2002	National Health Policy 2002 .	Emphasised IEC, decentralisation and increased public health spending.
2005	National Rural Health Mission (NRHM) launched (12 April 2005); creation of the ASHA — Accredited Social Health Activist.	The most significant recent development: one trained female community health volunteer per 1000 population acting as a health educator and change agent ; also the VHSNC , Village Health and Nutrition Days, Rogi Kalyan Samitis and a dedicated IEC/BCC budget in every state PIP.

Year	Development	Significance for health education
2013	National Urban Health Mission (NUHM); NRHM + NUHM = National Health Mission (NHM).	Extended organised health education to urban slums through ASHA and Mahila Arogya Samitis .
2014 onward	Swachh Bharat Mission (2 October 2014), Mission Indradhanush (2014), Poshan Abhiyaan (2018), Eat Right India, Fit India Movement (2019).	Mass behaviour-change campaigns on sanitation, immunisation, nutrition, safe food and physical activity — health education at national campaign scale.
2017	National Health Policy 2017.	Explicit goal of health promotion and prevention , school health programmes, 'Health in All Policies', and a target of 2.5% of GDP for health.
2018	Ayushman Bharat — Health and Wellness Centres (HWC) plus PM-JAY (23 September 2018).	HWCs deliver comprehensive primary health care with health promotion and wellness activities (yoga, exercise, screening); School Health and Wellness Ambassadors appointed; Community Health Officer (CHO) added as a health educator at HWC level.
2020–22	COVID-19 pandemic response.	The largest risk-communication and community-engagement exercise in Indian history — mass media, mobile caller tunes, Aarogya Setu, CoWIN, and the fight against the accompanying ' infodemic ' of misinformation.

9.4 The Central Health Education Bureau (CHEB)

DEFINITION

The **Central Health Education Bureau** was set up in **1956** at **New Delhi** under the **Directorate General of Health Services, Ministry of Health and Family Welfare**, on the recommendation of the **Bhore Committee (1946)**. It is the **apex national body for health education in India**.

► 9.4.1 Objectives

- To promote health education as an integral part of all health, medical and family welfare programmes.
- To develop and strengthen health education services at the national, state and district levels.
- To **train** health and allied personnel in health education methods.
- To produce, pre-test and distribute health education **material and media** in several languages.
- To conduct **research and evaluation** in health education and health behaviour.
- To coordinate the health education work of government, voluntary and international agencies.

► 9.4.2 Functions and Divisions

Function / division	Work done
Training	Courses and workshops for health education officers, medical and paramedical staff, teachers and field workers; the Diploma/certificate courses in health education .
Production of media and material	Posters, folders, flip books, flash cards, charts, exhibits, films, radio and television spots, slide sets and exhibition panels; a media production unit .
Research and evaluation	KAP studies, pre-testing of material, operational research, evaluation of IEC campaigns and studies of health behaviour.
Publication and documentation	Health education literature, technical reports, a library and documentation centre; the journal ' Swasth Hind '.
School health education	Development of school health education curricula, teachers' guides and training of teachers.
Technical advice and coordination	Advises the Ministry, states, national programmes and voluntary agencies; coordinates the observance of health days and national campaigns.
Community health education / field practice	Demonstration and field-practice projects; support to state and district bureaux.

9.5 Organisation of Health Education Services in India

TABLE 9.3 THE HEALTH EDUCATION MACHINERY, CENTRE TO VILLAGE

Level	Agency / officer
National / Centre	Central Health Education Bureau (CHEB) , New Delhi; the IEC Division of the Ministry of Health and Family Welfare; NIHFW ; the media divisions of the national programmes (NACO, NTEP, NVBDCP); Directorate of Advertising and Visual Publicity (DAVP/CBC) ; All India Radio, Doordarshan, Song and Drama Division and Field Publicity Directorate.
State	State Health Education Bureau / State IEC Bureau headed by a State Health Education Officer / Mass Media Officer, under the Directorate of Health Services.
District	District Health Education Officer / District Extension and Media Officer (DEMO) with a district mass education and information unit.
Block / PHC	Block Extension Educator (BEE) — the designated health educator at block level; Health Assistants (male and female).
Sub-centre	ANM / MPW (Female) and MPW (Male) .
Village / community	ASHA , anganwadi worker, VHSNC, Gram Panchayat, Mahila Mandal, self-help groups, school teachers, trained dais and local opinion leaders.

9.6 Institutions and Professional Bodies

Body	Note
CHEB, New Delhi (1956)	Apex national health education body.
NIHFW, New Delhi (1977)	National Institute of Health and Family Welfare — apex technical institute for training, research, evaluation, consultancy and specialised services in health and family welfare.
AIIPH, Kolkata (1932)	All India Institute of Hygiene and Public Health — oldest public health training institute in India.
NCDC, Delhi (formerly NICD, 1963; originally Malaria Institute of India, 1909)	National Centre for Disease Control — training and IEC on communicable disease.
ICMR, New Delhi (1911 as IRFA, renamed 1949)	Apex body for biomedical research; issues the national ethical guidelines.
IIHFW / SIHFW	Indian/State Institutes of Health and Family Welfare — state-level training in IEC and management.
IPHA	Indian Public Health Association — professional body; publishes the <i>Indian Journal of Public Health</i> .
IAPSM	Indian Association of Preventive and Social Medicine.
Indian Red Cross Society (1920), TB Association of India (1939), Hind Kusht Nivaran Sangh, Family Planning Association of India (1949), Indian Council for Child Welfare, Kasturba Health Society	Major voluntary agencies in health education.
IUHPE (1951) and SOPHE (1950)	International Union for Health Promotion and Education; Society for Public Health Education — the leading international professional bodies.
WHO (7 April 1948), UNICEF (1946), UNESCO, UNFPA, World Bank, USAID, Rockefeller and Ford Foundations	International agencies supporting health education in India. World Health Day is observed on 7 April.

9.7 Achievements and Continuing Constraints

Achievements	Constraints
Eradication of smallpox (last case 1975, India declared free 1977; world 1980) and of polio (last case 13 January 2011; India certified polio-free 27 March 2014) — both achieved largely through mass communication and community mobilisation.	Health education is still treated as a subsidiary activity with a small budget and few dedicated trained posts.
Elimination of guinea-worm (2000), yaws (2016) and maternal and neonatal tetanus (2015); leprosy elimination as a public health problem (2005).	Shortage of trained health educators ; BEE and DEMO posts often vacant or diverted to other work.

Achievements	Constraints
A very large trained community-level cadre — about a million ASHAs and anganwadi workers acting as health educators.	Material is often produced centrally and is not adapted to local language and culture ; little pre-testing.
Rise in institutional deliveries, immunisation coverage, ORS and contraceptive use, sanitation coverage and iodised-salt consumption.	Persistence of the KAP gap — knowledge has risen faster than practice.
Wide penetration of radio, television and mobile telephony giving unprecedented reach.	Misinformation on social media , commercial advertising of unhealthy products, and inadequate evaluation of IEC activity.

HIGH-YIELD FOR EXAM

- Dates that are asked most often: **CHEB – 1956, New Delhi**; **Bhore Committee report – 1946**; **AIIPH Kolkata – 1932**; **NIHFW – 1977**; **Alma-Ata – 1978**; **Ottawa Charter – 1986**; **NRHM and ASHA – 2005**; **ICDS – 1975**; **India's family planning programme – 1952 (world's first)**.
- The committee that recommended the establishment of CHEB was the **Bhore Committee (Health Survey and Development Committee, 1946)**.



STERILIZATION & DISINFECTION

Physical · Chemical · Mechanical Methods · Syringes, Glasswares & Apparatus · Waste Disposal

CHAPTER 10 FUNDAMENTALS AND TERMINOLOGY OF STERILIZATION AND DISINFECTION

Every question in this unit ultimately tests one of three things: **a definition, a temperature–time combination, or which method is used for which article**. This chapter fixes the definitions; the following chapters supply the rest.

10.1 The Essential Definitions

TABLE 10.1 CORE TERMINOLOGY

Term	Definition	Key discriminator
Sterilization	The process by which all forms of microbial life — including bacterial and fungal spores, vegetative bacteria, viruses and prions — are completely destroyed or removed from an article.	Absolute term. There is no such thing as 'partially sterile'. Spores must be killed.
Disinfection	The destruction or removal of pathogenic (vegetative) organisms capable of causing infection; spores are not necessarily destroyed .	Relative term. Applied to inanimate objects .
Antisepsis	The prevention of infection by destroying or inhibiting the growth of micro-organisms on living tissues — skin, mucous membranes, wounds.	Applied to living tissue ; an agent so used is an antiseptic .
Disinfectant	A chemical (germicide) used on inanimate surfaces and objects to destroy pathogens; usually too toxic or irritant for living tissue.	Object — disinfectant.
Antiseptic	A chemical applied to living tissue that destroys or inhibits micro-organisms.	Skin/wound — antiseptic. Some agents (e.g. 70% alcohol, povidone-iodine) serve both roles at different concentrations.
Sanitization	Reduction of the microbial population on an inanimate surface to a level judged safe by public health standards — it does not imply sterility.	Used for eating utensils, dairy and food equipment.
Decontamination	Any process that removes or reduces contamination so that an article is safe to handle ; the first step before cleaning and sterilization.	Protects the handler , not the patient.
Cleaning	Physical removal of visible soil, organic matter and micro-organisms by water, detergent and mechanical action.	Cleaning must always precede disinfection or sterilization — no sterilant works reliably through dirt.
Asepsis	The absence of pathogenic organisms; the set of practices that prevents micro-organisms from reaching the tissue or article.	Aseptic technique ; contrast with antisepsis, which destroys organisms already present.
Sterilant / chemisterilant	A chemical agent capable of producing complete sterilization , including the killing of spores.	Examples: ethylene oxide, glutaraldehyde (long contact), peracetic acid, formaldehyde gas, hydrogen peroxide plasma, beta-propiolactone.
Germicide / microbicide	Any agent that destroys micro-organisms; the suffix names the target — bactericide, virucide, fungicide, sporicide, tuberculocide, amoebicide .	-cidal = kills; -static = inhibits multiplication without killing (bacteriostatic, fungistatic, sporostatic).
Fumigation	Disinfection of an enclosed space, air and surfaces by exposure to a gas or vapour.	Classic agent: formaldehyde vapour for the operation theatre.
Preservative	A substance added to a product to prevent microbial spoilage or growth during storage and use.	Examples: benzoic acid, benzalkonium chloride, chlorocresol, thiomersal, parabens in pharmaceuticals and multi-dose vials.

HIGH-YIELD FOR EXAM

- **Sterilization kills spores; disinfection does not.** This is the single commonest MCQ in the whole unit.
- **Disinfectant — inanimate objects; Antiseptic — living tissue.**

▪ **Boiling is NOT sterilization** (spores survive) — it is high-level disinfection. Similarly **pasteurisation is not sterilization**.

10.2 Types and Levels of Disinfection

Classification	Description	Examples
Concurrent disinfection	Immediate disinfection of infectious discharges and contaminated articles while the patient is still the source of infection .	Disinfection of a tuberculosis patient's sputum, or of a cholera patient's stool, as they are produced.
Terminal disinfection	Disinfection carried out after the patient has been removed — by death, discharge or recovery — or after he has ceased to be a source.	Cleaning and disinfecting the bed, locker, linen, mattress and room after discharge; fumigation of an isolation room.
Prophylactic (pre-current) disinfection	Disinfection done in anticipation of infection, as a routine preventive measure.	Chlorination of drinking water, pasteurisation of milk, antiseptic preparation of the skin before injection or surgery.

► 10.2.1 Spaulding Classification of Instruments and the Level of Processing

TABLE 10.2 SPAULDING CLASSIFICATION (1957/1968)

Category	Definition	Processing required	Examples
Critical	Items that enter sterile tissue, the vascular system or sterile body cavities . Any contamination carries a high risk of infection.	STERILIZATION — steam under pressure, dry heat, ethylene oxide, hydrogen peroxide plasma, or a chemical sterilant with prolonged contact.	Surgical instruments, needles and syringes, implants, cardiac and urinary catheters, dental hand-pieces, arthroscopes and laparoscopes, intra-uterine devices.
Semi-critical	Items that contact intact mucous membranes or non-intact skin but do not penetrate sterile tissue.	HIGH-LEVEL DISINFECTION (HLD) as a minimum — kills all organisms except large numbers of bacterial spores.	Flexible endoscopes (gastroscope, colonoscope, bronchoscope), laryngoscope blades, respiratory therapy and anaesthesia equipment, oesophageal manometry probes, vaginal specula, thermometers.
Non-critical	Items that contact intact skin only , or do not touch the patient at all.	LOW-LEVEL DISINFECTION or simple cleaning with detergent and water.	Blood-pressure cuff, stethoscope, bed-pan, crutches, bed rails, linen, furniture, floors, walls, patient trolley.

TABLE 10.3 LEVELS OF DISINFECTION

Level of disinfection	Kills	Does not reliably kill	Typical agents
High (HLD)	Vegetative bacteria, mycobacteria , fungi, all viruses (lipid and non-lipid), and some spores	Large numbers of bacterial spores	2% glutaraldehyde (20–45 min), 0.55% ortho-phthalaldehyde, 6–7.5% hydrogen peroxide, 0.2% peracetic acid, pasteurisation at 75 °C, ≥ 1000 ppm free chlorine
Intermediate	Vegetative bacteria, mycobacteria (tuberculocidal), most viruses, most fungi	Bacterial spores	70–90% alcohols, phenolics, iodophors, chlorine compounds (≈ 500 ppm), some quaternary compounds with alcohol
Low	Most vegetative bacteria, some fungi, lipid (enveloped) viruses such as HIV, HBV and influenza	Mycobacteria, spores, non-lipid (non-enveloped) viruses such as polio and norovirus	Quaternary ammonium compounds, 3% hydrogen peroxide (surfaces), low-strength phenolics, detergents

10.3 Order of Microbial Resistance to Germicides

The processing level needed is decided by the **most resistant organism** likely to be present. Learn this ladder from most resistant at the top.

TABLE 10.4 DESCENDING ORDER OF MICROBIAL RESISTANCE

Rank	Organism group	Examples	Comment
1 (most resistant)	Prions	Creutzfeldt–Jakob disease agent, scrapie	Resist routine autoclaving; require 134 °C for 18 min (pre-vacuum) or 121 °C for 1 hour , or 1 N NaOH followed by autoclaving; incineration preferred.

Rank	Organism group	Examples	Comment
2	Bacterial spores	<i>Bacillus</i> , <i>Geobacillus stearothermophilus</i> , <i>Clostridium difficile</i> , <i>C. tetani</i> , <i>C. perfringens</i>	The benchmark for sterilization ; killing them defines the process.
3	Coccidia / protozoal cysts	<i>Cryptosporidium</i> , <i>Giardia</i> cysts	Highly resistant to chlorine — the reason for filtration of drinking water.
4	Mycobacteria	<i>M. tuberculosis</i> , <i>M. avium</i> , <i>M. terrae</i>	Waxy, lipid-rich cell wall; a tuberculocidal claim marks an intermediate level disinfectant .
5	Non-lipid / small non-enveloped viruses	Poliovirus, coxsackievirus, rhinovirus, norovirus, hepatitis A, human papilloma virus	No lipid envelope to dissolve — resist alcohols and quaternary compounds.
6	Fungi and fungal spores	<i>Candida</i> , <i>Aspergillus</i> , dermatophytes, <i>Trichophyton</i>	Moderately susceptible.
7	Vegetative bacteria	<i>Staphylococcus aureus</i> , <i>Pseudomonas aeruginosa</i> , <i>Salmonella</i> , <i>E. coli</i>	Readily killed, though <i>Pseudomonas</i> tolerates many weak disinfectants and can grow in them.
8 (least resistant)	Lipid / medium-sized enveloped viruses	HIV, hepatitis B and C, influenza, herpes simplex, SARS-CoV-2, measles	The lipid envelope is dissolved by alcohols, detergents and soaps — hence hand rub works well against them.

MNEMONIC

Top of the ladder downwards: **Prions – Spores – Coccidia – Mycobacteria – Non-lipid viruses – Fungi – Vegetative bacteria – Lipid viruses** — “Pretty Sure Cats Make Nice Furry Vet Labs”.

10.4 Kinetics of Microbial Death

Term	Meaning	Exam point
D-value (decimal reduction time)	The time (at a stated temperature) or the radiation dose required to reduce the microbial population by 90%, i.e. by one logarithm (1 log₁₀) .	For <i>G. stearothermophilus</i> at 121 °C, D ≈ 1.5–2 minutes .
Z-value	The temperature change (in °C) required to produce a tenfold change in the D-value .	For moist heat and <i>G. stearothermophilus</i> , Z ≈ 10 °C .
F-value / F₀	The equivalent time in minutes at 121 °C delivered by a process to a product, with Z = 10 °C. An F₀ of 8–12 minutes is generally accepted for pharmaceutical steam sterilization.	Allows a lower-temperature, longer cycle to be shown equivalent to 121 °C.
Thermal death point (TDP)	The lowest temperature that kills all the organisms in a standardised suspension in a fixed time (usually 10 minutes).	
Thermal death time (TDT)	The minimum time needed to kill all the organisms in a suspension at a given temperature.	
Sterility Assurance Level (SAL)	The probability that a single unit which has been through the process is non-sterile . The pharmacopoeial requirement is SAL of 10⁻⁶ — not more than one chance in a million of a viable organism surviving.	Very frequently asked. A 12-log reduction ('overkill') cycle is designed to achieve it.
Bioburden	The number and type of viable organisms present on an article before sterilization.	The higher the bioburden, the longer the process needed — hence cleaning first.

10.5 Factors Influencing the Efficacy of Sterilization and Disinfection

Factor	Effect
Number of organisms (bioburden)	Death is logarithmic; the larger the initial population, the longer the time required.
Type and form of organism	Spores, mycobacteria and non-enveloped viruses need far harsher treatment (Table 10.4). Biofilms are extremely resistant.

Factor	Effect
Concentration / potency of the agent	In general, activity rises with concentration — but there are important exceptions: 70% alcohol is more effective than absolute (100%) alcohol because water is needed for protein denaturation, and iodine and hydrogen peroxide have optimum concentrations.
Duration of exposure (contact time)	The agent must remain wet on the surface for the full stated contact time; this is the most commonly violated requirement in practice.
Temperature	Activity of chemical disinfectants generally increases 2- to 3-fold for each 10 °C rise ; heat sterilization time falls steeply as temperature rises.
pH	Glutaraldehyde needs alkaline pH (7.5–8.5) for sporicidal activity; hypochlorites and phenols are more active at acidic pH; quaternary ammonium compounds prefer alkaline pH.
Presence of organic matter (blood, pus, serum, faeces, sputum, milk)	Neutralises and physically shields the organisms. Hypochlorites, iodine and quaternary compounds are badly inactivated; glutaraldehyde and phenolics are relatively resistant to inactivation . Hence the rule: clean first .
Nature of the surface / article	Rough, porous, cracked, jointed, lumened and hinged items are hard to penetrate; narrow lumens and hinges must be opened and flushed .
Interfering substances	Soaps and anionic detergents inactivate quaternary ammonium compounds and chlorhexidine; hard water (Ca^{2+} , Mg^{2+}) reduces QAC activity; cork and some plastics adsorb preservatives.
Humidity	Critical for gaseous agents: ethylene oxide needs 30–60% relative humidity ; formaldehyde vapour needs > 60%; dry spores resist gases.
Presence of air	Trapped air prevents steam contact — air is the chief enemy of the autoclave ; it must be completely displaced or evacuated.

10.6 Mechanisms of Antimicrobial Action

Mechanism	How it kills	Agents acting this way
Protein denaturation and coagulation	Unfolds and precipitates structural and enzyme proteins.	Moist heat , alcohols, phenols, acids and alkalis.
Oxidative damage / oxidation of cell constituents	Oxidises proteins, lipids and nucleic acids; generates free radicals.	Dry heat , hydrogen peroxide, peracetic acid, halogens, potassium permanganate, ozone, plasma.
Damage to the cell membrane	Disorganises the lipid bilayer, causing leakage of cell contents.	Surface-active agents (QACs, soaps) , alcohols, phenols, chlorhexidine, polymyxins.
Alkylation of proteins and nucleic acids	Substitutes alkyl groups on $-\text{NH}_2$, $-\text{SH}$, $-\text{COOH}$ and $-\text{OH}$ groups, destroying function.	Ethylene oxide, formaldehyde, glutaraldehyde, beta-propiolactone (all alkylating agents).
Reaction with $-\text{SH}$ (sulphydryl) groups of enzymes	Inactivates essential enzymes.	Heavy metals — mercury, silver, copper ; oxidising agents.
Damage to nucleic acid	Forms thymine dimers or breaks the DNA strand, preventing replication.	Ultraviolet radiation (thymine dimers), ionising radiation (strand breaks, free radicals), acridine and aniline dyes.
Physical removal of organisms	The organism is not killed but separated from the fluid or air.	Filtration , washing, scrubbing, sedimentation, centrifugation.

10.7 Master Classification of Methods

TABLE 10.5 MASTER CLASSIFICATION OF METHODS OF STERILIZATION AND DISINFECTION

Category	Sub-group	Methods
A. PHYSICAL	Sunlight	Direct sunlight (natural method of sterilization)
	Drying / desiccation	Air drying — unreliable, bacteriostatic only

Category	Sub-group	Methods
	Dry heat	Red heat · Flaming · Incineration · Hot air oven · Infrared radiation · Microwave
	Moist heat below 100 °C	Pasteurisation · Inspissation (Koch–Henle) · Vaccine bath · Serum bath · Low-temperature steam formaldehyde (LTSF)
	Moist heat at 100 °C	Boiling · Steam at atmospheric pressure (Koch's or Arnold's steamer) · Tyndallisation (intermittent sterilization)
	Moist heat above 100 °C	Autoclave (steam under pressure) — gravity displacement and pre-vacuum (porous load) types
	Radiation	Non-ionising — ultraviolet , infrared · Ionising — gamma rays (Co-60) , X-rays, electron/cathode beams
	Sonic energy	Ultrasonic and sonic vibration (mainly used for cleaning)
B. MECHANICAL	Removal of organisms	Filtration — candle, asbestos, sintered glass, membrane, HEPA filters · Washing, scrubbing, mopping, dusting · Ultrasonic and washer-disinfector cleaning · Sedimentation and centrifugation · Laminar air flow
C. CHEMICAL	Liquid agents	Alcohols · Aldehydes · Phenols · Halogens · Oxidising agents · Heavy metals · Surface-active agents · Dyes · Acids and alkalies
	Gaseous agents	Ethylene oxide · Formaldehyde gas · Beta-propiolactone · Hydrogen peroxide vapour and plasma · Ozone · Chlorine dioxide

10.8 Control of Sterilization — Sterility Indicators

Every sterilizing process must be monitored. Three classes of monitor are used, and a fourth — sterility testing — tests the product itself.

Type of control	What it monitors	Examples
Physical (mechanical) indicators	That the machine reached and held the required physical conditions.	Thermometer and thermocouple readings, pressure gauge, timer, temperature–time–pressure chart recorder or printout, vacuum gauge, Browne's tube (a sealed glass tube whose indicator turns from red through amber to green when the correct temperature–time has been reached).
Chemical indicators	That a chemical change requiring the sterilizing conditions has occurred; graded as Class 1 to Class 6 by ISO 11140.	Class 1 – external process indicator (autoclave tape, indicator on the pack). Class 2 – specific-use test, notably the Bowie–Dick test which detects residual air and inadequate steam penetration in a pre-vacuum autoclave (done daily). Class 3 – single-variable. Class 4 – multi-variable. Class 5 – integrating indicator, reacting to all critical variables. Class 6 – emulating cycle-verification indicator.
Biological indicators (BI)	That standardised, highly resistant bacterial spores were actually killed — the most reliable proof of sterilization.	Spore strips, self-contained vials or ampoules. Placed in the most inaccessible part of the load, then incubated: no growth = process effective .
Sterility testing of the product	Whether the finished article is in fact sterile.	Sample units incubated in fluid thioglycollate medium at 30–35 °C (for anaerobes and aerobes) and soyabean-casein digest (tryptone soya) broth at 20–25 °C (for fungi and aerobes) for 14 days ; or membrane filtration of the whole volume followed by culture of the membrane.

TABLE 10.6 BIOLOGICAL INDICATORS — MUST BE MEMORISED

Sterilizing process	Biological indicator organism	Note
Autoclave / moist heat	<i>Geobacillus stearothermophilus</i> (formerly <i>Bacillus stearothermophilus</i>), ATCC 7953	Thermophilic; spores killed in about 12 minutes at 121 °C ; incubated at 55–60 °C .
Hot air oven / dry heat	<i>Bacillus atrophaeus</i> (formerly <i>B. subtilis</i> var. <i>niger</i> / <i>globigii</i>)	Incubated at 30–35 °C.
Ethylene oxide gas	<i>Bacillus atrophaeus</i>	Same organism as for dry heat — a common examination pairing.
Ionising (gamma) radiation	<i>Bacillus pumilus</i>	Highly radiation-resistant spores.
Hydrogen peroxide gas plasma	<i>Geobacillus stearothermophilus</i>	

Sterilizing process	Biological indicator organism	Note
Low-temperature steam formaldehyde	<i>Geobacillus stearothermophilus</i>	
Filtration (membrane integrity / bacterial challenge)	<i>Brevundimonas diminuta</i> (formerly <i>Pseudomonas diminuta</i>)	The standard challenge organism for validating a 0.22 µm sterilising filter.

HIGH-YIELD FOR EXAM

- *Geobacillus stearothermophilus* = autoclave; *Bacillus atrophaeus* = dry heat and ethylene oxide; *Bacillus pumilus* = gamma radiation; *Brevundimonas (Pseudomonas) diminuta* = 0.22 µm filter.
- **Bowie-Dick test** = detection of **air removal** / **steam penetration** failure in a **pre-vacuum** autoclave. **Browne's tube turning green** = correct temperature–time achieved.
- The **biological indicator is the most reliable** monitor of sterilization; physical and chemical indicators only show that conditions were probably met.



11.1 Sunlight

- Sunlight is the **natural method of sterilization** of water in tanks, rivers and lakes and of exposed surfaces.
- Its action is due mainly to the **ultraviolet component** together with heat and drying.
- In tropical countries such as India sunlight is appreciably **more effective** because of the greater intensity and duration of sunshine.
- **Uses:** drying and airing of bedding, mattresses, blankets, clothing, books and shoes in infectious disease; **SODIS** (solar disinfection) of drinking water in transparent PET bottles exposed for 6 hours of bright sun; solar drying of grain and chillies.
- **Limitations:** slow, unreliable, only surface action, no penetration through glass, cloth or dirt, and dependent on the weather — therefore **never used when true sterility is needed**.

11.2 Drying (Desiccation)

- Water is essential for bacterial growth; drying therefore has a definite **bacteriostatic (inhibitory) effect** on vegetative bacteria.
- **Spores are unaffected**, and many organisms survive drying for long periods — *M. tuberculosis* in dust, staphylococci in dried pus, and dried spore-formers indefinitely.
- **Freeze-drying (lyophilisation)** — rapid freezing followed by dehydration under high vacuum — is used not to kill but to **preserve** bacteria, vaccines, sera and heat-labile drugs for years.
- **Conclusion: drying is an unreliable method and is not a method of sterilization.**

11.3 Dry Heat — General Principles

DEFINITION

- **Mechanism of killing by dry heat:** oxidative damage to cell constituents, denaturation of proteins, a toxic rise in electrolyte concentration as water is lost, and finally charring and carbonisation.
- Dry heat **penetrates poorly and kills more slowly** than moist heat, so it needs a **higher temperature and a longer time** for the same effect.

TABLE 11.1 DRY-HEAT METHODS

Method	Temperature	Time	Articles sterilized	Notes
Red heat	Until the metal glows red (≈ 800 – 1000°C)	A few seconds	Bacteriological (inoculating) loops and wires, straight wires, tips of forceps, needles, searing spatulas	Held in the flame of a Bunsen burner until red-hot; the simplest and commonest laboratory method. The loop must be cooled before touching the culture.
Flaming	Passed through a Bunsen flame without letting the article become red hot	A few seconds	Mouths of culture tubes and flasks, glass slides, cover slips, scalpels, needles, cotton-wool plugs	Often the article is first dipped in spirit (alcohol) and then ignited — this is called flaming and is of doubtful efficacy ; it is not true sterilization.
Incineration	≈ 870 – 1200°C (primary chamber $800 \pm 50^\circ\text{C}$, secondary chamber $1050 \pm 50^\circ\text{C}$ in a hospital incinerator)	Until reduced to ash	Soiled dressings, cotton swabs, bandages, surgical pads, plaster casts, bedding, pathological and anatomical waste, animal carcasses, placenta, laboratory cultures and expired/cytotoxic drugs	An excellent method of safe disposal by complete destruction. Never incinerate PVC and chlorinated plastics (produce dioxins and furans), mercury, radioactive material or pressurised containers.
Hot air oven	160°C (also 170°C or 180°C)	2 hours at 160°C (1 hour at 170°C ; 30 min at 180°C ; $2\frac{1}{2}$ hours at 150°C) — holding time counted after the thermometer reaches the set temperature	Glassware (test tubes, petri dishes, pipettes, flasks), metal instruments, forceps, scalpels, scissors, glass syringes, swabs, oils, fats, waxes, greases, liquid paraffin, dusting and sulphonamide powders, ointment bases, glycerol, all-glass syringes	The most widely used method of dry-heat sterilization . See §11.4.

Method	Temperature	Time	Articles sterilized	Notes
Infrared radiation	≈ 180 °C	7½–10 minutes	Metal instruments, glass syringes, needles, catheters, packaged goods on a conveyor belt	Infrared rays from an electrically heated element are focused; the articles travel through a tunnel on a conveyor belt . Rapid and suited to mass industrial processing; radiation does not itself sterilize — it heats the article.
Microwave	Generated at 2450 MHz ; the article reaches roughly 97–100 °C	Varies (minutes)	Non-metallic items, laboratory waste, some biomedical waste, food, dentures	Kills by the heat produced when water molecules are agitated; needs moisture and gives uneven heating (hot and cold spots). It is not recommended for anatomical, cytotoxic or radioactive waste and cannot be used for metals.

11.4 The Hot Air Oven in Detail

► 11.4.1 Construction and Working

- An insulated, **double-walled metal chamber** heated by electrical elements in the walls or floor; the standard laboratory pattern is the **Hot Air Oven of Pasteur**.
- Fitted with **perforated shelves**, a **thermostat**, a **thermometer** (or thermocouple) projecting into the chamber, a **timer** and usually a **fan** or blower to circulate air and prevent hot and cold spots (a 'forced-air' or 'mechanical-convection' oven).
- **Loading**: the oven must be **loosely packed** — no article should touch the walls, and air must circulate freely between the packs, because **hot air is a bad conductor with poor penetrating power**.
- **Wrapping**: articles are wrapped in paper, placed in metal canisters or boxes, or plugged with cotton wool; test tubes and flasks are plugged and covered with paper caps.
- **Cycle**: load → heat to the set temperature → **hold for the full holding time** → switch off → allow to cool to below **60 °C before opening** — if opened while hot, **the sudden inrush of cool air cracks the glassware**.

► 11.4.2 Precautions, Sterility Control, Advantages and Limitations

Heading	Points
Precautions	Articles must be clean, dry and grease-free before loading · do not overload · leave space between articles · the holding time begins only when the whole load reaches the temperature · never place articles on the floor of the chamber · cotton wool, paper and cloth char above 160 °C · rubber, plastics and most fabrics cannot be used · cool before opening.
Sterility controls	Physical — thermocouples and a temperature chart; chemical — Browne's tube No. 3 (turns green) and heat-indicating tape; biological — spores of <i>Bacillus atrophaeus</i> (10 ⁶ spores on paper strips) placed in the centre of the load, which should be killed in 60 minutes at 160 °C.
Advantages	Simple, reliable, cheap to run, no corrosion of metal instruments or blunting of sharp edges compared with steam, no wetting of the load , suitable for articles that steam cannot penetrate (oils, fats, waxes, powders, glycerol) and for closed containers .
Limitations / disadvantages	Slow (a full cycle with heating-up and cooling takes several hours) · poor penetration · the high temperature destroys rubber, plastics, fabrics, cotton, paper, most culture media, surgical dressings and aqueous solutions · graduated and volumetric glassware may lose calibration · glass may crack on rapid cooling · unsuitable for heat-labile pharmaceuticals.

HIGH-YIELD FOR EXAM

- **Hot air oven: 160 °C for 2 hours** is the standard answer. Equivalents: **170 °C – 1 hour, 180 °C – 30 minutes, 150 °C – 2½ hours**.
- Articles sterilized **only by dry heat, never by autoclave: oils, fats, waxes, greases, liquid paraffin, dusting powders, glycerol and closed glass containers** — because **steam cannot penetrate them**.
- **Incineration temperature** in a hospital incinerator: primary chamber **800 ± 50 °C**, secondary chamber **1050 ± 50 °C** with a gas retention time of at least **2 seconds**.



CHAPTER 12 PHYSICAL METHODS — II : MOIST HEAT AND THE AUTOCLAVE

DEFINITION

- **Mechanism of killing by moist heat:** **denaturation and coagulation of proteins**, especially of essential enzymes, together with disruption of membranes and breakage of hydrogen bonds. Water vapour carries a large amount of **latent heat of condensation** (about 518–540 calories per gram at 100 °C), which is released directly onto the article the moment steam condenses on it.
- **Moist heat is far more efficient than dry heat** and therefore works at a lower temperature and in a shorter time. This is the single most important comparison in the chapter.

12.1 Moist Heat Below 100 °C

TABLE 12.1 MOIST HEAT BELOW 100 °C

Method	Temperature and time	Purpose and material treated
Pasteurisation of milk — Holder (LTLT) method	63 °C (145 °F) for 30 minutes , then rapid cooling to 13 °C or below	Kills the milk-borne pathogens <i>Mycobacterium bovis</i> and <i>M. tuberculosis</i> , <i>Brucella abortus</i> , <i>Salmonella</i> , <i>Streptococcus</i> , <i>Coxiella burnetii</i> , <i>Campylobacter</i> , <i>Listeria</i> without altering the taste or nutritive value. Spores and thermophilic organisms survive — it is not sterilization .
Pasteurisation — Flash / HTST method	72 °C (161 °F) for 15–20 seconds , then rapid cooling	The method used in modern large dairies; HTST = High Temperature Short Time. The most widely used commercially.
Ultra High Temperature (UHT)	125–150 °C for 1–5 seconds (commonly 135–140 °C for 2–4 seconds)	Gives commercially sterile, long-shelf-life packaged (tetra-pack) milk.
Inspissation (Koch–Henle method)	80–85 °C for 30 minutes on 3 successive days in an inspissator	Solidifies and disinfects egg- and serum-containing culture media — Löwenstein–Jensen medium , Dorset egg medium and Löffler's serum slope . Autoclaving would coagulate and ruin them.
Vaccine bath	60 °C for 1 hour in a water bath	Inactivates the organisms in a bacterial vaccine and destroys contaminating organisms without destroying antigenicity.
Serum / body-fluid bath	56 °C for 1 hour on several successive days (heating serum to 56 °C for 30 minutes also inactivates complement)	Disinfection of serum and body fluids , whose proteins coagulate at higher temperatures.
Low-Temperature Steam Formaldehyde (LTSF)	Dry saturated steam at 73–80 °C with formaldehyde vapour , ≈ 2 hours	A sterilizing process for heat-sensitive items — cystoscopes, plastic and rubber goods, some endoscopes. Requires efficient air removal and post-cycle aeration.
Low-temperature steam (LTS) / pasteurising washer-disinfector	73 °C for 10 minutes (or 80 °C for 1 min; 90 °C for 1 s)	High-level disinfection of respiratory and anaesthetic equipment, bed-pans and utensils in a washer-disinfector.

HIGH-YIELD FOR EXAM

- **Pasteurisation temperatures:** **63 °C/30 min (Holder)** and **72 °C/15–20 s (Flash)**. The organism whose thermal resistance sets the standard is *Coxiella burnetii* (the most heat-resistant milk pathogen); the classical target was *Mycobacterium tuberculosis/bovis*.
- The test used to check the efficiency of pasteurisation is the **phosphatase test** (alkaline phosphatase should be destroyed); the **standard plate count** and **coliform count** are also used, and the **methylene blue reduction test** assesses milk quality.
- **Inspissation** = **80–85 °C × 30 min × 3 days**, used for **Löwenstein–Jensen medium**. Do not confuse it with **Tyndallisation** = **100 °C × 20 min × 3 days**.

12.2 Moist Heat At 100 °C

TABLE 12.2 MOIST HEAT AT 100 °C

Method	Conditions	Uses	Comments
Boiling	100 °C for 10–30 minutes (at least 10 min; 20–30 min preferred). Boiling point falls with altitude, so time must be increased.	Emergency disinfection of metal instruments, syringes, needles, glass syringes, rubber goods, bed-pans, feeding utensils, linen and drinking water where no autoclave exists.	NOT sterilization — bacterial spores, hepatitis B virus and some thermophiles may survive . Adding 2% sodium bicarbonate (washing soda) raises the boiling point to about 105 °C, increases the killing power and prevents rusting of instruments . Articles must be fully immersed and the lid kept closed; timing starts when boiling begins.
Steam at atmospheric pressure (free steam)	100 °C for 90 minutes in a Koch's or Arnold's steamer (or an autoclave with the lid loose and the discharge tap open)	Culture media that are damaged by higher temperatures — media containing sugars, gelatin, milk, DCA and selenite F broth.	Steam at 100 °C kills vegetative organisms but not spores in a single exposure.
Tyndallisation (intermittent or fractional sterilization)	Free steam at 100 °C for 20–45 minutes on 3 successive days (classically 20 min × 3 days), the material being incubated at room temperature between exposures	Heat-labile culture media containing sugars, gelatin, egg, serum or milk ; some pharmaceutical preparations.	Principle: the first exposure kills the vegetative forms; surviving spores germinate into vegetative forms during the interval and are killed by the next exposure. Named after John Tyndall . Also called the Koch–Henle principle when applied at 80 °C (inspissation).

12.3 Moist Heat Above 100 °C — The Autoclave

DEFINITION

An **autoclave (steam sterilizer)** sterilizes by exposing articles to **saturated steam under pressure**. Raising the pressure raises the temperature at which water boils, so steam hotter than 100 °C is obtained. It is the **most widely used, most reliable, quickest and cheapest method of sterilization** in hospitals and laboratories.

► 12.3.1 Principle

- Water boils at 100 °C when the pressure above it is one atmosphere. If the pressure is raised, the boiling point and therefore the temperature of the steam rises.
- The killing is done by **saturated steam that condenses on the article**, giving up its **latent heat** and simultaneously wetting the surface — hence the two essentials are **direct steam contact** and **complete removal of air**.
- Air is the great enemy of the autoclave**. Air is heavier than steam, forms a layer at the bottom, prevents steam from contacting the load, and lowers the temperature actually achieved for a given pressure. Therefore all air must be displaced downwards through the discharge tap or evacuated by a pump.

TABLE 12.3 PRESSURE–TEMPERATURE–TIME RELATIONSHIPS (MUST BE MEMORISED)

Gauge pressure	Absolute pressure	Temperature of saturated steam	Holding time	Chief use
0 psi (atmospheric)	1 atm	100 °C	— (free steam)	Tyndallisation, delicate media
15 psi (1.05 kg/cm², ≈ 103 kPa, 1 bar)	2 atm	121 °C	15–20 minutes	The standard cycle — culture media, aqueous solutions, rubber, glassware, instruments, linen
20 psi (1.4 kg/cm²)	≈ 2.4 atm	126 °C	10 minutes	Faster routine cycle
30 psi (2.1 kg/cm²)	≈ 3 atm	134 °C	3–7 minutes (commonly 3–3½ min)	High-vacuum porous-load cycle — dressings, gowns, drapes, instruments; the standard hospital 'flash' cycle
30 psi	≈ 3 atm	134 °C	18 minutes	Prion (CJD) decontamination

► 12.3.2 Types of Autoclave

Type	Description	Suitable for
Laboratory (domestic / pressure-cooker) type	A vertical, cylindrical vessel of gunmetal or stainless steel in a supporting iron case, heated electrically or by gas; the lid is fastened by screw clamps and carries a pressure gauge, a safety (blow-off) valve, a steam discharge tap and a thermometer .	Small laboratory loads — culture media, glassware, discarded cultures.
Gravity (downward) displacement type — e.g. the hospital 'bowl and instrument' sterilizer	Steam, being lighter than air, enters at the top and drives the air downward and out through the discharge channel at the bottom , which contains a near-to-steam thermostatic trap that closes when steam reaches it.	Unwrapped solid instruments, bowls, fluids in vented containers, culture media. Poor at penetrating porous loads.
Pre-vacuum / high-vacuum porous-load type	A vacuum pump removes about 98% of the air before steam admission , so steam penetrates instantly and evenly; the cycle is very short (134 °C for 3–4 minutes) and the load is dried by a post-cycle vacuum.	Porous loads — surgical dressings, gowns, drapes, linen packs, wrapped instrument trays, hollow and lumened instruments. The standard in a modern CSSD. Must be checked daily by the Bowie–Dick test .
Horizontal rectangular (hospital) type	A large, jacketed, rectangular chamber with a door at one or both ends (double-ended / pass-through , allowing loading on the dirty side and unloading on the clean side).	Bulk hospital work; the basis of CSSD design.
Continuous / rotary and hydroclave types	Industrial designs for continuous processing of ampoules, bottles and biomedical waste.	Pharmaceutical manufacture; hydroclaving of biomedical waste.

► 12.3.3 Parts of a Laboratory Autoclave

- **Chamber (inner vessel)** — gunmetal or stainless steel, holding the load on a perforated tray above the water.
- **Outer jacket / case** — insulating and protective.
- **Lid** with a rubber gasket and radial screw clamps to make it airtight.
- **Heating element** (electric immersion heater) or gas burner beneath the water.
- **Pressure gauge** — shows the steam pressure inside.
- **Safety valve (blow-off valve)** — a spring-loaded or weighted valve that releases steam if the pressure exceeds the set limit, preventing explosion.
- **Steam discharge (air-vent) tap / stopcock** — to allow the air–steam mixture to escape at the start and to release steam at the end.
- **Thermometer or thermocouple** — the **temperature, not the pressure, is the true measure of sterilizing conditions**.
- **Whistle / pet-cock** and a **water-level indicator**.

► 12.3.4 Operating Procedure

1. Check that there is **sufficient water** in the chamber.
2. **Load loosely**, leaving space for steam circulation; place articles on the perforated tray, not in the water; bottles and tubes must have **loosened caps or vented closures**, and dressing drums must have their **vents open**.
3. Close the lid and **tighten the clamps evenly**, keeping the **discharge tap open**.
4. Switch on the heater. Allow steam to escape freely from the discharge tap until **all the air has been driven out** (a steady, forceful jet of steam with no spluttering; about 10 minutes, or until the mixture emerging is pure steam).
5. **Close the discharge tap**. The pressure now rises.
6. When the gauge reads **15 psi** and the thermometer reads **121 °C**, start timing the **holding period of 15–20 minutes**, keeping the pressure steady.
7. Switch off the heat and allow the autoclave to **cool until the pressure gauge returns to zero**.
8. Open the discharge tap **slowly** to admit air; then open the lid. For fluids, cool slowly to prevent the bottles boiling over and cracking.
9. Remove the load, check the indicators, and **allow packs to dry** before storage; a **wet pack is a contaminated pack**.

► 12.3.5 Articles Sterilized by Autoclave

Suitable (autoclavable)	NOT suitable
Culture media (except those with sugars, egg, serum or gelatin), aqueous solutions, parenteral (IV) fluids, buffers	Oils, fats, waxes, greases, liquid paraffin, petroleum jelly — steam cannot penetrate them (use dry heat)

Suitable (autoclavable)	NOT suitable
Surgical instruments, bowls, kidney trays, forceps, scissors (stainless steel)	Dry powders (dusting powder, talc, sulphonamide powder) — steam cannot penetrate (use dry heat)
Surgical dressings, gauze, cotton, gowns, drapes, linen, caps, masks, towels (pre-vacuum porous load cycle)	Sharp cutting instruments — scalpel blades, scissors, needles may be blunted and corroded by repeated steam (prefer dry heat or a chemical sterilant)
Rubber goods — gloves, tubing, catheters, bungs, closures (115–121 °C, shorter cycle)	Heat-labile plastics — polystyrene, PVC, most disposables (use ethylene oxide or gamma radiation)
Glassware, glass syringes, pipettes, petri dishes, flasks	Heat-labile solutions — sera, vaccines, antibiotic and vitamin solutions, urea broth, blood products (use filtration)
Discarded cultures, contaminated media, plates, pipettes and infectious laboratory waste (121 °C for 30–60 minutes)	Delicate instruments with lenses and fibre optics — endoscopes, cystoscopes (use chemical HLD or LTSF)
Autoclavable plastics — polypropylene, PTFE, polycarbonate	Volumetric/graduated glassware may lose calibration; sealed glass ampoules may burst

► 12.3.6 Causes of Failure of the Autoclave

- **Incomplete removal of air** — by far the commonest cause; produces an air–steam mixture whose temperature is below 121 °C at 15 psi.
- **Overloading or tight packing**, preventing steam circulation and penetration.
- **Use of superheated steam** (too dry) or **wet steam** (excessive water content) — only **dry saturated steam** sterilizes properly.
- **Holding time not observed** — timing begun before the whole load reached temperature, or the cycle cut short.
- **Failure to open vents** of drums and to loosen container caps.
- **Wrapping in impervious material** (e.g. aluminium foil, oil-cloth) which steam cannot penetrate.
- **Faulty gauges, thermostat, gasket or steam trap**; leakage; insufficient water.
- **Articles not cleaned** beforehand — organic matter and grease protect organisms.
- **Wet packs on removal**, which wick contamination from the environment.

12.4 Dry Heat versus Moist Heat — The Comparison Table

TABLE 12.4 DRY HEAT COMPARED WITH MOIST HEAT

Feature	Dry heat (hot air oven)	Moist heat (autoclave)
Mechanism of killing	Oxidation of cell constituents, protein denaturation, charring, and a toxic rise in electrolyte concentration	Coagulation and denaturation of proteins ; latent heat of condensation of steam
Temperature required	Higher — 160–180 °C	Lower — 121–134 °C
Time required	Longer — 30 min to 2 hours holding, plus long heating and cooling	Shorter — 3 to 20 minutes holding
Penetration	Poor (air is a bad conductor)	Excellent (steam penetrates and condenses)
Efficiency	Less efficient	More efficient
Effect on the load	Leaves the load dry ; may char paper and cloth, blunt and discolour instruments and crack glass on rapid cooling	Leaves the load moist ; may corrode and blunt steel and cannot be used for moisture-sensitive items
Best suited for	Oils, fats, waxes, powders, glassware, glass syringes, metal instruments, closed containers	Culture media, aqueous solutions, dressings, linen, gowns, rubber goods, instruments, laboratory waste
Biological indicator	<i>Bacillus atrophaeus</i>	<i>Geobacillus stearothermophilus</i>
Chemical indicator	Browne's tube No. 3, dry-heat indicator tape	Browne's tube No. 1, autoclave tape, Bowie–Dick test (pre-vacuum)

HIGH-YIELD FOR EXAM

- **121 °C · 15 psi · 15 minutes** — memorise this triad. 15 psi = 1.05 kg/cm² ≈ 103 kPa ≈ 1 bar ≈ 2 atmospheres absolute.
- **Moist heat kills by protein coagulation; dry heat kills by oxidation.**
- The **commonest cause of autoclave failure is incomplete removal of air**; the temperature (not the pressure) is the true index of sterilizing conditions.

- **Discarded cultures and contaminated media** are autoclaved at **121 °C for 30–60 minutes** before disposal — a longer time than the routine cycle because of the very high bioburden.



13.1 Classification of Radiation

Type	Examples	Mechanism	Effect
Non-ionising radiation	Ultraviolet (UV) rays, infrared rays, microwaves	Low energy, poor penetration ; UV acts on DNA, infrared and microwaves act by generating heat	Surface disinfection only
Ionising radiation	Gamma rays (from Cobalt-60 , caesium-137), X-rays , electron beams / cathode rays (accelerated electrons), alpha and beta particles	High energy; ejects electrons, producing ions and free radicals that break DNA strands	True sterilization , with excellent penetration

13.2 Ultraviolet Radiation

Heading	Details
Source	Low-pressure mercury vapour lamps (UV or 'germicidal' lamps); also present in sunlight.
Wavelength	The germicidal range is 200–280 nm (UV-C) ; maximum bactericidal activity is at 253.7 nm ($\approx$ 254 nm, often quoted as 260 nm) , which corresponds to the absorption maximum of nucleic acid (DNA) at 260 nm .
Mechanism	Absorbed by DNA, producing cross-linking between adjacent thymine bases — pyrimidine (thymine) dimers — which block replication and transcription; also causes mutation. Repair is possible by photo-reactivation and excision repair , so sub-lethal doses may allow recovery.
Uses	Air disinfection of operation theatres, laminar-flow hoods, biological safety cabinets, tissue-culture rooms, pharmaceutical filling rooms, bacteriological laboratories, entry vestibules and hospital isolation rooms · surface disinfection of work benches and safety cabinets · disinfection of drinking water and of waste water · disinfection of the inside of biosafety cabinets after work.
Advantages	No chemical residue, no toxicity to the article, cheap to run, effective against bacteria, fungi and viruses (including <i>M. tuberculosis</i> and SARS-CoV-2) on exposed surfaces.
Limitations / disadvantages	Very poor penetration — does not pass through glass, water films, paper, cloth, dust, plastics or grease, and acts only on the surface directly exposed; shadowed areas are unaffected · activity falls with distance, with the age of the lamp and at low temperature or high humidity · spores are relatively resistant · organic matter and dust protect organisms · lamps need regular cleaning and output monitoring.
Hazards to staff	Conjunctivitis, photokeratitis ('welder's flash'), erythema and burns of skin , and a long-term risk of skin cancer; ozone may be produced. Lamps must be switched off while people are in the room , and protective goggles and clothing worn.

HIGH-YIELD FOR EXAM

UV maximum germicidal wavelength = 253.7 nm ($\approx$ 254 nm); the mechanism is the formation of **thymine (pyrimidine) dimers in DNA**. UV is used for **air and surface disinfection, not for sterilization of instruments**, because it has **no penetrating power**.

13.3 Ionising Radiation — 'Cold Sterilization'

Heading	Details
Sources	Gamma rays from Cobalt-60 (^{60}Co) — the commercial standard; also caesium-137; high-energy electron (cathode) beams from linear accelerators; X-rays .
Dose	The internationally accepted sterilizing dose is 25 kGy = 2.5 Mrad (2.5 megarad); 1 Gy = 1 J/kg and 1 kGy = 0.1 Mrad. Lower doses (radurisation, radicidation at 1–10 kGy) are used for food preservation, and radappertisation (> 10 kGy) for commercial sterility.
Mechanism	Ejects electrons from atoms, generating ions and highly reactive free radicals (especially hydroxyl radicals from water) that cause single- and double-strand breaks in DNA , cross-linking and damage to enzymes and membranes. It is a non-thermal process.
Uses	Industrial, large-scale sterilization of pre-packed single-use disposables: plastic syringes and needles, IV infusion sets, blood-collection bags and transfusion sets, catheters, urine bags, surgical gloves, cannulae, sutures and surgical dressings, petri dishes and plastic labware, dental items, bone and tissue grafts, heart valves, prosthetic implants, adhesive dressings ; also sterilization of some pharmaceuticals, ophthalmic ointments, antibiotics, hormones, vitamins and food , and the production of some vaccines .

Heading	Details
Advantages	Excellent penetration — sterilizes articles already sealed inside their final packaging , so there is no risk of recontamination · no rise in temperature , therefore ideal for heat-labile plastics and biological materials · no toxic residue and no need for aeration (unlike ethylene oxide) · continuous, automated, high-throughput, reliable and easily validated · highly sporicidal .
Disadvantages	Very high capital cost and elaborate shielding — available only in large industrial plants, never in a hospital · needs trained staff and strict radiation protection , licensing and monitoring · may cause discoloration, embrittlement and degradation of some plastics (PTFE, polypropylene) and of glass , and may produce free radicals or off-flavours in some drugs and foods · public apprehension about irradiated products.
Biological indicator	Spores of <i>Bacillus pumilus</i> ; dosimetry is monitored with radiochromic and perspex dosimeters and colour-change dose indicators.

HIGH-YIELD FOR EXAM

- **Disposable plastic syringes, needles, IV sets, gloves, catheters and sutures are sterilized by GAMMA RADIATION from Cobalt-60 at a dose of 2.5 Mrad (25 kGy) at the factory.** This is one of the most frequently asked facts in the entire unit.
- Ionising radiation is called '**cold sterilization**' because there is no significant rise in temperature.
- **Order of increasing resistance to radiation:** vegetative bacteria < moulds and yeasts < bacterial spores < **viruses (most resistant)**. Note this is **different** from the order of resistance to heat and chemicals.

13.4 Sonic and Ultrasonic Vibration

- High-frequency sound waves (**> 20 000 cycles per second**) passed through a liquid produce **cavitation** — the rapid formation and collapse of microscopic bubbles — which disrupts cells mechanically.
- **It is not a reliable method of sterilization or disinfection:** many organisms, and all spores, survive.
- Its real and important use is **mechanical cleaning**: the **ultrasonic cleaner (sonicator)** removes blood, tissue debris and cement from the **joints, hinges, serrations and lumens of surgical and dental instruments** before they are sterilized. It is therefore a **pre-sterilization cleaning device** and a **mechanical method**.
- Ultrasonics are also used in the laboratory to **disrupt bacterial cells** in order to release intracellular enzymes and antigens.

REMEMBER

Associate ultrasonic waves with **cleaning of instruments, not sterilization**. They are a **pre-sterilization** step, and they belong to the **mechanical** group of methods.



CHAPTER 14 MECHANICAL METHODS — FILTRATION AND PHYSICAL REMOVAL

Mechanical methods **remove** micro-organisms rather than killing them. The most important of them, and the one always asked about, is **filtration**.

14.1 Filtration — Principle and Indications

DEFINITION

- **Principle:** a liquid or a gas is passed through a filter of pore size small enough to **retain the micro-organisms mechanically** (by sieving), assisted by **adsorption** onto the filter material and by the electrostatic charge of the membrane. The organisms are **removed, not killed**.
- **Chief indication:** **sterilization of heat-labile (thermolabile) fluids** that would be destroyed by autoclaving.

- **Materials sterilized by filtration:** sera and serum-containing media, **vaccines and toxoids**, bacterial toxins and culture filtrates, **antibiotic solutions, vitamin and hormone solutions, urea broth**, sugar solutions, **heat-labile parenteral (large- and small-volume) solutions**, ophthalmic solutions, radiopharmaceuticals, tissue-culture media, blood products, **enzyme preparations**, and **air** (by HEPA filters).
- Also used to **separate soluble products of bacterial growth** (exotoxins, bacteriophage) from the organisms, to **clarify** liquids, and to **concentrate organisms from large volumes of water, urine or beverages for culture** (the membrane-filter technique used in water bacteriology).

14.2 Types of Filters

TABLE 14.1 TYPES OF FILTERS

Filter	Material	Description and grades	Uses and remarks
Earthenware candle filters — Chamberland filter	Porcelain (unglazed kaolin / clay)	Hollow candle-shaped filters; graded L1 to L13 (Chamberland–Pasteur), L1 being the coarsest. French in origin.	Water purification and filtration of fluids; largely historical in the laboratory but the principle survives in domestic candle water filters.
Earthenware candle filters — Berkefeld filter	Kieselguhr (diatomaceous earth)	Graded by porosity as V (Veil – coarse), N (Normal – medium) and W (Wenig – fine/dense) . German in origin.	Water purification; the grades V, N, W are a classic MCQ .
Mandler filter	Kieselguhr, asbestos and plaster of Paris	—	Filtration of fluids.
Seitz filter	Asbestos (magnesium silicate) pads	A single-use compressed asbestos disc clamped in a metal mount; the pad is discarded after use.	Was widely used for sterilizing sera and sugar solutions. Its great disadvantages are that asbestos is carcinogenic , and the pad adsorbs protein and alters the fluid ; largely abandoned in favour of membranes.
Sintered (fritted) glass filter	Finely ground borosilicate glass fused into a disc	Grades numbered by porosity (e.g. No. 5 for bacteria); fused into a funnel.	Minimal adsorption of protein , chemically inert, re-usable and cleanable; but brittle and expensive .
Membrane filter (the modern standard)	Cellulose acetate, cellulose nitrate, mixed cellulose esters, polycarbonate, PVDF, PTFE, nylon	Thin ($\approx 150\ \mu\text{m}$) discs with a very high proportion of uniform pores. Common pore sizes: 0.8, 0.45 and 0.22 μm ; a 0.22 μm membrane is the accepted 'sterilising grade' filter , while 0.45 μm is the standard for water bacteriology (coliform counting). Pre-filters and 0.1 μm membranes are used for mycoplasma.	The method of choice today for sterilizing heat-labile pharmaceutical solutions, and for counting organisms in water. Advantages: no adsorption/alteration of the fluid, disposable, cheap, rapid, allows the organisms retained to be cultured, and can be integrity-tested (bubble-point test). Disadvantages: low capacity and clogs easily, and membrane rupture would be unnoticed without integrity testing.
Syringe filter	Membrane in a plastic housing	0.22 or 0.45 μm , attached to the nozzle of a syringe.	Bedside and small-volume sterilization of drug solutions and of solutions for cell culture.
HEPA filter — High Efficiency Particulate Air	Pleated glass-fibre / borosilicate microfibre paper	Removes 99.97% of particles of 0.3 μm and above (H13/H14 grades remove 99.95–99.995%); works by interception, impaction and diffusion.	Air sterilization in operation theatres, laminar air flow benches , biological safety cabinets, pharmaceutical clean rooms, and isolation rooms for immunocompromised and transplant patients. ULPA filters are still finer.

HIGH-YIELD FOR EXAM

- **0.22 µm** = the sterilising-grade membrane pore size. **0.45 µm** = used for water bacteriology / coliform counts. **HEPA = 99.97% of particles ≥ 0.3 µm.**
- **Berkefeld grades: V, N, W** (coarse – fine). **Chamberland: L1–L13.** **Seitz = asbestos.** **Sintered glass = ground borosilicate glass.**
- **Filtration removes bacteria and fungi but does NOT remove viruses, mycoplasma, prions, bacterial toxins, endotoxins (pyrogens) or L-forms,** since these are smaller than the pore size. Therefore a filtered fluid is **not necessarily virus-free** — a very common MCQ.

14.3 Laminar Air Flow and Biological Safety Cabinets

- **Laminar air flow (LAF)** is the flow of HEPA-filtered air in **parallel, unidirectional streams at a uniform velocity** ($\approx 0.45 \text{ m/s} \pm 20\%$) so that particles are swept away without turbulence.
- **Horizontal LAF** — filtered air blows from the back of the cabinet **towards the operator**. It protects the **product only**, not the worker; it must **never** be used for infectious material, but is suitable for aseptic dispensing of sterile non-hazardous preparations and media pouring.
- **Vertical LAF** — filtered air flows downward from the top. Protects both the product and, in properly designed cabinets, the operator.

TABLE 14.2 BIOLOGICAL SAFETY CABINETS

Biological safety cabinet	Air flow and features	Protects	Used for
Class I	Room air is drawn in across the open front at $\geq 0.38 \text{ m/s}$ and exhausted through a HEPA filter. Open-fronted.	Operator and environment (not the product)	Low- and moderate-risk work; Biosafety Level 1–3 agents.
Class II (A1, A2, B1, B2)	About 70% of the air is recirculated through a HEPA filter as a vertical downflow over the work surface, with an inward air curtain at the front aperture; 30% exhausted through HEPA.	Operator, environment AND product	The commonest type in hospitals and laboratories — tissue culture, microbiology, and aseptic compounding of sterile and cytotoxic drugs (Class II B2 for cytotoxics).
Class III (glove box)	Totally enclosed, gas-tight, maintained under negative pressure ; access only through attached rubber gloves; supply and exhaust air both HEPA-filtered (exhaust often double-HEPA or incinerated).	Maximum containment — absolute protection	Biosafety Level 4 agents — highly dangerous pathogens such as the Ebola, Marburg, Nipah and Lassa viruses.

14.4 Other Mechanical Methods

Method	How it works and where it is used
Washing, scrubbing and cleaning	The most basic and most important mechanical method. Water, detergent and friction physically remove organic matter and the great majority of organisms. Hand-washing with soap and water for 40–60 seconds is the single most effective measure for preventing hospital-acquired infection. Cleaning is always the first step before any disinfection or sterilization.
Wet mopping, damp dusting and wiping	Used for floors, walls, furniture and surfaces. Dry sweeping and dry dusting are forbidden in hospitals because they raise infected dust into the air; always use damp or wet methods, ideally with a detergent–disinfectant, and clean from the cleanest to the dirtiest area.
Ultrasonic cleaner (sonicator) and washer-disinfector	Removes debris from hinges, box-joints, serrations and lumens of instruments before sterilization; the automated washer-disinfector combines washing with thermal disinfection (e.g. 90°C for 1 min) and is the standard first stage in a modern CSSD.
Sedimentation and centrifugation	Sedimentation (with or without coagulants such as alum) removes suspended matter and a large proportion of organisms from water in the plain and chemical sedimentation stages of water purification; centrifugation concentrates or removes cells in the laboratory.

Method	How it works and where it is used
Slow and rapid sand filtration of water	The biological (slow sand / Jewel) filter removes 98–99% of bacteria , largely through the <i>Schmutzdecke</i> ('dirt blanket') — a slimy, biologically active zoogeleal layer on the sand surface; rapid sand (mechanical) filters are faster but less efficient bacteriologically and must be followed by chlorination.
Straining, air filtration and dilution	Straining through cloth (e.g. of drinking water to remove <i>Cyclops</i> in guinea-worm control); dilution and ventilation reduce the concentration of airborne organisms.

REMEMBER

In classifications, remember that **filtration is placed under MECHANICAL methods** (some textbooks list it under physical methods) because it acts by **physical removal**, not by killing. Washing, scrubbing, dusting, mopping, sedimentation and ultrasonic cleaning are the other mechanical methods.



15.1 Properties of an Ideal Disinfectant

- **Wide (broad) spectrum** — active against bacteria, spores, mycobacteria, fungi, viruses and protozoa.
- **Rapid action** at a low, economical concentration, with a **long shelf life** and stability in storage and in dilution.
- **Not inactivated by organic matter**, soap, hard water, plastics or rubber.
- **Non-toxic, non-irritant, non-allergenic and non-carcinogenic** to man and animals.
- **Non-corrosive** to metals and not damaging to rubber, plastics, fabrics, paint or cement.
- **Soluble in water**, able to penetrate crevices and organic matter, and with good **wetting (surface-active)** properties.
- **Odourless or pleasant smelling**, colourless and non-staining.
- **Cheap, readily available and easy to use; environmentally safe and biodegradable.**
- Ideally **leaves a residual effect** and has a **visible indicator** of activity.

REMEMBER

No single disinfectant possesses all of these properties — the ideal disinfectant does not exist, and the choice always involves a compromise between efficacy, safety, material compatibility and cost.

15.2 Classification of Chemical Agents

TABLE 15.1 CLASSIFICATION OF CHEMICAL DISINFECTANTS

Group	Members
1. Alcohols	Ethyl alcohol (ethanol), isopropyl alcohol, methyl alcohol (methanol), benzyl alcohol, trichlorobutanol
2. Aldehydes	Formaldehyde (formalin), glutaraldehyde , ortho-phthalaldehyde (OPA), glyoxal
3. Phenols and related compounds	Phenol (carbolic acid), cresol, Lysol , chloroxylenol (Dettol) , hexachlorophene, thymol, chlorhexidine (a bisbiguanide), triclosan, resorcinol, creosote
4. Halogens	Chlorine and chlorine-releasing agents (bleaching powder, sodium hypochlorite, chloramine, sodium dichloroisocyanurate, chlorine dioxide, halazone); Iodine and iodophors (povidone-iodine)
5. Oxidising agents (peroxygens)	Hydrogen peroxide , peracetic acid , potassium permanganate, benzoyl peroxide, ozone, sodium perborate
6. Heavy metals and their salts	Mercury — mercuric chloride, mercurochrome, thiomersal (merthiolate), phenyl mercuric nitrate; Silver — silver nitrate, silver sulphadiazine, colloidal silver; Copper — copper sulphate; zinc salts
7. Surface-active agents (surfactants / detergents)	Cationic — quaternary ammonium compounds (QACs) : cetrimide, cetylpyridinium chloride, benzalkonium chloride (Zephiran), Savlon (cetrimide + chlorhexidine); Anionic — soaps, alkyl sulphates, sodium lauryl sulphate; Non-ionic — Tweens (mostly inactive, may even support growth); Amphoteric (ampholytic) — 'Tego' compounds
8. Dyes	Aniline dyes — crystal violet, gentian violet, brilliant green, malachite green; Acridine dyes — proflavine, acriflavine, euflavine, aminacrine
9. Acids and alkalies	Mineral acids (HCl, H ₂ SO ₄), organic acids (benzoic, acetic, salicylic, propionic, sorbic, undecylenic acid), sodium hydroxide , calcium oxide (quicklime) , lime , washing soda
10. Gaseous (vapour-phase) agents	Ethylene oxide (EtO) , formaldehyde gas, beta-propiolactone (BPL) , hydrogen peroxide vapour and plasma , ozone, chlorine dioxide, propylene oxide, methyl bromide
11. Miscellaneous	Ethylene glycol and propylene glycol vapour (air disinfection), 8-hydroxyquinoline, salts of heavy metals in paints, essential oils

15.3 Alcohols

Heading	Details
Agents and strength	Ethyl alcohol (ethanol) 60–90%, optimally 70% (v/v); isopropyl alcohol 60–80% (slightly more potent and less volatile, but more toxic and with a stronger odour); methyl alcohol (methanol) — used only for fungicidal action on incubators and inoculation hoods , since it is toxic and neuro-/retinotoxic.
Mechanism	Denaturation and precipitation of proteins , dissolution of membrane lipids and dehydration of the cell. Water is essential for protein denaturation , which is why 70% alcohol is far more effective than absolute (100%) alcohol — a favourite MCQ.
Spectrum	Rapidly kills vegetative bacteria, <i>M. tuberculosis</i> , fungi and lipid-enveloped viruses (HIV, HBV, influenza, SARS-CoV-2) . NOT sporicidal and poorly active against non-enveloped viruses (poliovirus, norovirus, hepatitis A) and protozoal cysts . Therefore an intermediate-level disinfectant .
Uses	Skin antiseptic before injection and venepuncture · the basis of alcohol-based hand rub (WHO formulation: 80% ethanol or 75% isopropanol with hydrogen peroxide and glycerol) · disinfection of clinical thermometers, stethoscope diaphragms, rubber stoppers of multi-dose vials, ampoule necks and small surfaces · 70% alcohol with 0.5–2% chlorhexidine or with iodine (tincture of iodine) for surgical skin preparation · as a solvent for other antiseptics.
Limitations	Not sporicidal · inactivated by organic matter and must be applied to clean skin · evaporates rapidly , so the required contact time may not be achieved · inflammable — a fire hazard , and must be dry before diathermy or cautery · no residual activity · dries, cracks and irritates the skin, and stings open wounds · hardens and damages rubber, some plastics and the cement of lensed instruments.

15.4 Aldehydes

► 15.4.1 Formaldehyde

Heading	Details
Nature and preparations	A gas at room temperature, highly soluble in water. Formalin = a 37–40% w/v aqueous solution of formaldehyde (often stabilised with methanol). 10% formalin (i.e. ≈ 4% formaldehyde) is the standard histological fixative ; formaldehyde gas is used for fumigation .
Mechanism	Alkylation of the amino, carboxyl, sulphydryl and hydroxyl groups of proteins and of the amino groups of nucleic acid bases.
Spectrum	Bactericidal, sporicidal, fungicidal and virucidal — a true chemosterilant , though it acts slowly.
Uses	Fumigation of operation theatres, wards, laboratories, safety cabinets, sick rooms and ambulances · disinfection of bedding, mattresses, blankets, furniture, books, leather and clothing · inactivation of viruses and toxins in vaccine production — the preparation of toxoids (diphtheria and tetanus), inactivated polio (Salk) vaccine and rabies vaccine · preservation of anatomical specimens · destruction of anthrax spores in hides and wool · disinfection of instruments (8% formaldehyde in 70% alcohol) and of renal dialysis equipment · 1–2% formaldehyde as a surface disinfectant.
Method of fumigation	The room is sealed (windows, doors, ventilators and exhausts taped) and cleaned. Then either: (a) boil 500 ml of formalin with 1000 ml of water per 1000 cubic feet of room space in an electric boiler; or (b) the potassium permanganate method — add 280 ml of formalin to 150 g of KMnO₄ per 1000 cubic feet in a deep, heat-resistant vessel placed on the floor (the reaction is violently exothermic and may spatter , so the operator leaves immediately). Relative humidity must be > 60–70% and the temperature > 20 °C . The room is kept sealed for 12–24 hours (at least 12 h), then the residual vapour is neutralised with ammonia (ammonium bicarbonate/ammonium carbonate, ≈ 250 ml of strong ammonia per litre of formalin used) and the room ventilated for several hours before it is re-occupied.
Disadvantages and hazards	Pungent, intensely irritating vapour causing lacrimation, rhinitis, conjunctivitis, cough, bronchospasm and asthma · skin irritation and contact dermatitis/sensitisation · classified as a human carcinogen (nasopharyngeal carcinoma) · poor penetration — acts only on surfaces reached by the vapour · activity falls sharply at low temperature and low humidity · leaves a white residue of paraformaldehyde · prolonged sealing and ventilation make the theatre unusable for a day. Because of these hazards, formaldehyde fumigation has largely been replaced by hydrogen peroxide vapour/fogging and by improved ventilation and HEPA filtration.

► 15.4.2 Glutaraldehyde — The Most Important Liquid Chemosterilant

Heading	Details
Preparation	A 2% aqueous solution (trade name Cidex). Supplied acidic and 'activated' just before use by adding the sodium bicarbonate activator to bring the pH to 7.5–8.5 (alkaline) , which is necessary for sporicidal activity. Once activated its shelf life is about 14 days (longer for stabilised formulations); the concentration must be checked with test strips before each use.

Heading	Details
Mechanism	Alkylation of amino, sulphhydryl, hydroxyl and carboxyl groups; it also cross-links proteins of the cell wall and membrane. It is a dialdehyde , more potent than formaldehyde.
Contact times	High-level disinfection: 20–45 minutes at 20–25 °C (20 min is the common Indian/CDC figure; 45 min at 25 °C by some standards). Sterilization (sporicidal): 3–10 hours (often quoted as 10 h to kill all spores). M. tuberculosis requires at least 20 minutes.
Uses	The agent of choice for heat-sensitive instruments — flexible fibre-optic endoscopes (gastrosopes, colonoscopes, bronchoscopes, cystoscopes, laparoscopes), arthroscopes, anaesthetic and respiratory equipment, face masks and corrugated tubing, plastic and rubber goods, dialysers, dental instruments, thermometers, polythene tubing, metal instruments including sharp cutting edges and lensed instruments.
Advantages	Sporicidal, tuberculocidal, virucidal and fungicidal · relatively unaffected by organic matter (blood, protein) — a major advantage over hypochlorites · non-corrosive to metals, and does not damage rubber, plastics, lenses or the cement of instruments · active over a wide temperature range.
Disadvantages and precautions	Irritant and sensitising — causes dermatitis, conjunctivitis, rhinitis, epistaxis and occupational asthma ; must be used in a covered container in a well-ventilated area or fume cupboard , with gloves, gown, eye protection and a mask · slow sporicidal action · expensive ; unstable once activated · coagulates blood and fixes tissue , so instruments must be scrupulously cleaned first · instruments must be thoroughly rinsed with sterile water before use, as residues cause chemical colitis and keratitis · may leave a sticky residue and stain the hands brown · <i>Mycobacterium chelonae</i> and some glutaraldehyde-resistant mycobacteria have caused outbreaks.
Ortho-phthalaldehyde (OPA) 0.55%	A newer aromatic dialdehyde (Cidex OPA); achieves high-level disinfection in 5–12 minutes at 20–25 °C , needs no activation, is less irritant with a milder odour and is more stable, but is more expensive and stains skin, clothing and surfaces grey-green .

HIGH-YIELD FOR EXAM

- **2% alkaline glutaraldehyde (Cidex) is the disinfectant of choice for endoscopes** — HLD in **20 minutes**, sterilization in **3–10 hours**.
- It must be **activated with sodium bicarbonate** to pH 7.5–8.5; activated shelf life ≈ **14 days**.
- Both formaldehyde and glutaraldehyde act by **alkylation**; so do ethylene oxide and beta-propiolactone.

15.5 Phenols and Related Compounds

TABLE 15.2 PHENOLS AND RELATED ANTISEPTICS

Agent	Strength and features	Uses
Phenol (carbolic acid)	The first widely used surgical antiseptic , introduced by Joseph Lister (1867) , who sprayed carbolic acid in the operating theatre and is regarded as the father of antiseptic surgery . Used as a 1–5% solution ; it is the reference standard against which other disinfectants are compared (phenol coefficient) . Bactericidal and fungicidal but not sporicidal ; poorly active against non-enveloped viruses.	Disinfection of sputum, faeces, urine, pus and laboratory discard jars (5%) ; disinfection of drains and floors. Too caustic and toxic for skin use .
Cresol (and Lysol)	Cresol is a methyl phenol, 3–10 times more powerful than phenol ; Lysol is a 50% solution of cresol in soap (a saponified cresol), used at 2–5% . Cheap, with a strong smell, and retains activity in the presence of organic matter better than most agents. Caustic to skin.	Disinfection of floors, walls, drains, sinks, bed-pans, urinals, spittoons, furniture, ambulances and excreta ; the classic hospital 'carbolicisation'.
Chloroxylenol (Dettol)	A halogenated phenol; much less toxic and irritant than phenol and cresol, with a pleasant smell; usually used as a 4.8% concentrate diluted for use . Its great weakness is that it is relatively weak, is inactivated by hard water and organic matter, and <i>Pseudomonas aeruginosa</i> can survive and even multiply in dilute solutions .	Domestic antiseptic; skin and wound cleansing, first aid, floors and surfaces. Not a hospital-grade disinfectant.
Hexachlorophen e	A bisphenol, highly active against Gram-positive organisms, especially staphylococci ; used in 3% soaps and detergents . It accumulates on the skin giving a persistent effect, but is absorbed through the skin and is neurotoxic , having caused vacuolar encephalopathy in neonates — hence routine bathing of newborns with it is prohibited .	Surgical hand scrub; control of staphylococcal outbreaks in nurseries (restricted use only).

Agent	Strength and features	Uses
Chlorhexidine (a bisbiguanide, not a true phenol)	Available as the gluconate or acetate: 0.5% in 70% alcohol for pre-operative skin preparation, 2–4% detergent scrub (Hibiscrub) for surgical hand disinfection, 0.2% mouthwash, 0.05–0.1% for bladder irrigation and wound cleaning , and 0.5–1% dusting powder. Most active at alkaline pH ; has excellent substantivity (persistent residual activity on skin) and low irritancy . Active against Gram-positive and (less so) Gram-negative bacteria; NOT sporicidal, NOT tuberculocidal , poor against non-enveloped viruses; inactivated by soaps, anionic detergents and hard water . Ototoxic (avoid in the middle ear) and neurotoxic (avoid contact with brain, meninges and eye) ; can cause anaphylaxis rarely.	The first choice for surgical hand scrub, pre-operative skin preparation, central-line and umbilical cord care, oral hygiene and bladder irrigation. 2% chlorhexidine in 70% alcohol is now the recommended skin antiseptic before surgery and central-line insertion.
Triclosan	A chlorinated bisphenol, 0.3–2% ; active mainly against Gram-positive organisms; used in medicated soaps, hand washes, toothpastes and deodorants. Concerns exist about resistance and environmental persistence .	Hand hygiene products; control of MRSA carriage.
Thymol, resorcinol, creosote	Weaker phenolic derivatives.	Mouth washes, dental preparations, preservation of timber.

► 15.5.1 Testing the Potency of Disinfectants

Test	Principle	Key point
Phenol coefficient — Rideal–Walker (RW) test	The highest dilution of the test disinfectant that kills a standard suspension of <i>Salmonella typhi</i> in a given time (2½, 5, 7½ and 10 minutes) is divided by the corresponding dilution of phenol .	Carried out in the ABSENCE of organic matter — therefore it overestimates the value of a disinfectant in real use. A phenol coefficient > 1 means more potent than phenol .
Chick–Martin test	The same comparison with phenol, but carried out in the PRESENCE of organic matter (dried human faeces or yeast suspension) and at a fixed contact time of 30 minutes.	More realistic than the RW test and gives a lower coefficient . This contrast — RW without organic matter, Chick–Martin with organic matter — is a very common MCQ.
Kelsey–Sykes capacity (in-use) test	Measures the capacity of a disinfectant to retain its activity when repeatedly challenged with successive additions of a bacterial suspension, with and without organic matter.	Assesses re-use life of a working solution.
In-use test (Kelsey–Maurer)	A sample of the disinfectant actually in use in the ward is diluted and cultured to count survivors.	Detects contaminated disinfectant solutions in the hospital — e.g. <i>Pseudomonas</i> growing in dilute chloroxylenol or QAC.
AOAC use-dilution test	Standard American (Association of Official Analytical Chemists) method using dried inocula on carriers; also the AOAC sporicidal and tuberculocidal tests.	Basis of regulatory label claims.
Minimum inhibitory / minimum bactericidal concentration (MIC / MBC)	Lowest concentration inhibiting or killing the organism.	Used mainly for antibiotics and preservatives.

15.6 Halogens

► 15.6.1 Chlorine and Chlorine-Releasing Agents

Heading	Details
Mechanism	In water, chlorine forms hypochlorous acid (HOCl) , which liberates nascent oxygen and chlorinates the cell; it is a powerful oxidising agent that destroys enzymes, membranes and nucleic acid. Activity is expressed as available (free) chlorine in ppm or % and is greater at acidic pH (undissociated HOCl is the active form).
Preparations	Bleaching powder (chlorinated lime, calcium hypochlorite) — contains about 33% available chlorine when fresh, but loses it on storage; sodium hypochlorite solution (liquid bleach) — commonly supplied at 4–6% and diluted for use; stabilised bleach, bleach powder, chloramine-T, halazone tablets, sodium dichloroisocyanurate (NaDCC) tablets (convenient and stable, widely used in hospitals), chlorine dioxide, chlorine gas (for water works), double bleaching powder .

Heading	Details
Working strengths (learn these)	1% sodium hypochlorite (10 000 ppm available chlorine) — for blood and body-fluid spills and for discard jars for plastic waste · 5% hypochlorite — laboratory discard jars, highly infectious material · 0.5% (5000 ppm) — general surface decontamination in laboratories and for viral haemorrhagic fevers · 0.1% (1000 ppm) — routine environmental surfaces, floors, and non-critical items · 0.05–0.1% — clean surfaces and toys · 0.5–5 ppm — chlorination of drinking water (minimum free residual chlorine of 0.5 mg/L for one hour contact; 0.5 ppm is the standard, raised to ≥ 0.7 ppm during an epidemic) · 1–2 ppm — swimming pools · Horrock's apparatus is used to determine the chlorine demand/dose of a well.
Uses	Disinfection of drinking water and swimming pools · blood and body-fluid spills · disinfection of floors, sinks, toilets, bed-pans and surfaces · dairy, food and catering equipment · laboratory discard jars and pipette jars · disinfection of plastic biomedical waste before shredding · dental and endodontic irrigation (2.5–5.25% NaOCl) · disinfection of milk cans and haemodialysis water systems · Dakin's solution and Eusol for wound irrigation.
Advantages	Cheap, widely available, rapid and very broad spectrum — bactericidal, virucidal (including HIV, HBV and non-enveloped viruses), fungicidal, tuberculocidal and, at high strength, sporicidal ; leaves no toxic residue on surfaces; effective against <i>C. difficile</i> spores at 1000–5000 ppm.
Disadvantages	Rapidly inactivated by organic matter (blood, pus, faeces) — so the surface must be cleaned or a higher concentration used · corrosive to metals and damages fabrics, rubber and some plastics · unstable — loses chlorine on exposure to light, heat and air, so solutions must be freshly prepared daily and stored in dark, closed containers · irritant to skin, eyes and the respiratory tract ; bleaches fabrics · must NEVER be mixed with acids (releases toxic chlorine gas), with ammonia (releases chloramine vapour) or with formaldehyde (forms a carcinogen) · activity falls at alkaline pH · chlorination of organic-rich water produces trihalomethanes .

► 15.6.2 Iodine and Iodophors

Preparation	Composition and strength	Uses and comments
Aqueous iodine solution (Lugol's iodine)	5% iodine + 10% potassium iodide in water.	Skin antiseptic; also used in microbiology (Gram's stain, Lugol's iodine for stool microscopy).
Tincture of iodine	2% iodine + 2.4% sodium iodide in 50% alcohol (strong tincture contains 7% iodine).	Powerful and rapid skin antiseptic before surgery and venepuncture; but stains the skin and clothing, stings, and causes irritation, burns under occlusion and hypersensitivity ; must be removed with alcohol after the procedure.
Iodophors — povidone-iodine (PVP-I, Betadine)	Iodine complexed with the carrier polyvinylpyrrolidone , which releases free iodine slowly. 10% povidone-iodine contains 1% available iodine ($\approx 0.1\%$ free iodine); also supplied as a 7.5% surgical scrub, 5% solution for mucosa and 2.5% ophthalmic preparation.	The most widely used skin and mucous-membrane antiseptic today — pre-operative skin preparation and painting, surgical hand scrub, wound and burn dressing, umbilical cord care, gargles and mouthwash, vaginal pessaries, and 2.5% drops for prevention of ophthalmia neonatorum . Advantages over tincture : much less irritant and staining , water-miscible, sustained release of iodine, pleasant to use. Disadvantages : slower action (needs 2 minutes' contact and must be allowed to dry), inactivated by organic matter and by dilution beyond the optimum, may cause hypersensitivity and, in neonates or over large burns, iodine absorption with transient hypothyroidism ; discolours surfaces.

- **Mechanism of iodine**: iodinates and oxidises proteins, nucleotides and fatty acids, and blocks hydrogen bonding — killing rapidly.
- **Spectrum**: bactericidal, **tuberculocidal**, virucidal, fungicidal, amoebicidal and, with prolonged contact, **sporicidal** — the broadest spectrum of the common skin antiseptics.
- **Contraindications/cautions**: known iodine allergy, pregnancy and lactation (large areas), neonates, thyroid disease; **do not use on the eye except the specific ophthalmic preparation**.

15.7 Oxidising Agents (Peroxygens)

Agent	Strength	Uses	Notes
Hydrogen peroxide	3% (10 volumes) for wounds and surfaces; 6% and 7.5% for high-level disinfection; 10% for soft contact lenses; 30–35% (vaporised, VHP) and plasma for sterilization; 1.5–3% mouthwash	Wound and ear cleaning (mechanical loosening of slough and pus by effervescence), mouthwash for oral ulcers and gingivitis, disinfection of contact lenses, ventilators and soft plastics, decontamination of rooms and safety cabinets by vapour or fogging , and (as gas plasma) sterilization of heat-sensitive instruments	Acts by liberating free hydroxyl radicals and nascent oxygen . Bactericidal, virucidal, fungicidal and, at high concentration, sporicidal . Decomposed by catalase in tissues and blood (the fizzing), which limits its activity on tissue; unstable — must be stored cool, dark and in a vented container ; concentrated solutions are corrosive and cause skin blanching and burns ; can cause surgical emphysema if instilled into closed cavities.
Peracetic acid (peroxyacetic acid)	0.2–0.35% (automated systems); 0.08% with hydrogen peroxide	Automated sterilization of flexible endoscopes, dialysers and dental instruments (e.g. the STERIS system, ≈ 30–40 °C for 12–30 minutes); food-industry and aseptic-packaging sanitisation	Very rapid, sporicidal at low temperature and in the presence of organic matter ; decomposes to harmless acetic acid, water and oxygen — environmentally friendly and leaves no toxic residue . But it is unstable, corrosive to soft metals (copper, brass, bronze, plain steel), pungent and irritant , and expensive.
Potassium permanganate (KMnO₄)	1:1000 to 1:10 000 (a pink solution); crystals for fumigation	Disinfection of water in wells (the classic Indian practice) , fruits and vegetables , gargles, urethral irrigation, wet dressings and sitz baths for weeping skin lesions , snake-bite (obsolete), and as the oxidiser in formaldehyde fumigation	A powerful oxidising agent; astringent and deodorising . Rapidly inactivated by organic matter, stains skin, nails and fabrics brown , irritant in strong solution, and the crystals are a fire/explosion hazard with glycerol and organic material.
Ozone	Generated on site; 1–4 mg/L in water	Disinfection of drinking water and of bottled water , air and surface decontamination, sterilization chambers	Extremely powerful oxidiser; leaves no taste, odour or residue and does not form trihalomethanes , but has no residual protective effect , must be generated on site, is costly , and is toxic and irritant to the lungs .
Benzoyl peroxide, sodium perborate	2.5–10%	Acne treatment; oral rinses and denture cleaning	Mild oxidising agents.

15.8 Heavy Metals and Their Salts

Heavy metals act by combining with the **sulphydryl (–SH) groups** of enzymes and by precipitating proteins. Very low concentrations are inhibitory — the **oligodynamic action**. Most are **bacteriostatic rather than bactericidal**, and are **toxic**, so their use has declined greatly.

Metal	Compounds and strength	Uses	Limitations
Mercury	Mercuric chloride (perchloride of mercury) 1:500–1:1000 ; mercurochrome ; merthiolate (thiomersal) 1:10 000 ; phenyl mercuric nitrate/acetate 1:50 000 ; ammoniated mercury ointment	Formerly a general disinfectant; thiomersal and phenylmercuric salts as preservatives in vaccines, sera, eye drops and biological products; mercurochrome as a skin antiseptic	Highly toxic (nephrotoxic, neurotoxic), corrosive to metals, inactivated by organic matter, and only bacteriostatic . Now largely abandoned, and mercury is being phased out of health care (Minamata Convention) ; mercury waste must go to yellow/special collection, never to incineration .
Silver	Silver nitrate 1% (Credé's prophylaxis) ; 0.5% silver nitrate for burns; 1% silver sulphadiazine cream ; silver-impregnated catheters and dressings ; colloidal silver	1% silver nitrate eye drops for the prevention of ophthalmia neonatorum (gonococcal conjunctivitis of the newborn) — Credé's method ; silver sulphadiazine is a standard topical agent for burns ; antimicrobial dressings, water purification	Causes chemical conjunctivitis (hence largely replaced by erythromycin ointment or povidone-iodine for Credé's prophylaxis); argyria (grey pigmentation) on prolonged use; stains; silver nitrate is caustic.
Copper	Copper sulphate 1:100 000 ; copper–silver ionisation	Algicide in water tanks, swimming pools and reservoirs ; molluscicide against snails in schistosomiasis control; fungicide in agriculture (Bordeaux mixture); control of <i>Legionella</i> in water systems	More fungicidal and algicidal than bactericidal ; toxic to fish and plants.

Metal	Compounds and strength	Uses	Limitations
Zinc	Zinc sulphate, zinc oxide, zinc undecylenate	Astringent and mild antiseptic in calamine and barrier creams; antifungal preparations	Weak activity.

15.9 Surface-Active Agents (Surfactants / Detergents)

These molecules have a **hydrophilic (water-loving) head** and a **hydrophobic (lipid-loving) tail**, so they concentrate at interfaces, reduce surface tension and **disorganise the cell membrane**, causing leakage of cell contents.

Class	Examples	Activity and uses	Limitations
Cationic — quaternary ammonium compounds (QACs)	Cetrimide (cetyltrimethylammonium bromide), benzalkonium chloride (Zephiran), cetylpyridinium chloride, Savlon (cetrimide + chlorhexidine), domiphen bromide. Used at 0.1–2% .	The most active surfactant group. Good against Gram-positive bacteria, fungi and lipid (enveloped) viruses ; also detergent, deodorant, low-irritant and non-corrosive with a pleasant smell. Used for cleaning and disinfecting floors, walls, furniture and non-critical equipment , skin and wound cleansing, pre-operative skin preparation, and as preservatives in eye drops and nasal sprays (benzalkonium 0.01%). Cetrimide is also a selective medium for <i>Pseudomonas aeruginosa</i> .	LOW-LEVEL disinfectants only. NOT sporicidal, NOT tuberculocidal, and inactive against non-enveloped viruses (polio, noro) and <i>Pseudomonas</i> · INACTIVATED BY SOAPS AND ANIONIC DETERGENTS, hard water, organic matter, cotton gauze, cork and some plastics · <i>Pseudomonas</i> and <i>Serratia</i> can survive and multiply in QAC solutions, causing outbreaks — hence solutions must never be topped up and must be freshly made.
Anionic	Soaps (sodium and potassium salts of fatty acids), alkyl and aryl sulphates, sodium lauryl sulphate , bile salts	Weak antibacterial action, but excellent cleansing and mechanical removal of organisms — the value of hand-washing with soap and water is chiefly mechanical . Active against some Gram-positives, <i>Neisseria</i> , pneumococci and enveloped viruses.	Very weak germicides; incompatible with cationic agents .
Non-ionic	Tweens (polysorbates), Spans, Triton X	Essentially no antimicrobial activity ; used as emulsifiers and wetting agents in pharmacy.	May actually protect organisms and even support their growth ; they neutralise QACs and are used in culture media to inactivate residual disinfectant .
Amphoteric (ampholytic)	'Tego' compounds	Combine the detergency of anionic with some of the germicidal power of cationic agents; active over a wide pH range and relatively non-toxic; used in the food industry.	Expensive; intermediate activity.

15.10 Dyes

Group	Members	Action and uses
Aniline dyes	Crystal violet, gentian violet, brilliant green, malachite green	Much more active against Gram-positive bacteria than Gram-negative, and active against some fungi; weak and readily inactivated by pus . Uses: 1% gentian violet paint for oral candidiasis, impetigo, superficial fungal infection and burns ; selective agents in culture media (crystal violet in MacConkey and in blood agar; malachite green in Löwenstein-Jensen medium ; brilliant green in Wilson-Blair). They stain the skin and clothing.
Acridine dyes	Proflavine, acriflavine, euflavine, aminacrine, aminoacridine	Act by intercalating into DNA and displacing the purine bases; more active against Gram-positive organisms; retain activity in the presence of pus and serum better than aniline dyes. Uses: antiseptic for wounds, burns and ulcers (0.1% acriflavine) , urethral and bladder irrigation, and as a mutagen and a curing agent for plasmids in the laboratory. They stain yellow and may delay wound healing.

15.11 Acids and Alkalies

- Act by **altering the pH**, denaturing proteins and, for organic acids, by undissociated molecules penetrating the cell.
- Mineral acids** (hydrochloric, sulphuric) are strongly germicidal but **corrosive**; used for disinfecting hides in anthrax and for cleaning.
- Organic acids** — **benzoic acid, sorbic acid, propionic acid, acetic acid, salicylic acid, undecylenic acid, lactic acid** — are widely used as **food and pharmaceutical preservatives** (benzoates in syrups, propionates in bread, sorbates in foods), as **antifungals** (salicylic and benzoic acid in Whitfield's ointment, undecylenic acid for tinea pedis), and **1–2% acetic acid for *Pseudomonas* in burns, ear and urinary infections**. Lactic acid vapour has been used for air disinfection.
- Alkalies** — **sodium hydroxide, lime (calcium oxide/quicklime), sodium carbonate (washing soda), soda ash** — are used for the disinfection of **animal houses, cattle sheds, drains, floors and premises in anthrax and foot-and-mouth disease**, for **whitewashing** with lime, and for treating **prion-contaminated material (1 N NaOH for 1 hour, followed by autoclaving)**.
- Boric acid** (weak antiseptic, eye washes) and **quicklime/bleaching powder** for **deep burial and excreta disposal** are also included here.

15.12 Gaseous Sterilization

► 15.12.1 Ethylene Oxide (EtO) — The Most Important Gaseous Sterilant

Heading	Details
Physical nature	A colourless gas with a sweet, ethereal (fruity) odour ; boiling point 10.7 °C , so it is stored as a liquid under pressure. Highly inflammable and explosive above 3% in air , therefore supplied diluted with an inert carrier — carbon dioxide (10:90) or hydrochlorofluorocarbon — or used as 100% EtO in small single-use cartridges within a sealed chamber.
Mechanism	Alkylation of the amino, carboxyl, sulphhydryl and hydroxyl groups of proteins and of the bases of nucleic acids, which blocks metabolism and replication.
Spectrum	Highly effective against all micro-organisms including spores, mycobacteria, fungi, all viruses and even some prions — a true sterilant.
Cycle parameters (learn these)	Concentration 450–1200 mg/L · temperature 37–63 °C (commonly 55–60 °C) · relative humidity 30–60% (essential — dry spores are resistant) · exposure time 1–6 hours (typically 3–5 h at 55 °C; up to 12–24 h at lower temperature) , followed by aeration/degassing for 8–12 hours or more (up to 7 days at room temperature) to remove residues.
Uses	Sterilization of heat- and moisture-sensitive articles: plastic and rubber disposables (syringes, catheters, endotracheal and tracheostomy tubes), heart-lung machine and dialyser components, respirators and anaesthetic equipment, prosthetic implants and heart valves, pacemakers, fibre-optic endoscopes, cystoscopes, delicate and lensed instruments, surgical sutures and dressings, single-use medical devices, bone and tissue grafts, powders, some pharmaceuticals, glassware, clothing, blankets, mattresses, books, documents, laboratory equipment and cardiac catheters . It is the standard low-temperature industrial and hospital-CSSD gaseous process.
Advantages	Sterilizes at low temperature — nothing is damaged by heat · excellent penetration through paper, cardboard, plastic wrapping, fabric and lumens, so items can be sterilized inside their final packaging · non-corrosive to metals and lenses · compatible with almost all materials · highly reliable and validatable.
Disadvantages and hazards	Highly toxic, mutagenic and CARCINOGENIC (leukaemia, lymphoma; strict occupational exposure limits) · irritant to skin, eyes and airways, causing burns, blistering and vesication, headache, nausea and vomiting · explosive and inflammable · very slow — a long cycle plus a long compulsory aeration period , so turnaround is poor · leaves toxic residues (EtO, ethylene chlorohydrin and ethylene glycol) that have caused haemolysis, tracheal and laryngeal burns and anaphylaxis, so aeration is mandatory · absorbed by many plastics and rubber · expensive equipment and gas · must not be used for materials that adsorb it (e.g. some polystyrene) · requires trained staff, monitoring and good ventilation.
Sterility control	Biological indicator: spores of <i>Bacillus atrophaeus</i> (formerly <i>B. subtilis</i> var. <i>niger</i>), placed inside a test pack — the same organism used for dry heat. Chemical indicators and residual-gas analysis are also used.

► 15.12.2 Other Gaseous Agents

Agent	Conditions	Uses	Comments
Formaldehyde gas	See §15.4.1; > 60% RH, > 20 °C, 12–24 hours	Fumigation of theatres, wards, laboratories, safety cabinets; disinfection of bedding and books; LTSF sterilization at 73–80 °C	Sporicidal but poor penetration , intensely irritant, carcinogenic, and leaves a paraformaldehyde residue; needs ammonia neutralisation.

Agent	Conditions	Uses	Comments
Beta-propiolactone (BPL)	0.2% (2 mg/L or $\approx$ 4–5 ml per m³) for fumigation; 0.05–0.2% in solution; $\approx$ 2 hours at 24–30 °C with high humidity	Fumigation of rooms and laboratories; inactivation of viruses in vaccine production — rabies, influenza and some inactivated vaccines; sterilization of tissue grafts, bone and arterial grafts, and of serum	More potent than formaldehyde and EtO and leaves less residue , and is rapidly hydrolysed to harmless lactic acid; but it has poor penetrating power and is CARCINOGENIC , which limits its use.
Hydrogen peroxide gas plasma	Vaporised H₂O₂ ($\approx$ 58%) excited by radiofrequency or microwave energy into a low-temperature plasma of free radicals; 45–75 minutes at 40–50 °C	Sterilization of heat- and moisture-sensitive instruments — rigid and flexible endoscopes, cameras, cables, batteries, plastics and delicate instruments (the STERRAD system)	Rapid, low temperature, no toxic residue (breaks down to water and oxygen), no aeration needed , environmentally safe — an excellent modern alternative to EtO. Cannot be used for cellulose (paper, linen, gauze), powders, liquids or long narrow lumens , and is expensive; needs special non-cellulosic (Tyvek) wrap.
Hydrogen peroxide vapour (VHP) / fogging	30–35% H ₂ O ₂ aerosolised into a sealed room	Terminal decontamination of operation theatres, ICUs, isolation rooms and safety cabinets — the modern replacement for formaldehyde fumigation	Short cycle, rapid re-occupancy, no carcinogenic residue; needs a sealed room and a validated machine.
Ozone	6–12% in oxygen, $\approx$ 30–40 °C	Low-temperature sterilization chambers; water treatment	Powerful oxidiser, no residue; oxidises and damages some polymers, rubber and copper; toxic to inhale.
Chlorine dioxide gas	Generated on site	Room and equipment decontamination, water treatment	Sporicidal, effective at low concentration; explosive at high concentration, must be generated on site.
Propylene oxide, methyl bromide	Similar to EtO	Fumigation of foodstuffs, spices, grain and packaging	Less active than EtO; residues are a concern; methyl bromide is being phased out (ozone-depleting).
Propylene / ethylene glycol vapour	Aerosolised into air	Air disinfection of enclosed spaces	Low toxicity; largely historical.

15.13 Summary — Choosing a Chemical Agent

TABLE 15.3 AGENT OF CHOICE FOR EACH PURPOSE

Purpose	Agent of choice
Hand hygiene — routine (clean hands)	Alcohol-based hand rub (60–80%) for 20–30 seconds
Hand hygiene — visibly soiled hands	Soap and water for 40–60 seconds (alcohol does not work through dirt); also mandatory for <i>C. difficile</i> and spore risk
Surgical hand scrub	4% chlorhexidine gluconate or 7.5% povidone-iodine detergent scrub, or an alcohol-based surgical rub
Pre-operative skin preparation	2% chlorhexidine in 70% alcohol (first choice) or 10% povidone-iodine
Skin before injection / venepuncture	70% alcohol swab , allowed to dry
Endoscopes and other heat-sensitive semi-critical instruments	2% alkaline glutaraldehyde (20 min) , 0.55% OPA or 0.2% peracetic acid
Blood and body-fluid spills	1% sodium hypochlorite (10 000 ppm) — cover, leave 10 minutes, then clean
Floors, walls, sinks, bed-pans and general surfaces	Phenolic (Lysol 2–5%) or 0.1% hypochlorite or a QAC-detergent
Laboratory discard jars	5% phenol or 1–5% hypochlorite , with overnight contact
Drinking water	Chlorine — minimum free residual 0.5 mg/L after 1 hour contact
Operation theatre / room decontamination	Hydrogen peroxide vapour or fogging (formaldehyde fumigation is the older method)
Heat-sensitive plastics, rubber, prostheses and lensed instruments (sterilization)	Ethylene oxide or hydrogen peroxide gas plasma

Purpose	Agent of choice
Wounds and burns	Povidone-iodine , silver sulphadiazine (burns), acriflavine, hydrogen peroxide for cleaning; 1–2% acetic acid for <i>Pseudomonas</i>
Mucous membranes and eye	Povidone-iodine (2.5–5%) , 0.2% chlorhexidine mouthwash; never alcohol or phenol

HIGH-YIELD FOR EXAM

- **70% alcohol > 100% alcohol** because water is needed for protein denaturation.
- **QACs are inactivated by soap; chlorhexidine and hypochlorites by organic matter; glutaraldehyde and phenolics are relatively resistant to organic matter.**
- **Alkylating agents** = formaldehyde, glutaraldehyde, ethylene oxide, beta-propiolactone. **Oxidising agents** = hydrogen peroxide, peracetic acid, halogens, KMnO₄, ozone, plasma.
- **Not sporicidal:** alcohols, QACs, chlorhexidine, phenols (ordinary strengths), dyes. **Sporicidal:** glutaraldehyde (prolonged), formaldehyde, EtO, beta-propiolactone, peracetic acid, high-strength hypochlorite and hydrogen peroxide.



CHAPTER 16 STERILIZATION OF SYRINGES, GLASSWARES, APPARATUS AND HOSPITAL ARTICLES

This chapter answers the practical question the examiner most often asks: '**which method for which article?**' The governing rule is always the same — **clean first, then choose the mildest method that will reliably sterilize without damaging the article.**

16.1 The Universal Sequence of Reprocessing

TABLE 16.1 THE SEVEN STEPS OF REPROCESSING

Step	What is done	Why
1. Decontamination / pre-cleaning	Immediately after use, immerse the article in water with a detergent or an enzymatic cleaner (or 0.5% hypochlorite where indicated). Never allow blood and protein to dry on the instrument.	Protects the handler and prevents organic matter from drying and hardening.
2. Cleaning	Manual scrubbing with a brush under water, or an ultrasonic cleaner or automated washer-disinfector . Dismantle all detachable parts, open hinges and box-joints, and flush all lumens .	The most important single step. No sterilant can be relied upon through dirt; organic matter shields organisms and neutralises chemicals.
3. Rinsing and drying	Rinse in clean (preferably distilled or de-ionised) water and dry completely.	Residual detergent interferes with disinfectants; water dilutes chemical agents and causes corrosion and wet packs.
4. Inspection and maintenance	Check for cleanliness, damage, cracks, blunt edges and function; lubricate hinged instruments.	A damaged or dirty instrument must not be packed.
5. Packing and labelling	Wrap in the appropriate material — paper, muslin, crepe, non-woven wrap, peel pouches or metal drums/canisters. Label with the contents, the date of sterilization, the expiry date and the load/batch number, and attach a chemical indicator .	Maintains sterility until use and provides traceability.
6. Sterilization	By the method appropriate to the article (see §16.6).	—
7. Cooling, drying and storage	Allow to cool and dry; store in a clean, dry, dust-free, closed cupboard, above floor level, away from sinks and traffic; practise first-in first-out stock rotation.	A wet or torn pack is a contaminated pack. Sterility is event-related , not merely time-related.

16.2 Sterilization of Syringes and Needles

► 16.2.1 Glass (Re-usable) Syringes

- Immediately after use**, flush the syringe and needle several times with cold water (**cold, not hot** — hot water coagulates blood and protein inside the barrel), then immerse in a detergent solution.
- Dismantle** the syringe — separate the **barrel from the plunger** and remove the needle. A syringe must **never be sterilized assembled**, because differential expansion of barrel and plunger can crack the glass and steam or hot air cannot reach the ground-glass surfaces.
- Clean** thoroughly with a brush and detergent, paying attention to the nozzle; use an ultrasonic cleaner if available. Rinse in distilled water and **dry completely**.
- Pack** the barrel and plunger separately — in a **paper wrap, a muslin pack, a metal syringe canister or an aluminium/metal box**, together with the needle in its own container.
- Sterilize** by either method:
 - Hot air oven — 160 °C for 1–2 hours** (the preferred method for glass syringes, since it leaves them dry and does not corrode the needle), **or**
 - Autoclave — 121 °C at 15 psi for 15–20 minutes** (barrel and plunger separated, wrapped in muslin).
- Allow the oven to **cool below 60 °C before opening**, or let the autoclaved pack dry, before removing the syringes.
- Assemble aseptically** only at the moment of use, handling the plunger by its end and the needle by its hub.
- In an emergency only**, syringes and needles may be **boiled at 100 °C for at least 20–30 minutes** (add 2% sodium bicarbonate to prevent rusting). This is **disinfection, not sterilization** — spores and hepatitis B virus may survive, so it is **not acceptable for routine injections**.

HIGH-YIELD FOR EXAM

- **Glass syringes: hot air oven at 160 °C for 1–2 hours, with the barrel and plunger separated** — the classic answer.
- **Disposable plastic syringes and needles: gamma radiation (Cobalt-60, 2.5 Mrad / 25 kGy) or ethylene oxide, done by the manufacturer inside the sealed pack.**
- **Boiling is NOT sterilization** and must not be used for syringes as a routine.

► 16.2.2 Disposable Plastic Syringes and Needles

- Made of polypropylene or polyethylene, which are **destroyed by heat**; they are therefore sterilized **at the factory, inside their sealed individual packs**, by **gamma irradiation from Cobalt-60 at 2.5 Mrad (25 kGy)** or by **ethylene oxide**.
- They are **strictly single-use**. **Auto-disable (AD) syringes**, which lock after one use, and **re-use prevention (RUP)** and **sharps-injury-prevention** syringes are mandated for immunisation and therapeutic injections under the **WHO/Government of India injection-safety policy**.
- The pack must be checked before use for **intactness, dryness, the sterilization indicator and the expiry date**; a torn, wet or expired pack must be discarded.
- **After use: never recap, bend or break the needle by hand**. Cut the hub with a **needle/hub cutter or needle destroyer at the point of generation**, put the **needle and sharps into the white translucent puncture-proof container** and the **plastic barrel into the red bag/container** after disinfection (see Chapter 17).
- **Never re-sterilize a disposable syringe** — heat distorts it, and it is impossible to clean reliably.

16.3 Sterilization of Glasswares

► 16.3.1 Cleaning of Glassware

- **New glassware** is alkaline; it should be soaked in **1–2% hydrochloric acid or dilute nitric acid overnight**, then rinsed thoroughly in tap water and finally in distilled water.
- **Used glassware** should first be **decontaminated** — discarded cultures and contaminated tubes are **autoclaved at 121 °C for 30–60 minutes** or immersed in a **disinfectant discard jar overnight** — **before** being washed.
- Wash with a **detergent or soap solution** and a brush, then rinse repeatedly in tap water and finally **two or three times in distilled water**.
- Greasy or heavily soiled glassware is soaked in **chromic acid (dichromate) cleaning solution** — potassium dichromate in concentrated sulphuric acid — or in an alkaline hypochlorite solution, then rinsed exhaustively. **Chromic acid is highly corrosive: wear gloves, apron and eye protection, and always add acid to water, never water to acid.**
- **Pipettes** are dropped **tip upwards** into a **pipette jar containing disinfectant** immediately after use, then washed in a pipette washer.
- **Dry** completely in a drying oven or inverted on a rack before sterilization.
- Glassware is considered clean when water **drains evenly without leaving droplets or streaks** on the inner surface.

► 16.3.2 Sterilization of Glassware

TABLE 16.2 GLASSWARE AND ITS STERILIZATION

Article	Preparation	Method
Test tubes, culture tubes	Plug the mouth with non-absorbent cotton wool , or fit a metal/plastic cap; bundle and wrap in paper.	Hot air oven, 160 °C for 2 hours (or autoclave 121 °C/15 min if they contain media).
Petri dishes	Stack in metal canisters or cylindrical tins , or wrap in stout paper (a set of 4–6 dishes per pack).	Hot air oven, 160 °C for 2 hours . (Disposable plastic petri dishes are gamma-irradiated by the maker.)
Pipettes (graduated, Pasteur, bacteriological)	Plug the mouthpiece end with cotton wool; place in a copper or aluminium pipette canister (tip first) or wrap in paper.	Hot air oven, 160 °C for 2 hours .
Flasks, conical flasks, bottles, measuring cylinders (non-graduated), funnels	Plug or cap the mouth; wrap the neck in paper; caps must be loosened if autoclaved.	Hot air oven, 160 °C for 2 hours , or autoclave if they contain aqueous media/solutions.
Glass syringes	Barrel and plunger separated (see §16.2.1).	Hot air oven 160 °C/1–2 h , or autoclave.

Article	Preparation	Method
Glass slides and cover slips	Clean, degrease in alcohol or ether, dry.	Flaming , or hot air oven; for used slides, immerse in disinfectant and then autoclave/discard.
Volumetric and graduated glassware — volumetric flasks, burettes, graduated cylinders, calibrated pipettes	—	Do NOT use the hot air oven — the high temperature can distort the glass and destroy the calibration . Use autoclaving at 121 °C, chemical disinfection (70% alcohol, glutaraldehyde) or aseptic rinsing with a sterile solvent, according to the requirement.
Sintered glass filters, glass filter holders	Clean by back-flushing and soaking in cleaning solution.	Autoclave, 121 °C for 15–20 minutes.
Glass ampoules and vials (sealed, containing solutions)	—	Autoclave in the final container, or filter-sterilize and fill aseptically if the contents are heat-labile; empty vials by dry heat (depyrogenation at 250 °C for 30 minutes or 180 °C for 3 hours to destroy endotoxin/pyrogens).

REMEMBER

- **Why is glassware sterilized by dry heat rather than steam?** Because it is **heat-stable and moisture-sensitive** — dry heat leaves it **completely dry** and ready for use, avoids water spotting, and penetrates the closed canisters and plugged tubes in which glassware is packed.
- **Depyrogenation** (removal of bacterial endotoxin) requires much harsher dry heat than sterilization — **250 °C for 30 min** — because endotoxin is far more heat-stable than the organism itself. Autoclaving does **not** destroy pyrogens.

16.4 Sterilization of Apparatus, Instruments and Other Hospital Articles

TABLE 16.3 ARTICLE-WISE METHOD OF CHOICE

Article / apparatus	Recommended method	Notes and cautions
Stainless-steel surgical instruments — forceps, artery clamps, retractors, bowls, kidney trays	Autoclave, 121 °C/15 psi/15–20 min , or 134 °C/3 ½ min in a pre-vacuum sterilizer	Open all hinges and box-joints; do not nest bowls tightly; lubricate hinged instruments after cleaning.
Sharp cutting instruments — scalpels, blades, scissors, needles, trocars, dental burs	Hot air oven (160 °C/2 h) , or a chemical sterilant (2% glutaraldehyde, 10 h) ; autoclaving is acceptable in modern practice but repeated cycles dull the edge	Repeated moist heat blunts and corrodes cutting edges — a classic examination point. Protect tips with silicone guards.
Rubber goods — surgical gloves, tubing, catheters, drains, bungs, closures, mackintosh, hot-water bottles	Autoclave at 115–121 °C for 15–30 min (a lower temperature and shorter cycle than for metal), or ethylene oxide ; new disposables are gamma-irradiated	Rubber perishes, becomes sticky and loses elasticity with repeated or excessive heat; powder gloves lightly, pack them without folding sharply, and never use dry heat . Do not exceed 5–6 cycles.
Plastic and polythene articles — tubing, cannulae, endotracheal and tracheostomy tubes, suction catheters, urine bags, three-way taps	Ethylene oxide, gamma radiation (industrial), hydrogen peroxide gas plasma , or 2% glutaraldehyde / LTSF for re-usable items; autoclavable plastics (polypropylene, PTFE, polycarbonate) may be autoclaved at 121 °C	Polystyrene, PVC and polyethylene melt or distort with heat . Check the manufacturer's instructions for every device.
Flexible fibre-optic endoscopes — gastroscopes, colonoscopes, bronchoscopes, cystoscopes, duodenoscopes	High-level disinfection with 2% alkaline glutaraldehyde for 20 min, 0.55% OPA for 5–12 min, or 0.2% peracetic acid in an automated endoscope reprocessor; EtO or H₂O₂ plasma when sterilization is required	Never autoclave — heat destroys the fibre-optic bundles, lens cement and seals. Meticulous manual cleaning and brushing of all channels first , then a leak test ; afterwards rinse with sterile water, flush with alcohol and dry , and hang vertically uncoiled. Inadequately dried endoscopes have caused <i>Pseudomonas</i> and <i>Mycobacterium</i> outbreaks.
Rigid endoscopes, laparoscopes, arthroscopes and their cables	Autoclave if labelled autoclavable ; otherwise EtO or hydrogen peroxide plasma , or glutaraldehyde	Check with the manufacturer; lensed instruments tolerate EtO and plasma well.

Article / apparatus	Recommended method	Notes and cautions
Anaesthetic and respiratory equipment — face masks, corrugated tubing, reservoir bags, laryngoscope blades, airways, ventilator circuits	Autoclave (121 °C) or pasteurising washer-disinfector at 73–80 °C or 2% glutaraldehyde; EtO for delicate parts; single-use circuits and bacterial/viral filters are now preferred	Laryngoscope blades are semi-critical — they need HLD as a minimum ; the handle needs low-level disinfection. Dry thoroughly before reassembly.
Surgical dressings, gauze, cotton, gowns, drapes, linen, caps, masks, towels	Autoclave — pre-vacuum porous-load cycle, 134 °C for 3–3½ min, or 121 °C for 15–30 min in drums with the vents open	Pack loosely; check the Bowie–Dick test daily; packs must be dry on removal. Do not use dry heat — cloth and paper char above 160 °C.
Culture media — nutrient broth and agar, blood agar base, MacConkey, peptone water	Autoclave, 121 °C/15 psi/15–20 min	Special cases: media with sugars, gelatin, milk or DCA — free steam at 100 °C or tyndallisation; egg/serum media (LJ, Dorset, Löffler's) — inspissation at 80–85 °C × 30 min × 3 days; urea broth, sera and antibiotic-containing media — membrane filtration (0.22 µm); blood, serum and antibiotic supplements — added aseptically to the cooled sterile base at 45–50 °C.
Heat-labile solutions — sera, vaccines, toxins, antibiotic, vitamin and hormone solutions, IV fluids with labile drugs	Membrane filtration through 0.22 µm, with aseptic filling under laminar air flow	Validate the filter (bubble-point test); remember that viruses, mycoplasma and pyrogens pass through .
Oils, fats, waxes, greases, liquid paraffin, ointment bases, glycerol, dusting and sulphonamide powders	Hot air oven, 160 °C for 2 hours (or 150 °C for 1 h for some oils)	Steam cannot penetrate anhydrous and oily materials — therefore dry heat is the only heat option. This is a very common MCQ.
Bed-pans, urinals, sputum mugs, kidney dishes, feeding utensils, measuring jugs	Washer-disinfector at 80–90 °C, boiling, or chemical disinfection with 1% hypochlorite or a phenolic	Non-critical items; cleaning is the main requirement.
Thermometers (clinical)	Wipe clean, then immerse in 70% alcohol or 2% glutaraldehyde for 10–20 minutes, rinse and dry; use individual or disposable-sheath thermometers	Never boil or autoclave a mercury thermometer — it will burst. Mercury thermometers are being replaced by digital ones (Minamata Convention).
Blood-pressure cuff, stethoscope, patient trolley, bed rails, furniture, floors, walls	Low-level disinfection — detergent cleaning, 70% alcohol wipe for the stethoscope diaphragm, phenolic or QAC for surfaces	Non-critical items; damp dusting only — never dry sweeping .
Mattresses, pillows, blankets, bedding, books, leather, documents	Ethylene oxide, formaldehyde fumigation, sunlight and airing; washable covers laundered at ≥ 71 °C	Impervious mattress covers are strongly preferred so that surface disinfection suffices.
Operation theatre — room, air and surfaces	Routine wet mopping and cleaning with a detergent-disinfectant ; terminal cleaning weekly; HEPA-filtered laminar air flow with 20–25 air changes per hour and positive pressure ; UV lamps when unoccupied; hydrogen peroxide vapour/fogging or formaldehyde fumigation for terminal decontamination; environmental (settle-plate) surveillance	Modern practice relies chiefly on ventilation, HEPA filtration, discipline and cleaning rather than on routine fumigation.
Hands of the surgeon and staff	Surgical scrub with 4% chlorhexidine or 7.5% povidone-iodine for 3–5 minutes, or an alcohol-based surgical rub; routine care by alcohol-based hand rub (20–30 s) or soap and water (40–60 s)	WHO 'My 5 Moments for Hand Hygiene': (1) before touching a patient, (2) before a clean/aseptic procedure, (3) after body-fluid exposure risk, (4) after touching a patient, (5) after touching patient surroundings.
Patient's skin and mucosa	Povidone-iodine, chlorhexidine–alcohol, 70% alcohol; never phenol, formaldehyde, glutaraldehyde or hypochlorite on tissue	See Table 15.3.

Article / apparatus	Recommended method	Notes and cautions
Excreta, sputum, pus, urine, vomit, blood spills	Cover a spill with absorbent material, pour 1% hypochlorite (or bleaching powder), leave 10 minutes, then remove with gloves and clean; sputum and faeces — 5% phenol or 1–2% hypochlorite for 1–2 hours , or autoclave/incinerate	Use gloves, mask, apron and eye protection; treat all body fluids as potentially infectious (standard precautions).
Drinking water	Chlorination (free residual chlorine ≥ 0.5 mg/L after 1 h), boiling for 5–10 minutes (the most reliable household method), filtration, UV , ozonation, SODIS	For wells, use bleaching powder with Horrock's apparatus to find the dose.
Milk	Pasteurisation — $63^{\circ}\text{C}/30$ min or $72^{\circ}\text{C}/15$ – 20 s; UHT	Checked by the phosphatase test .
Air	HEPA filtration and laminar flow , ventilation and air changes, UV lamps , chemical aerosols (glycol vapour), dust control	Air-borne organisms travel on dust and droplet nuclei.

16.5 The Central Sterile Supply Department (CSSD)

- A **single, centralised department** that receives, cleans, packs, sterilizes, stores and issues all sterile supplies for the whole hospital. Centralisation gives **better quality control, trained staff, proper monitoring, economy and safety**.
- **Layout** — **strict one-way (unidirectional) flow from dirty to clean, with physical separation of zones:** receiving & decontamination (dirty) → washing/ultrasonic/washer-disinfector → drying & inspection → packing & assembly (clean) → sterilization → sterile storage → issue. Air pressure is **negative in the decontamination zone and positive in the clean and sterile zones**.
- **Equipment:** washer-disinfectors, ultrasonic cleaners, drying cabinets, sealing machines, **pre-vacuum steam sterilizers**, hot air ovens, low-temperature sterilizers (EtO or H_2O_2 plasma), and trolleys.
- **Quality assurance:** daily **Bowie–Dick test**, physical printouts for every cycle, **chemical indicators in and on every pack, weekly (or daily) biological indicators**, load documentation and **traceability of each pack to the patient**, plus planned preventive maintenance and validation of every sterilizer.
- **Shelf life** depends on the wrap and the storage conditions and is now regarded as **event-related** — a pack remains sterile until its integrity is breached (wet, torn, dropped, opened). Indicative periods: single muslin wrap ≈ 1 week; double wrap ≈ 7 – 12 weeks; **sealed peel pouches ≈ 6 – 12 months**.
- **Staff safety:** personal protective equipment in the decontamination zone, **hepatitis B immunisation**, sharps safety, EtO and glutaraldehyde exposure monitoring, and training.

16.6 Ready Reckoner — Article versus Method

TABLE 16.4 METHOD → ARTICLE READY RECKONER (REVISE THIS TABLE LAST)

Method	Articles
Red heat	Inoculating loops and wires, tips of forceps, needles, searing spatulas
Flaming	Mouths of culture tubes, glass slides, cover slips, scalpels
Incineration	Soiled dressings, cotton swabs, anatomical and pathological waste, animal carcasses, placenta, laboratory cultures, expired and cytotoxic drugs, plaster casts
Hot air oven ($160^{\circ}\text{C}/2$ h)	Glassware (test tubes, petri dishes, pipettes, flasks), glass syringes , metal instruments, sharp cutting instruments, oils, fats, waxes, liquid paraffin, glycerol, dusting and sulphonamide powders , ointment bases, swabs
Autoclave ($121^{\circ}\text{C}/15$ psi/15 min)	Culture media , aqueous and parenteral solutions, surgical instruments, rubber goods and gloves (115–121°C) , glassware, autoclavable plastics, contaminated/discarded cultures and media (30–60 min) , laboratory waste, sintered glass filters
Autoclave ($134^{\circ}\text{C}/3\frac{1}{2}$ min, pre-vacuum)	Surgical dressings, gowns, drapes, linen, wrapped instrument trays (porous loads); 134°C for 18 min for prions
Free steam 100°C / tyndallisation	Culture media containing sugars, gelatin, milk, DCA, selenite F
Inspissation (80–$85^{\circ}\text{C} \times 30$ min $\times 3$ days)	Löwenstein–Jensen medium, Dorset egg medium, Löffler's serum slope
Pasteurisation	Milk ($63^{\circ}\text{C}/30$ min or $72^{\circ}\text{C}/15$ – 20 s); respiratory equipment (73 – 80°C)

Method	Articles
Gamma radiation (2.5 Mrad / 25 kGy)	Disposable plastic syringes and needles, IV and blood-transfusion sets, catheters, urine bags, surgical gloves, sutures, petri dishes, bone and tissue grafts, heart valves, prostheses, adhesive dressings (industrial, in the final pack)
Ultraviolet radiation	Air and exposed surfaces — operation theatres, laminar-flow hoods, biological safety cabinets, entry vestibules, water treatment
Filtration (0.22 µm membrane)	Sera, vaccines, toxins, antibiotic, vitamin and hormone solutions, urea broth, heat-labile parenteral and ophthalmic solutions, tissue-culture media
HEPA filtration	Air — operation theatres, laminar air flow, biological safety cabinets, clean rooms
Ethylene oxide	Plastics and rubber disposables, heart–lung machine, dialysers, prosthetic implants and heart valves, pacemakers, endoscopes, delicate and lensed instruments, catheters, sutures, powders, books, blankets, mattresses
Hydrogen peroxide gas plasma	Heat- and moisture-sensitive endoscopes, cameras, cables, batteries, delicate instruments (not cellulose, liquids or powders)
2% glutaraldehyde	Flexible endoscopes, cystoscopes, anaesthetic and respiratory equipment, plastic and rubber goods, dialysers, thermometers, lensed and sharp instruments
70% alcohol	Skin before injection , clinical thermometers, stethoscope diaphragm, rubber stoppers of vials, small clean surfaces
1% hypochlorite	Blood and body-fluid spills , plastic biomedical waste, surfaces, laboratory discard jars (1–5%)
5% phenol / 2–5% Lysol	Sputum, faeces, urine, pus , laboratory discard jars, floors, drains, bed-pans, ambulances



CHAPTER 17 DISPOSAL OF CONTAMINATED MEDIA AND BIOMEDICAL WASTE MANAGEMENT

17.1 Disposal of Contaminated Culture Media and Laboratory Material

Used culture media, plates and specimens carry an enormous bioburden of live, often pathogenic organisms. The cardinal rule is: **decontaminate FIRST, then discard**. Contaminated material must **never** be washed, thrown away or poured down a sink before it has been made safe.

TABLE 17.1 DISPOSAL OF CONTAMINATED MEDIA AND LABORATORY MATERIAL

Material	Method of decontamination	Then
Used culture plates, slopes, broths and contaminated media	Autoclave at 121 °C (15 psi) for 30–60 minutes — a longer holding time than the routine 15-minute cycle because the bioburden is very high and the load is bulky. Place the plates in an autoclavable bag or a leak-proof metal discard bin , not stacked tightly.	After autoclaving: the agar is melted and poured off (into a collection container, not down the drain, since agar solidifies and blocks the pipes); glass petri dishes are washed and re-sterilized; disposable plastic plates are shredded and disposed of as red-category waste (or, if the rules of the establishment require, as yellow-category microbiology waste for incineration).
Media that cannot be autoclaved, or where no autoclave is available	Incineration (the method of choice for microbiology and biotechnology waste under the Indian rules), or chemical disinfection by immersing in 1–5% sodium hypochlorite or 5% phenol for a minimum of 1–2 hours, preferably overnight .	Drain the disinfectant to the effluent treatment plant; dispose of the solids in the appropriate colour-coded bag.
Discard jars for pipettes, slides, tubes and small articles	Jars kept at every bench containing 1–5% hypochlorite, 5% phenol or 2% Lysol , into which the article is completely immersed immediately after use (pipettes tip-upward) and left overnight . The solution must be changed daily and must never be merely topped up.	Autoclave the jar and contents, then wash the glassware.
Specimens — blood, serum, urine, stool, sputum, CSF, swabs	Autoclave (121 °C/30–60 min) or add a disinfectant (1% hypochlorite or 5% phenol, 1–2 hours); highly infectious specimens are incinerated .	Liquid residues to the ETP; containers as per colour code.
Stock cultures, reference strains and vaccine residues; cultures of Biosafety Level 3–4 agents	Autoclave (121 °C for 60 min) on site — mandatory pre-treatment before the waste leaves the laboratory ; then incinerate.	The BMW Rules require that laboratory and microbiology waste, blood samples and blood bags be disinfected by autoclaving, microwaving or hydroclaving ON SITE before being handed over for final disposal.
Sharps contaminated in the laboratory — broken glass, slides, capillary tubes, scalpels	Never handled by hand; swept up with a brush and dustpan or forceps into a puncture-proof container , then autoclaved or chemically disinfected .	Contaminated broken glass — blue container; metal sharps and needles — white translucent puncture-proof container.
Liquid waste from the laboratory and from washing	Chemical pre-treatment with hypochlorite (for blood-containing effluent) and/or autoclaving , then discharged to the effluent treatment plant (ETP) and sewer, conforming to the prescribed effluent standards (pH 6.5–9.0, BOD, COD, suspended solids, oil and grease, bio-assay limits).	Never discharge untreated infectious liquid, or chemicals such as chromic acid, mercury or cytotoxic residues, into the sewer.
Animal carcasses, tissues and anatomical waste	Incineration (or deep burial where permitted).	Yellow category.

HIGH-YIELD FOR EXAM

- Contaminated / discarded culture media are autoclaved at 121 °C for 30–60 minutes before disposal — note that this is longer than the 15-minute cycle used to sterilize fresh media.
- Laboratory and microbiology waste must be pre-treated (autoclaved, microwaved or hydroclaved) on site before it leaves the laboratory — a specific requirement of the BMW Rules, 2016.
- Molten agar must never be poured down the sink — it solidifies and blocks the drain.

17.2 Biomedical Waste — Definition and Magnitude

Biomedical waste (BMW Rules, 2016) — Any waste which is generated during the diagnosis, treatment or immunisation of human beings or animals, or in research activities pertaining thereto, or in the production or testing of biologicals, or in any health camp, including the categories mentioned in Schedule I of the Rules.

- The governing law in India is the **Bio-Medical Waste Management Rules, 2016** (notified on 28 March 2016 under the **Environment (Protection) Act, 1986**), as amended in 2018 and 2019. They replaced the Bio-Medical Waste (Management and Handling) Rules, 1998.
- Of the total waste generated in a hospital, only about **15–25% is hazardous/infectious**; the remaining **75–85% is general, non-hazardous waste** similar to domestic refuse. **Mixing the two converts the whole lot into hazardous waste** — which is why **segregation at the point of generation** is the single most important step in the whole system.
- The regulatory authorities are the **Central Pollution Control Board (CPCB)** and the **State Pollution Control Boards / Pollution Control Committees**, which grant **authorisation**; the prescribed authority for a health-care facility is the SPCB/PCC.
- **Hazards of poor management:** transmission of **hepatitis B, hepatitis C and HIV** through needle-stick injury (the greatest occupational risk), other infections, injuries to waste handlers and rag-pickers, **repackaging and illegal resale of used syringes and gloves**, environmental pollution with dioxins, furans and mercury, and antimicrobial resistance.

17.3 Colour Coding and Categories — BMW Rules, 2016

The 1998 rules had **10 categories**; the 2016 rules simplified these into **four colour-coded categories**. This table is the most examinable single item in the chapter.

TABLE 17.2 THE FOUR COLOUR-CODED CATEGORIES (BMW RULES, 2016)

Colour / container	Waste included	Treatment and disposal
YELLOW (yellow non-chlorinated plastic bag)	(a) Human anatomical waste — tissues, organs, body parts, placenta. (b) Animal anatomical waste . (c) Soiled waste — items contaminated with blood, body fluids, or blood-soaked cotton, dressings, plaster casts, linen, beddings. (d) Expired or discarded medicines including cytotoxic drugs. (e) Chemical waste (solid and liquid) — discarded disinfectants, chemicals used in production of biologicals. (f) Discarded linen, mattresses and beddings contaminated with blood or body fluid . (g) Microbiology, biotechnology and other clinical laboratory waste — blood bags, laboratory cultures, stocks of micro-organisms, live or attenuated vaccines, culture dishes, devices used to transfer cultures.	Incineration ($\geq 1200^\circ\text{C}$ for cytotoxic waste), plasma pyrolysis or deep burial (deep burial only in towns with a population below 5 lakh and in rural areas). Laboratory, microbiology, blood-sample and blood-bag waste must first be pre-treated on site by autoclaving/microwaving/hydroclaving . Chemical liquid waste — pre-treatment then ETP. Cytotoxic waste — incineration at $\geq 1200^\circ\text{C}$, or return to the manufacturer, or encapsulation . Never autoclave or land-fill cytotoxic waste .
RED (red non-chlorinated plastic bag or container)	Contaminated recyclable waste generated from disposable items — tubing, bottles, intravenous tubes and sets, catheters, urine bags, syringes (without needles and with the fixed needle cut off), vacutainers with the needle cut, and gloves.	Autoclaving, microwaving or hydroclaving , followed by mutilation / shredding , and then sent to a registered / authorised recycler or for energy recovery. Treated waste must not be sent to a landfill .
WHITE (translucent) (puncture-proof, leak-proof, tamper-proof container)	Waste sharps including metals — needles, syringes with fixed needles, needles from needle-tip cutters or burners, scalpels, blades, and any contaminated sharp object that may cause puncture or cut, including both used and discarded/unused sharps.	Autoclaving or dry-heat sterilization followed by shredding, mutilation, or encapsulation in a metal container or cement concrete , and finally disposal in a sharps pit, a concrete sharps pit, or sent to a metal smelter/authorised recycler .
BLUE (cardboard box with a blue marking, containing a puncture-proof container)	(a) Glassware — broken, discarded and contaminated glass, including medicine vials and ampoules (except those contaminated with cytotoxic drugs, which go to yellow). (b) Metallic body implants .	Disinfection by soaking in 1% hypochlorite, or autoclaving / microwaving / hydroclaving , then sent for recycling to an authorised glass or metal recycler.

MNEMONIC

- **Yellow = burn it** (incinerate) · **Red = Recycle the plastic** · **White = the sharps that make you bleed** · **Blue = Bottles, Broken glass and Body implants.**
- **Key discriminators:** a syringe **with its needle still attached** → **WHITE**; a syringe **whose needle has been cut off** → **RED**. A **glass vial** → **BLUE**; a **cytotoxic-contaminated vial** → **YELLOW**. **Gloves** → **RED**. **Placenta and soiled dressings** → **YELLOW**. **Laboratory cultures** → **YELLOW** (after on-site pre-treatment).

17.4 Treatment Technologies and Their Standards

TABLE 17.3 TREATMENT TECHNOLOGIES AND PRESCRIBED STANDARDS

Technology	Operating standard	Used for / remarks
Incineration	Primary chamber 800 ± 50 °C; secondary chamber 1050 ± 50 °C with a gas residence time of at least 2 seconds; stack height ≥ 30 m; continuous emission monitoring; ≥ 1200 °C for cytotoxic waste	Yellow category — anatomical, soiled, expired medicines, laboratory waste. Chlorinated plastics (PVC), mercury, radioactive material, halogenated chemicals and pressurised containers must NEVER be incinerated — they generate dioxins and furans or explode. Ash is disposed of in a secured landfill.
Autoclaving	Gravity flow: 121 °C & 15 psi for 60 min , or 135 °C & 31 psi for 45 min , or 149 °C & 52 psi for 30 min . Vacuum type: 121 °C & 15 psi for 45 min . Validated weekly with <i>Geobacillus stearothermophilus</i> spores (a 4-log ₁₀ reduction is required) and monitored with chemical indicators for every batch.	Red and white categories, and on-site pre-treatment of laboratory waste. Not for anatomical, cytotoxic, volatile chemical or radioactive waste.
Microwaving	Frequency 2450 MHz , wavelength 12.24 cm; the waste must reach a minimum of 97.5 °C throughout for at least 30 minutes ; a 4-log reduction of <i>B. atrophaeus</i> spores must be demonstrated	Alternative to autoclaving for red/white waste. Not for anatomical, cytotoxic, radioactive or large metal items.
Hydroclaving	Similar to autoclaving but with continuous mechanical mixing and fragmentation of the waste (e.g. 121 °C for 20 min, 111 °C for 30 min)	Combined treatment and volume reduction.
Chemical disinfection	1% sodium hypochlorite (or an equivalent), with adequate contact time and full immersion	Blue-category glassware, liquid waste, spills, and surfaces. Note that chemical treatment of plastics before shredding is now discouraged in favour of autoclaving.
Deep burial	A pit 1.5–2 m deep , half filled with waste, then covered with lime and finally with at least 50 cm of soil ; sited away from habitation and from any water source, on impermeable soil with a low water table, fenced and with records maintained	Permitted ONLY for towns with a population of less than 5 lakh and for rural areas , and only for yellow-category anatomical waste where no CBWTF is accessible.
Plasma pyrolysis / gasification	Very high temperature (thousands of °C) in an oxygen-starved environment, which breaks the waste into elemental gases and slag	An environmentally superior alternative to incineration; permitted for yellow-category waste.
Shredding / mutilation / encapsulation / inertisation	Mechanical destruction of form so that items cannot be reused ; encapsulation in a metal drum or cement concrete; inertisation by mixing with cement and lime	Mandatory after disinfection of red and white waste , to prevent illegal repackaging and resale. Encapsulation and sharps pits are used for sharps and for cytotoxic residues.
Sharps pit	A circular or rectangular concrete-lined, covered pit into which disinfected sharps are dropped through a pipe	For sharps in small facilities without access to a CBWTF.
Common Bio-medical Waste Treatment Facility (CBWTF)	A shared, authorised facility to which member health-care establishments hand over their segregated waste; it must be located within a prescribed distance and collect waste daily	Every health-care facility must use a CBWTF where one is available within 75 km; only then may it install its own captive treatment equipment.

17.5 Handling Rules, Duties and Documentation

► 17.5.1 Segregation, Packaging, Transport and Storage

- **Segregation at the point of generation** into the correct colour-coded container is the **responsibility of the person generating the waste** — never of the sweeper. Bins must be placed **at the bedside/point of use**, foot-operated and lidded, lined with the correct non-chlorinated bag, and **filled only to three-quarters**.
- Bags and containers must carry the **biohazard symbol** (yellow, red and white categories) or the **cytotoxic hazard symbol** for cytotoxic waste, together with a label showing the **date, waste category, sender's name, address and contact details**.
- **Bar-coding and global positioning system (GPS) tracking** of bags and of collection vehicles were made **mandatory** by the 2016 Rules.
- **No untreated biomedical waste may be stored beyond 48 hours**; if longer storage is unavoidable, the prescribed authority must be informed and the waste kept refrigerated.
- **Transport** within the facility is by **covered, dedicated, labelled, leak-proof wheeled trolleys** (never by hand or chute) along a **defined route away from patient-care areas**; off-site transport is only in **authorised, labelled vehicles** with the manifest system.
- **Untreated waste must not be mixed with municipal solid waste**, and **no biomedical waste may be given to a rag-picker or an unauthorised person**.

► 17.5.2 Needle and Sharps Safety

- **Never recap, bend, break or manipulate a used needle by hand, and never pass a sharp from hand to hand.** Use a **needle/hub cutter or needle destroyer at the point of generation**.
- Dispose of sharps **immediately after use, by the user**, into a **puncture-proof, leak-proof, tamper-proof white container** placed within arm's reach; **never overfill beyond three-quarters** and never reach into the container.
- Prefer **auto-disable (AD), retractable and safety-engineered devices and needle-free connectors**.
- **All health-care staff must be immunised against hepatitis B** and trained annually.
- **After a needle-stick injury**: do **not** squeeze or suck the wound; **wash immediately with soap and running water** (flush mucosa/eyes with water or saline); **report at once** and record the incident; assess the source patient; and start **post-exposure prophylaxis (PEP)** — for **HIV, ideally within 2 hours and certainly within 72 hours**, a 28-day course of antiretrovirals; for **hepatitis B**, hepatitis B immunoglobulin plus vaccine as indicated by the vaccination and antibody status; follow-up testing at 6 weeks, 3 months and 6 months.

► 17.5.3 Duties of the Occupier and Records

- Obtain and renew **authorisation** from the SPCB/PCC; every facility with beds must be authorised, and a bed-less clinic requires a one-time authorisation.
- Ensure **segregation, pre-treatment, safe storage, transport and handing over to a CBWTF**; **establish a Bar-Code system**; provide **training to all staff at induction and annually**, and maintain training records.
- Provide **personal protective equipment** (gloves, masks, aprons, gowns, boots, goggles), **immunise all workers against hepatitis B and tetanus**, and arrange **health check-ups**.
- Constitute a **Bio-medical Waste Management Committee** (mandatory for facilities with 30 beds or more) which meets at least **every six months**.
- Maintain **daily records** of waste generated category-wise, and submit an **annual report to the prescribed authority by 30 June** for the preceding calendar year.
- **Report any major accident** involving biomedical waste immediately, and maintain an accident register; display the **monthly record on the website** where applicable.
- **Phase out chlorinated plastic bags, gloves and blood bags** (a requirement introduced by the 2016 Rules) and **stop the use of mercury equipment**.
- Ensure that liquid waste is **pre-treated and meets the effluent discharge standards** before release.

17.6 Spill Management and Standard Precautions

1. **Restrict access** to the area and put on **PPE** — gloves, mask, apron, and eye protection.
2. **Cover the spill** with absorbent material (paper towels, sawdust or absorbent granules) to prevent aerosolisation and spread.
3. **Pour a disinfectant** — **1% sodium hypochlorite (10 000 ppm)** — from the **periphery towards the centre** and leave for **at least 10 minutes** (20–30 minutes for a large spill).
4. **Remove** the absorbed material with forceps or a scoop (never with bare hands; pick up any glass with a dustpan and brush) and place it in the **yellow bag**.

5. **Clean** the area with detergent and water, then **disinfect again** and allow to dry.
 6. **Discard** the PPE appropriately, **wash hands**, and **record the incident**; replenish the spill kit.
- **Standard (universal) precautions** apply to **every patient, at all times, regardless of diagnosis**, on the principle that **all blood and body fluids are potentially infectious**. They comprise: **hand hygiene, appropriate PPE, safe injection practice and sharps handling, safe handling of contaminated equipment, linen and surfaces, respiratory hygiene and cough etiquette, patient placement, and waste management**.
 - The **single most effective measure** for preventing health-care-associated infection remains **hand hygiene**.

17.7 Pharmacy-Specific Disposal

Item	Correct disposal
Expired, unused, damaged and returned medicines	Yellow category — incineration (or encapsulation/inertisation). They must never be resold, given away, flushed down a drain or thrown into municipal waste. Records of destruction must be maintained and, for scheduled drugs, the prescribed procedure and witnesses followed.
Cytotoxic and antineoplastic drugs, their vials, spillage and contaminated PPE	Yellow category with the cytotoxic hazard label — incineration at $\geq 1200^\circ\text{C}$, or return to the manufacturer , or encapsulation. Never autoclave, never landfill, never put a cytotoxic vial in the blue bin . Prepare only in a Class II B2 biological safety cabinet with full PPE, using a closed-system transfer device, and keep a spill kit.
Narcotic and psychotropic (NDPS) drugs	Destroyed only in the presence of the authorised officer/committee under the NDPS Rules, with a certificate of destruction; recorded in the narcotic register.
Empty vials and ampoules (non-cytotoxic)	Blue category — disinfect and send for glass recycling. Vials that contained live or attenuated vaccines are yellow (laboratory/biological waste).
Half-used vials with residual drug, and reconstituted antibiotics	Discard the residue as chemical/pharmaceutical waste — yellow ; the glass then goes to blue after rinsing where permissible.
Mercury thermometers, sphygmomanometers and dental amalgam	Never incinerate . Collect spilled mercury with a mercury spill kit into a sealed, labelled, unbreakable container under water and hand over to an authorised hazardous-waste handler; phase out mercury devices.
Chemicals — chromic acid, solvents, reagents, disinfectant concentrates	Yellow category chemical waste ; neutralise or pre-treat where specified and hand over to an authorised handler. Never pour concentrated chemicals down the sink .
Radiopharmaceuticals	Governed not by the BMW Rules but by the Atomic Energy Regulatory Board (AERB) — decay-in-storage, then disposal under AERB authorisation.
Batteries, e-waste and pressurised containers (aerosol inhalers, cylinders)	Separate streams under the E-Waste and Battery Waste Management Rules ; pressurised containers must never be incinerated — they explode.

HIGH-YIELD FOR EXAM

- **BMW Rules 2016: 4 colour categories — Yellow, Red, White (translucent), Blue**. The 1998 rules had **10 categories**.
- **Storage limit for untreated waste = 48 hours**. Annual report due by **30 June**. BMW committee mandatory for **≥ 30 beds**.
- **Incinerator: primary $800 \pm 50^\circ\text{C}$, secondary $1050 \pm 50^\circ\text{C}$, gas retention ≥ 2 seconds; cytotoxic $\geq 1200^\circ\text{C}$.**
- **BMW autoclave: $121^\circ\text{C}/15\text{ psi}/60\text{ min}$ (gravity) or 45 min (vacuum). Microwave: 2450 MHz , $\geq 97.5^\circ\text{C}$, 30 min .**
- **Deep burial: only for population < 5 lakh and rural areas; pit $1.5\text{--}2\text{ m}$ deep with lime and $\geq 50\text{ cm}$ soil cover.**



CHAPTER 18 QUICK REVISION — MASTER TABLES AND READY RECKONER

Revise this chapter last, on the day before the examination. Everything here has already been explained in the preceding chapters; this is pure recall.

18.1 Temperature and Time — The Numbers That Must Be Automatic

TABLE 18.1 MASTER TEMPERATURE-TIME TABLE

Process	Temperature	Time / condition
Autoclave — standard	121 °C at 15 psi (1.05 kg/cm ²)	15–20 minutes
Autoclave — rapid	126 °C at 20 psi	10 minutes
Autoclave — pre-vacuum porous load	134 °C at 30 psi	3–3½ minutes
Autoclave — prions (CJD)	134 °C	18 minutes
Autoclave — discarded cultures / contaminated media	121 °C	30–60 minutes
Autoclave — biomedical waste (gravity)	121 °C at 15 psi	60 minutes
Autoclave — rubber goods	115–121 °C	15–30 minutes
Hot air oven — standard	160 °C	2 hours
Hot air oven — alternatives	170 °C / 180 °C / 150 °C	1 hour / 30 minutes / 2½ hours
Hot air oven — glass syringes	160 °C	1–2 hours
Depyrogenation (destruction of endotoxin)	250 °C	30 minutes
Incineration	Primary 800 ± 50 °C; secondary 1050 ± 50 °C	Gas retention ≥ 2 seconds; cytotoxic ≥ 1200 °C
Infrared	180 °C	7½–10 minutes
Pasteurisation — Holder (LTLT)	63 °C	30 minutes
Pasteurisation — Flash (HTST)	72 °C	15–20 seconds
Pasteurisation — UHT	125–150 °C	1–5 seconds
Inspissation (LJ medium)	80–85 °C	30 min × 3 successive days
Tyndallisation	100 °C (free steam)	20–45 min × 3 successive days
Steam at atmospheric pressure (Koch/Arnold)	100 °C	90 minutes
Boiling	100 °C	10–30 minutes (+2% NaHCO ₃)
Vaccine bath	60 °C	1 hour
Serum bath / complement inactivation	56 °C	1 hour / 30 minutes
Low-temperature steam formaldehyde (LTSF)	73–80 °C	≈ 2 hours
Microwave (BMW)	≥ 97.5 °C at 2450 MHz	30 minutes
Ethylene oxide	37–63 °C (usually 55–60 °C), RH 30–60%, 450–1200 mg/L	1–6 hours + aeration 8–12 h or more
Hydrogen peroxide gas plasma	40–50 °C	45–75 minutes
Gamma radiation	Ambient ('cold')	2.5 Mrad = 25 kGy from Cobalt-60
UV radiation	—	253.7 nm, surface/air only
Glutaraldehyde 2% — HLD	20–25 °C	20–45 minutes
Glutaraldehyde 2% — sterilization	20–25 °C	3–10 hours
Ortho-phthalaldehyde 0.55%	20–25 °C	5–12 minutes
Formaldehyde fumigation	> 20 °C, RH > 60–70%	Sealed 12–24 hours, then neutralise with ammonia
Hand rub / hand wash	—	Alcohol rub 20–30 s; soap and water 40–60 s; surgical scrub 3–5 min

18.2 Concentrations to Remember

TABLE 18.2 CONCENTRATIONS OF CHEMICAL AGENTS

Agent	Concentration	Use
Ethyl alcohol	70% (range 60–90%)	Skin antiseptic, hand rub
Isopropyl alcohol	60–80%	Skin, surfaces
Formalin	37–40% formaldehyde solution; 10% formalin for fixation	Fixative, fumigation
Glutaraldehyde	2% alkaline (pH 7.5–8.5)	Endoscopes
Ortho-phthalaldehyde	0.55%	Endoscopes
Phenol	1–5% (5% for discard jars)	Excreta, discard jars
Lysol / cresol	2–5%	Floors, drains, bed-pans
Chloroxylonol (Dettol)	4.8% concentrate, diluted	Domestic antiseptic
Chlorhexidine gluconate	0.5% in alcohol ; 2–4% scrub ; 0.2% mouthwash	Skin, hands, mouth
Hexachlorophene	3%	Surgical scrub (restricted)
Sodium hypochlorite	1% (10 000 ppm) spills; 5% discard jars; 0.1% surfaces	Spills, waste, surfaces
Bleaching powder	33% available chlorine (fresh)	Water, excreta
Drinking-water chlorination	Free residual 0.5 mg/L after 1 h (≥ 0.7 in epidemics)	Water supply
Tincture of iodine	2% iodine in 50% alcohol	Skin
Povidone-iodine	10% (= 1% available iodine) ; 7.5% scrub; 2.5% ophthalmic	Skin, mucosa, wounds
Hydrogen peroxide	3% wounds; 6–7.5% HLD; 30–35% vapour	Wounds, surfaces, rooms
Peracetic acid	0.2–0.35%	Endoscopes, dialysers
Potassium permanganate	1:1000 – 1:10 000	Wells, dressings, gargles
Silver nitrate	1% (Credé's); 0.5% burns	Ophthalmia neonatorum
Mercuric chloride	1:500 – 1:1000	Obsolete disinfectant
Thiomersal	1:10 000	Preservative in biologicals
Cetrimide / benzalkonium (QAC)	0.1–2%	Surfaces, skin, preservative
Gentian violet	1%	Oral thrush, skin
Acriflavine	0.1%	Wounds, burns
Beta-propiolactone	0.2%	Fumigation, vaccines
Acetic acid	1–2%	<i>Pseudomonas</i> in burns, ear, urine

18.3 Who's Who — Eponyms and Names

Name	Contribution
Joseph Lister (1867)	Introduced carbolic acid (phenol) into surgery — the father of antiseptic surgery .
Louis Pasteur	Pasteurisation ; disproved spontaneous generation; the hot air oven bears his name; rabies and anthrax vaccines.
Robert Koch	Koch's steamer (free steam at 100 °C); Koch's postulates; the Koch–Henle method of inspissation.
John Tyndall	Tyndallisation — intermittent sterilization.
Charles Chamberland	The Chamberland (porcelain) candle filter , graded L1–L13.
Ignaz Semmelweis	Introduced hand washing with chlorinated lime in obstetrics — the pioneer of hand hygiene.
Earle H. Spaulding	The critical / semi-critical / non-critical classification of instruments.
Rideal and Walker	The phenol coefficient test (without organic matter).
Chick and Martin	The phenol coefficient test in the presence of organic matter .
Kelsey and Sykes / Kelsey and Maurer	The capacity test and the in-use test for disinfectants.

Name	Contribution
Credé	1% silver nitrate eye drops for ophthalmia neonatorum.
Florence Nightingale	Hospital hygiene, ventilation and sanitation; founder of modern nursing.
Berkefeld	The kieselguhr candle filter — grades V, N, W.
Seitz	The asbestos pad filter .
Bowie and Dick	The Bowie–Dick test for steam penetration in a pre-vacuum autoclave.
Browne	Browne's tubes — chemical indicators that turn green.
Horrock	Horrock's apparatus — determination of the chlorine demand of well water.

18.4 One-Line Facts Most Often Asked

- **Most reliable and most widely used method of sterilization** → autoclaving (moist heat under pressure).
- **Most reliable monitor of sterilization** → biological indicator.
- **Most resistant form of microbial life** → prions, then bacterial spores.
- **Most resistant organism to radiation** → viruses.
- **Most heat-resistant milk pathogen** → *Coxiella burnetii*.
- **Commonest cause of autoclave failure** → failure to remove air.
- **True index of autoclave performance** → temperature, not pressure.
- **Method for oils, fats, waxes and powders** → dry heat (hot air oven).
- **Method for disposable plastic syringes** → gamma radiation.
- **Method for glass syringes** → hot air oven 160 °C/1–2 h, barrel and plunger separated.
- **Method for heat-labile sera and vaccines** → membrane filtration (0.22 µm).
- **Method for endoscopes** → 2% glutaraldehyde.
- **Method for prosthetic implants, pacemakers and heart–lung machine** → ethylene oxide.
- **Method for Löwenstein–Jensen medium** → inspissation.
- **Method for media containing sugar or gelatin** → tyndallisation / free steam.
- **Best method for blood spills** → 1% sodium hypochlorite.
- **Best single measure to prevent hospital infection** → hand hygiene.
- **First and most important step before any sterilization** → cleaning.
- **Sterility assurance level required** → 10⁻⁶.
- **Sterilising-grade filter pore size** → 0.22 µm.
- **HEPA filter efficiency** → 99.97% of particles ≥ 0.3 µm.
- **Alcohol concentration of choice** → 70%, because water is needed for protein denaturation.
- **Disinfectant inactivated by soap** → quaternary ammonium compounds (and chlorhexidine).
- **Disinfectant least affected by organic matter** → glutaraldehyde (and phenolics).
- **Agent that is carcinogenic yet widely used as a gaseous sterilant** → ethylene oxide (also formaldehyde and beta-propiolactone).
- **Colour bag for a used syringe with the needle cut off** → red; with the needle attached → white.
- **Colour bag for a placenta** → yellow; for a glass vial → blue; for gloves → red.
- **Maximum storage time for untreated biomedical waste** → 48 hours.

18.5 Glossary of Terms

Term	One-line meaning
Asepsis	Practices that keep organisms away from the article or tissue.
Antisepsis	Destroying or inhibiting organisms on living tissue.
Bactericidal	Kills bacteria.
Bacteriostatic	Inhibits multiplication without killing.
Bioburden	Number and type of organisms present before treatment.
Biofilm	Community of organisms in a polysaccharide matrix on a surface; highly resistant.
Cleaning	Physical removal of soil and organisms.
Decontamination	Making an article safe to handle.

Term	One-line meaning
Depyrogenation	Destruction of bacterial endotoxin (250 °C/30 min).
Disinfection	Destruction of vegetative pathogens; spores may survive.
D-value	Time or dose for a 1-log (90%) reduction.
Fumigation	Disinfection of an enclosed space by a gas or vapour.
Germicide	Any agent that kills micro-organisms.
HLD	High-level disinfection — kills all but large numbers of spores.
Oligodynamic action	Antimicrobial action of very low concentrations of heavy metals.
Pasteurisation	Heating below 100 °C to kill specified pathogens without sterilizing.
Preservative	Prevents microbial spoilage of a product during storage.
Pyrogen	Fever-producing substance, chiefly bacterial endotoxin; heat-stable and not removed by filtration.
SAL	Sterility assurance level; required value 10^{-6} .
Sanitisation	Reduction of organisms to a publicly safe level.
Sporicidal	Kills bacterial spores.
Sterilant	Chemical capable of complete sterilization.
Sterilization	Complete destruction or removal of ALL microbial life, including spores.
Substantivity	Persistence of an antiseptic's activity on the skin (e.g. chlorhexidine).
Terminal disinfection	Disinfection after the patient has been removed.
Tyndallisation	Free steam 100 °C for 20–45 min on 3 successive days.
Z-value	Temperature change producing a tenfold change in D-value.

REMEMBER

Final revision strategy: on the last day read only **Table 10.1 (definitions)**, **Table 10.4 (resistance ladder)**, **Table 10.6 (biological indicators)**, **Table 12.3 (pressure–temperature–time)**, **Table 12.4 (dry vs moist heat)**, **Table 15.3 (agent of choice)**, **Table 16.4 (article → method)**, **Table 17.2 (colour coding)**, and **Tables 18.1–18.2 (numbers)**. For Part I, read Table 3.1 (principles), Tables 8.1–8.4 (methods and AV aids) and Tables 9.1–9.2 (dates).

